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(54) **AMYLOID PRECURSOR PROTEIN AS A DIAGNOSTIC MARKER FOR BILIARY ATRESIA**

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CPC G01N 33/6893; G01N 2333/705; G01N 2333/4709; G01N 2800/085; C12Q 1/6883; C12Q 2600/158
See application file for complete search history.

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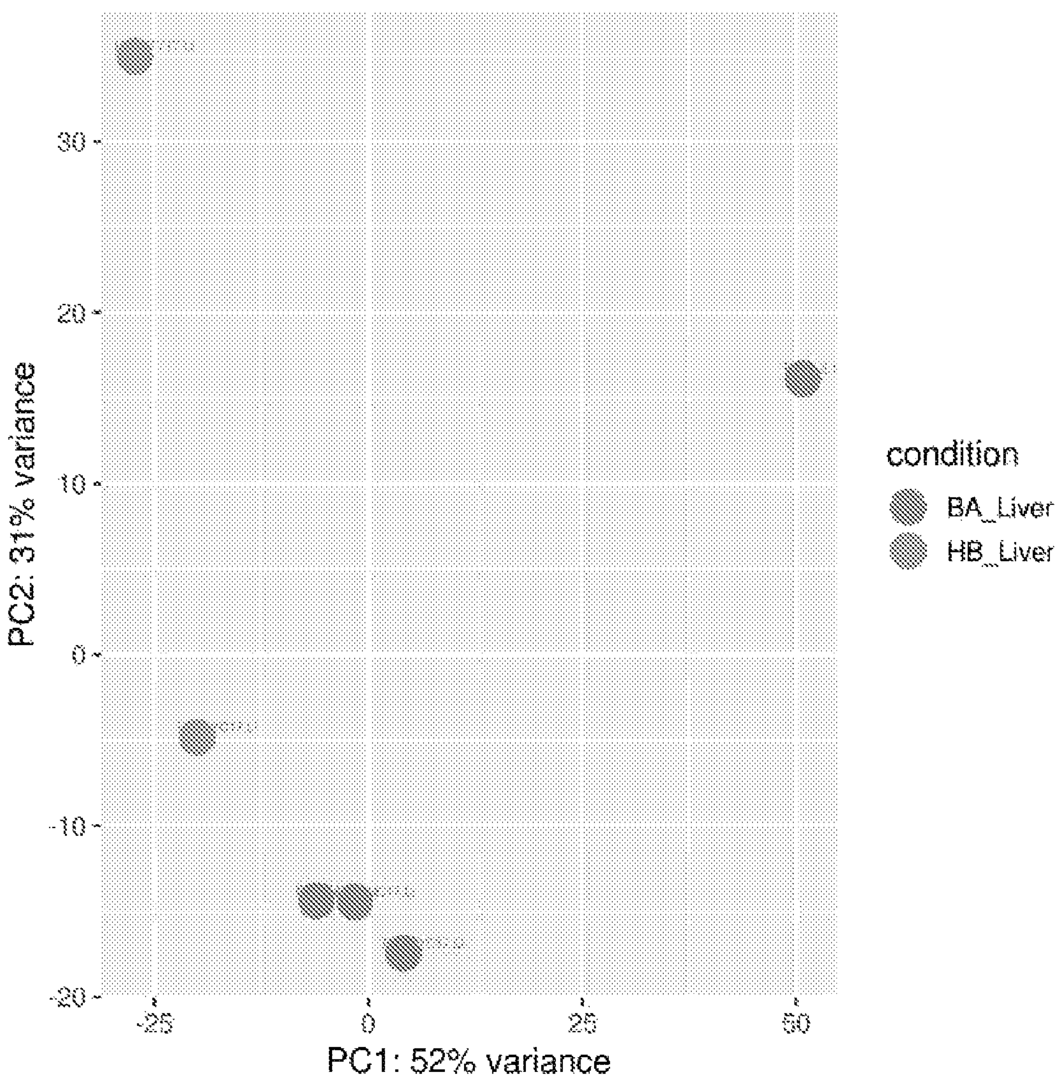
(57) **ABSTRACT**

Provided are novel diagnostic and prognostic methods for biliary atresia using amyloid precursor protein.

15 Claims, 2 Drawing Sheets

Specification includes a Sequence Listing.

Figure 1 PCA plot for liver biopsies RNA-seq analysis



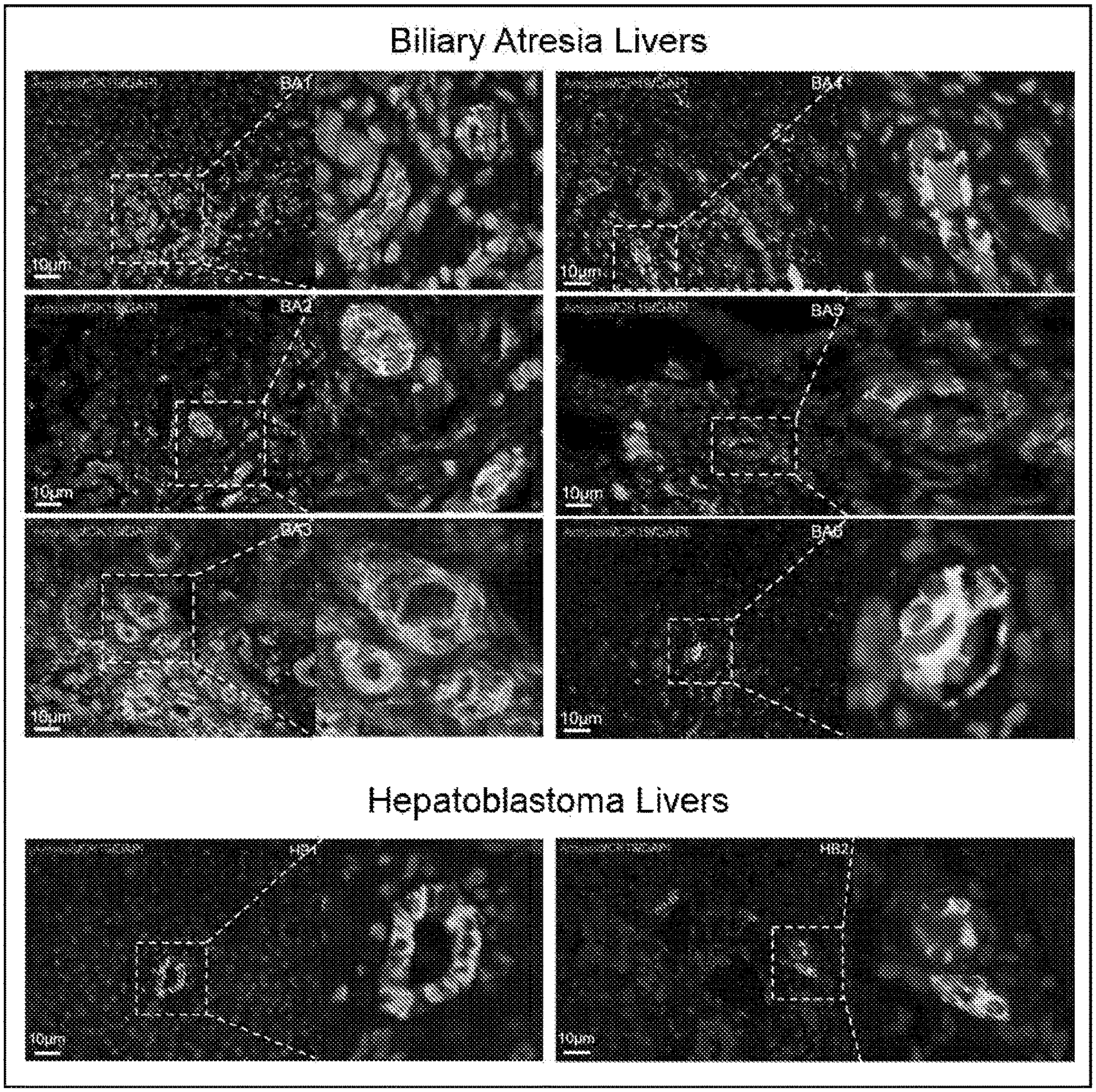

Biliary Atresia Livers
BA1
BA4
BA2
BA5
BA3
BA6
10μm
Hepatoblastoma Livers
HB1
HB2
10μm

Figure 2

AMYLOID PRECURSOR PROTEIN AS A DIAGNOSTIC MARKER FOR BILIARY ATRESIA

SEQUENCE LISTING

A Sequence Listing conforming to the rules of for Application filed prior to Jul. 1, 2022 is hereby incorporated by reference. Said Sequence Listing has been filed as an electronic document via the USPTO patent electronic filing system in ASCII formatted text. The electronic document, created on May 17, 2025 is entitled "104525-1253597-000310US_ST25", and is 82,065 bytes in size.

BACKGROUND

Biliary atresia is a disease of the liver and bile ducts that occurs in infants. Typically, symptoms of biliary atresia appear or develop about two to eight weeks after birth. Cells within the liver produce bile, a digestive liquid that helps digest fat. It also carries waste products from the liver to the intestines for excretion. This network of channels and ducts in which bile is produced, digests fat, and eventually moves to intestines is called the biliary system. Under the normal physiological conditions, the biliary system allows the bile to drain from the liver into the intestines. When a baby has biliary atresia, however, bile flow from the liver to the gallbladder is blocked. This causes the bile to be trapped inside the liver, which can quickly cause damage and scarring of the liver cells, resulting in liver cirrhosis and eventually liver failure.

While the exact cause of biliary atresia is yet to be fully illustrated, it is believed that some infants develop biliary atresia due to their bile ducts not formed properly during pregnancy. For most other children with biliary atresia, their bile ducts may be damaged by the body's immune system in response to a viral infection acquired after birth. Although biliary atresia is a rare disease, it affects the Asian population at a notably higher rate than other ethnicities. Diagnosis of biliary atresia relies on blood tests, ultrasound of the abdomen, liver biopsy, as well as diagnostic surgery (which may include an operative cholangiogram). Once diagnosed, biliary atresia is treated by Kasai procedure or hepatoportoenterostomy. With a 65-85% success rate routinely achieved by this procedure, failure still can result in a small percentage of infants, who will then likely require a liver transplant for long term survival. Because of this life-threatening potential of biliary atresia, there exists an urgent need for new and more reliable methods for the early diagnosis of infants suspected of having biliary atresia and for risk assessment among those who have undergone Kasai operation. This invention fulfills this and other related needs.

SUMMARY OF THE INVENTION

The present inventors observed increased expression of amyloid precursor protein in the liver tissue of an infant who is at risk of developing biliary atresia or is suffering from biliary atresia and has a poor outcome of biliary atresia treatment by Kasai operation. This discovery thus allows new methods to be devised for use in the diagnosis or prognosis of biliary atresia.

In the first aspect, the present invention provides a method for diagnosing or assessing the risk of biliary atresia (BA) in an infant by detecting in a liver sample taken from the infant the level of amyloid precursor protein (APP). The method includes these steps: (i) determining the expression level of

APP in a liver sample taken from the infant; (ii) detecting an increase in the APP expression level in comparison to a standard control level; and (iii) determining the infant as having BA or at an increased risk of developing biliary atresia.

In some embodiments, the APP expression level is APP mRNA level or protein level. In some embodiments, step (i) comprises a reverse transcription polymerase chain reaction (RT-PCR) or an immunoassay. In some embodiments, the liver sample is a liver biopsy. In some embodiments, the method further includes after step (iii) a step of performing at least one additional diagnostic test for BA to confirm the diagnosis. In some embodiments, the method may further include after step (iii) a step of treating the infant with Kasai procedure.

In the second aspect, the present invention provides a method for assessing the likelihood of poor outcome (e.g., ineffective Kasai procedure) in an infant after the infant has been diagnosed with BA and has received Kasai operation. The method includes these steps: (i) determining the expression level of APP in a liver sample taken from the infant; (ii) detecting an increase in the APP expression level in comparison to a standard control level; and (iii) determining that the Kasai procedure is likely a failure.

In some embodiments, the APP expression level is APP mRNA level or protein level. In some embodiments, step (i) comprises a reverse transcription polymerase chain reaction (RT-PCR) or an immunoassay. In some embodiments, the liver sample is a liver biopsy. In some embodiments, the method further includes after step (iii) a step of performing at least one additional diagnostic test for BA to confirm the prognosis-if the additional test for BA indicates presence of BA, one may conclude the treatment has likely failed. In some embodiments, the method may further include after step (iii) a step of providing the infant with further treatment such as liver transplant.

In a third aspect, the present invention provides a kit for diagnosing BA, assessing risk of developing BA, or prognosing unsuccessful Kasai procedure in an infant. The kit comprises a first container containing at least one reagent for detecting the mRNA or protein level of APP and a second container containing a standard control sample having an average level of APP expression in healthy liver tissue.

In some embodiments, the APP expression level is APP mRNA level or APP protein level. In some embodiments, the first container contains one or more reagents for an RT-PCR. In some embodiments, the first container contains one or more reagents for an immunoassay. In some embodiments, one or more reagents contained in the first container include a set of primers for PCR. In some embodiments, one or more reagents contained in the first container include an antibody against APP. In some embodiments, the kit further includes user instructions for using the kit.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**. Principal component analysis of control and BA liver.

FIG. **2**. Expression of APP in BA bile ducts. Sections of livers of biliary atresia (A) and normal controls (B; Hepatoblastoma) were immuno-fluorescent stained for APP (red) and CK19 (green; bile duct marker). Photos of liver sections of six BA patients (BA1 to BA6) and two HB patients (HB1 and HB2) were shown for comparison. Regions highlighted were enlarged and shown on the right.

DEFINITIONS

As used herein, an "amyloid precursor protein (APP)" refers to a membrane protein encoded by a gene located on chromosome 21 (in human). A highly conserved protein found in many species, there are several known isoforms of human APP due to alternative splicing, ranging in length from 639 to 770 amino acids, with certain isoforms preferentially expressed in neurons. An integral membrane protein expressed in many tissues and concentrated in the synapses of neurons, APP has been implicated as a regulator of synapse formation, neural plasticity, and iron export, although its precise functions are yet to be fully understood. The term "amyloid precursor protein" or "APP" is used herein to broadly encompass any isoforms of the protein such as naturally occurring homologues or orthologues or mutants, including proteins or peptides derived from the full-length APP such as peptides generated by its proteolysis, for instance, beta amyloid (Aβ), a polypeptide containing about 37 to about 49 amino acid residues, whose amyloid fibrillar form is the primary component of amyloid plaques found in the brains of Alzheimer's disease patients. Preferably, an APP is specifically recognized by an antibody that specifically recognizes the full-length APP, e.g., a polypeptide encoded by any one of the polynucleotide sequences set forth in Table 3.

Biliary atresia (BA) is a disease of blocked biliary system in the liver and bile ducts that typically occurs in infants in the age range of 2 weeks to 2 months. Symptoms of biliary atresia include jaundice, dark urine, acholic stools (clay-colored stools), weight loss and irritability. The preferred treatment method for BA is the Kasai procedure or operation, a surgical bypass procedure that allows bile to drain from the liver thus avoiding permanent damage to the liver. As used herein, a subject suffering from BA is an infant (for examples, aged about 1 week or about 2 weeks or about 3 weeks to about 1 month or about 2 months or about 3 months) who has been diagnosed with BA but is yet to receive the Kasai operation, or who has been diagnosed with BA, received the Kasai operation, yet still has symptoms relevant to the disease despite having received the Kasai operation.

In this disclosure the term "biological sample" or "sample" includes sections of tissues such as biopsy and autopsy samples, and frozen sections taken for histologic purposes, or processed forms of any of such samples. Biological samples include blood and blood fractions or products (e.g., serum, plasma, platelets, red blood cells, and the like), sputum or saliva, lymph and tongue tissue, oral swab, cultured cells, e.g., primary cultures, explants, transformed cells, stool, urine, and biopsy taken from a pre-selected organ or tissue (such as liver cells or hepatic tissue sample) as well as cells or tissue derived from biopsy. A biological sample is typically obtained from a eukaryotic organism, which may be a mammal, may be a primate and may be a human subject.

The term "nucleic acid" or "polynucleotide" refers to deoxyribonucleotides or ribonucleotides and polymers thereof in either single- or double-stranded form. Unless specifically limited, the term encompasses nucleic acids containing known analogues of natural nucleotides which have similar binding properties as the reference nucleic acid and are metabolized in a manner similar to naturally occurring nucleotides. Unless otherwise indicated, a particular nucleic acid sequence also implicitly encompasses conservatively modified variants thereof (e.g., degenerate codon substitutions) and complementary sequences as well as the sequence explicitly indicated. Specifically, degenerate codon substitutions may be achieved by generating sequences in which the third position of one or more selected (or all) codons is substituted with mixed-base and/or deoxyinosine residues (Batzer et al., *Nucleic Acid Res.*, 19:5081 (1991); Ohtsuka et al., *J. Biol. Chem.*, 260:2605-2608 (1985); and Cassol et al., (1992); Rossolini et al., *Mol. Cell. Probes*, 8:91-98 (1994)). The terms nucleic acid and polynucleotide are used interchangeably with gene, cDNA, and mRNA encoded by a gene.

The terms "polypeptide," "peptide," and "protein" are used interchangeably herein to refer to a polymer of amino acid residues. The terms apply to amino acid polymers in which one or more amino acid residue is an artificial chemical mimetic of a corresponding naturally occurring amino acid, as well as to naturally occurring amino acid polymers and non-naturally occurring amino acid polymers. As used herein, the terms encompass amino acid chains of any length, including full length proteins (i.e., antigens), wherein the amino acid residues are linked by covalent peptide bonds.

The term "amino acid" refers to naturally occurring and synthetic amino acids, as well as amino acid analogs and amino acid mimetics that function in a manner similar to the naturally occurring amino acids. Naturally occurring amino acids are those encoded by the genetic code, as well as those amino acids that are later modified, e.g., hydroxyproline, γ-carboxyglutamate, and O-phosphoserine. Amino acid analogs refers to compounds that have the same basic chemical structure as a naturally occurring amino acid, i.e., an a carbon that is bound to a hydrogen, a carboxyl group, an amino group, and an R group, e.g., homoserine, norleucine, methionine sulfoxide, methionine methyl sulfonium. Such analogs have modified R groups (e.g., norleucine) or modified peptide backbones, but retain the same basic chemical structure as a naturally occurring amino acid. "Amino acid mimetics" refers to chemical compounds that have a structure that is different from the general chemical structure of an amino acid, but that functions in a manner similar to a naturally occurring amino acid.

Amino acids may be referred to herein by either their commonly known three letter symbols or by the one-letter symbols recommended by the IUPAC-IUB Biochemical Nomenclature Commission. Nucleotides, likewise, may be referred to by their commonly accepted single-letter codes.

A "polynucleotide hybridization method" as used herein refers to a method for detecting the presence and/or quantity of a pre-determined polynucleotide sequence based on its ability to form Watson-Crick base-pairing, under appropriate hybridization conditions, with a polynucleotide probe of a known sequence. Examples of such hybridization methods include Southern blot, Northern blot, and in situ hybridization.

"Primers" as used herein refer to oligonucleotides that can be used in an amplification method, such as a polymerase chain reaction (PCR), to amplify a nucleotide sequence based on the polynucleotide sequence corresponding to a gene of interest, e.g., the polynucleotide sequence encoding an APP or a derivative thereof. Typically at least one, possibly two, of the PCR primers for amplification of a polynucleotide sequence is sequence-specific for that polynucleotide sequence. The exact length of the primer will depend upon many factors, including temperature, source of the primer, and the method used. For example, for diagnostic and prognostic applications, depending on the complexity of the target sequence, the oligonucleotide primer typically contains at least 10, or 15, or 20, or 25 or more nucleotides, although it may contain fewer nucleotides or more nucleotides. The factors involved in determining the appropriate length of primer are readily known to one of ordinary skill in the art. The primers used in particular embodiments may be designed to specifically amplify a coding sequence for an APP or its variant or derivative. Also, the primers may be designed for specifically amplify only one segment of the APP coding sequence for a fragment of APP. In this disclosure the term "primer pair" means a pair of primers that hybridize to opposite strands a target DNA molecule or to regions of the target DNA which flank a nucleotide sequence to be amplified (e.g., encoding an APP for the purpose of detection such as for assessing BA risk or disease prospect, especially after Kasai operation). In this disclosure the term "primer site" means the area of the target DNA or other nucleic acid to which a primer hybridizes.

A "label," "detectable label," or "detectable moiety" is a composition detectable by spectroscopic, photochemical, biochemical, immunochemical, chemical, or other physical means. For example, useful labels include radioactive isotopes, fluorescent dyes, electron-dense reagents, enzymes (e.g., as commonly used in an ELISA), biotin, digoxigenin, or haptens and proteins that can be made detectable, e.g., by incorporating a radioactive component into the peptide or used to detect antibodies specifically reactive with the peptide. Typically a detectable label is attached to a probe or a molecule with defined binding characteristics (e.g., a polypeptide with a known binding specificity or a polynucleotide), so as to allow the presence of the probe (and therefore its binding target, e.g., a coding sequence for an APP or a fragment or variant thereof) to be readily detectable.

The term "treat" or "treating," as used in this application, describes to an act that leads to the elimination, reduction, alleviation, reversal, or prevention or delay of onset or recurrence of any symptom of a relevant condition (e.g., BA). In other words, "treating" a condition encompasses both therapeutic and prophylactic intervention against the condition: for example, upon testing positive to have elevated APP expression either at mRNA or protein level, an infant will be given additional diagnostic tests (such as blood tests, ultrasound of the abdomen, liver biopsy, as well as diagnostic surgery, which may include an operative cholangiogram) and, upon confirmation of the diagnosis of BA, undergoes therapeutic or prophylactic regimen under the supervision of a medical professional including Kasai operation. For an infant who has received treatment for BA by Kasai operation but still has BA symptoms without improvement or even with worsening symptoms, subsequent treatment will likely require liver transplant depending on physicians' decision upon careful monitoring of the infant. Administration of steroids and/or antibiotics may serve as additional/adjuvant treatment post-Kasai operation with an aim to slow down or halt further inflammation.

The term "effective amount" as used herein refers to an amount of a given substance that is sufficient in quantity to produce a desired effect. For example, when an effective amount of a therapeutic agent for treating BA by way of suppressing or eliminating the expression and/or activity of APP especially in an infant's liver is administered to the infant, the symptoms of BA are reduced, reversed, eliminated, prevented, or delayed of the onset in the infant. An amount adequate to accomplish this is defined as the "therapeutically effective dose" when administration takes place after BA symptoms have become detectable, or is defined as the "prophylactically effective dose" when administration takes place before any symptom has arisen. The dosing range varies with the nature of the therapeutic agent being administered and other factors such as the route of administration and the severity of a patient's BA and related condition.

The term "subject" or "subject in need of treatment," as used herein, includes individuals who seek medical attention due to risk of, or actual suffering from, biliary atresia. Subjects also include individuals currently undergoing therapy that seek manipulation of the therapeutic regimen. Subjects or individuals in need of treatment include those that demonstrate symptoms of biliary atresia, or those at are at risk of suffering from biliary atresia or its symptoms, or those have been tested positive for amyloid precursor protein expression but may not may not have any known risk and/or may or may not have any potentially relevant symptoms. For example, subjects in need of treatment include individuals with a genetic predisposition or family history for biliary atresia, those that are acutely suffering from relevant symptoms, those that have been exposed to a triggering substance or event, as well as those have received treatment for BA (e.g., Kasai operation) but are continuing to suffer from symptoms of the condition. A subject in need of treatment for BA may be any gender and typically an infant at the age of 3 months or younger, for example, between about 1 week to about 3 months, or between about 2 weeks to about 2 months.

The term "about" when used in reference to a predetermined value denotes a range encompassing ±10% of the value.

DETAILED DESCRIPTION

I. General

The present invention relates to the detection of APP expression in the liver cells or tissue of infants who are suffering from or at risk of suffering from biliary atresia, or who are at risk of suffering from undesirable clinical outcome after receiving the Kasai operation. This invention thus provides methods for diagnosis of biliary atresia and prognosis for clinical outcome post-Kasai operation in infants based on detection of APP expression in a liver sample obtained from infants suspected of suffering from biliary atresia or having undergone the Kasai procedure.

Basic texts disclosing general methods and techniques in the field of recombinant genetics include Sambrook and Russell, *Molecular Cloning, A Laboratory Manual* (3rd ed. 2001); Kriegler, *Gene Transfer and Expression: A Laboratory Manual* (1990); and Ausubel et al., eds., *Current Protocols in Molecular Biology* (1994).

For nucleic acids, sizes are given in either kilobases (kb) or base pairs (bp). These are estimates derived from agarose or acrylamide gel electrophoresis, from sequenced nucleic acids, or from published DNA sequences. For proteins, sizes are given in kilodaltons (kDa) or amino acid residue numbers. Proteins sizes are estimated from gel electrophoresis, from sequenced proteins, from derived amino acid sequences, or from published protein sequences.

Oligonucleotides that are not commercially available can be chemically synthesized, e.g., according to the solid phase phosphoramidite triester method first described by Beaucage & Caruthers, *Tetrahedron Lett.* 22:1859-1862 (1981), using an automated synthesizer, as described in Van Devanter et. al., *Nucleic Acids Res.* 12:6159-6168 (1984). Purification of oligonucleotides is performed using any art-recognized strategy, e.g., native acrylamide gel electrophoresis or anion-exchange HPLC as described in Pearson & Reanier, *J. Chrom.* 255:137-149 (1983).

The polynucleotide sequence encoding a polypeptide of interest, e.g., an amyloid precursor protein or a variant or derivative thereof, and synthetic oligonucleotides can be verified after cloning or subcloning using, e.g., the chain termination method for sequencing double-stranded templates of Wallace et al., *Gene* 16:21-26 (1981).

II. Acquisition of Samples and Analysis of mRNA

The present invention relates to detecting the expression level of APP, either at the mRNA or protein level, in the liver cells or tissue taken from an infant suspected of suffering from biliary atresia (BA) or an infant who has been diagnosed with BA and has received the Kasai operation, as a means to detect the presence of BA, to assess the risk of developing of BA, and/or to provide prognosis of BA treatment such as the Kasai operation in an infant, so as to allow the correct treatment and/or follow-up strategies after Kasai operation. Thus, the first steps of practicing this invention are to obtain an appropriate sample from a test subject and extract mRNA from the sample.

A. Acquisition and Preparation of Samples

An appropriate sample such as a liver cell or tissue sample obtained from an infant, who may or may not exhibit BA symptoms, to be tested or monitored for BA or treatment outcome using a method of the present invention. Collection of a liver biopsy sample from an individual is performed in accordance with the standard protocol hospitals or clinics generally follow. An appropriate amount of tissue sample, including biopsy samples, is collected and may be stored according to standard procedures prior to further preparation.

B. Extraction of RNA

There are numerous methods for extracting RNA from a biological sample. The general methods of RNA preparation (e.g., described by Sambrook and Russell, *Molecular Cloning: A Laboratory Manual* 3d ed., 2001) can be followed; various commercially available reagents or kits, such as Trizol reagent (Invitrogen, Carlsbad, CA), Oligotex Direct mRNA Kits (Qiagen, Valencia, CA), RNeasy Mini Kits (Qiagen, Hilden, Germany), and PolyATtract® Series 9600™ (Promega, Madison, WI), may also be used to obtain RNA from a liver sample from an infant. Combinations of more than one of these methods may also be used.

It is preferable in some applications that all or most of the contaminating DNA be eliminated from the RNA preparations. Thus, careful handling of the samples, thorough treatment with DNase, and proper negative controls in the amplification and quantification steps should be used.

C. PCR-Based Quantitative Determination of RNA Level

Once RNA is extracted from a biological sample, the amount of RNA derived from a genetic locus of interest, e.g., and encoding for a protein of interest such as APP, may be quantified. The preferred method for determining the RNA level is an amplification-based method, e.g., by PCR.

Prior to the amplification step, a DNA copy (cDNA) of the RNA of interest must be synthesized. This is achieved by reverse transcription, which can be carried out as a separate step, or in a homogeneous reverse transcription-polymerase chain reaction (RT-PCR), a modification of the polymerase chain reaction for amplifying RNA. Methods suitable for PCR amplification of ribonucleic acids are described by Romero and Rotbart in *Diagnostic Molecular Biology: Principles and Applications* pp. 401-406; Persing et al., eds., Mayo Foundation, Rochester, MN, 1993; Egger et al., *J. Clin. Microbiol.* 33:1442-1447, 1995; and U.S. Pat. No. 5,075,212.

The general methods of PCR are well known in the art and are thus not described in detail herein. For a review of PCR methods, protocols, and principles in designing primers, see, e.g., Innis, et al., *PCR Protocols: A Guide to Methods and Applications*, Academic Press, Inc. N.Y., 1990. PCR reagents and protocols are also available from commercial vendors, such as Roche Molecular Systems.

PCR is most usually carried out as an automated process with a thermostable enzyme. In this process, the temperature of the reaction mixture is typically cycled through a denaturing region, a primer annealing region, and an extension reaction region automatically. In some protocols, the annealing region and the extension reaction region are merged. Machines specifically adapted for this purpose are commercially available.

Although PCR amplification of the target RNA is typically used in practicing the present invention. One of skill in the art will recognize, however, that amplification of these RNA species in a maternal blood sample may be accomplished by any known method, such as ligase chain reaction (LCR), transcription-mediated amplification, and self-sustained sequence replication or nucleic acid sequence-based amplification (NASBA), each of which provides sufficient amplification. More recently developed branched-DNA technology may also be used to quantitatively determining the amount of RNA markers in maternal blood. For a review of branched-DNA signal amplification for direct quantitation of nucleic acid sequences in clinical samples, see Nolte, Adv. *Clin. Chem.* 33:201-235, 1998.

C. Other Quantitative Methods

The RNA species of interest (such as the APP mRNA) can also be detected using other standard techniques, well-known to those of skill in the art. Although the detection step is typically preceded by an amplification step, amplification is not required in the methods of the invention. For instance, the RNA species of interest may be identified by size fractionation (e.g., gel electrophoresis), whether or not proceeded by an amplification step. After running a sample in an agarose or polyacrylamide gel and labeling with ethidium bromide according to well-known techniques (see, e.g., Sambrook and Russell, supra), the presence of a band of the same size as the standard control is an indication of the presence of a target RNA, the amount of which may then be compared to the control based on the intensity of the band. Alternatively, oligonucleotide probes specific to RNA transcribed from a genetic locus, e.g., the APP gene, can be used to detect the presence of such RNA species and indicate the amount of RNA in comparison to the standard control, based on the intensity of signal imparted by the probe.

Sequence-specific probe hybridization is a well-known method of detecting a particular nucleic acid comprising other species of nucleic acids. Under sufficiently stringent hybridization conditions, the probes hybridize specifically only to substantially complementary sequences. The stringency of the hybridization conditions can be relaxed to tolerate varying amounts of sequence mismatch.

A number of hybridization formats are well-known in the art, including but not limited to, solution phase, solid phase, or mixed phase hybridization assays. The following articles provide an overview of the various hybridization assay formats: Singer et al., *Biotechniques* 4:230, 1986; Haase et al., *Methods in Virology*, pp. 189-226, 1984; Wilkinson, *In situ Hybridization*, Wilkinson ed., IRL Press, Oxford University Press, Oxford; and Hames and Higgins eds., *Nucleic Acid Hybridization: A Practical Approach*, IRL Press, 1987.

The hybridization complexes are detected according to well-known techniques and the detection is not a critical aspect of the present invention. Nucleic acid probes capable of specifically hybridizing to a target nucleic acid, i.e., the APP mRNA species or the amplified DNA, can be labeled by any one of several methods typically used to detect the presence of hybridized nucleic acids. One common method of detection is the use of autoradiography using probes labeled with ^3H, ^{125}I, ^{35}S, ^{14}C, or ^{32}P, or the like. The choice of radioactive isotope depends on research preferences due to ease of synthesis, stability, and half-lives of the selected isotopes. Other labels include compounds (e.g., biotin and digoxigenin), which bind to antiligands or antibodies labeled with fluorophores, chemiluminescent agents, and enzymes. Alternatively, probes can be conjugated directly with labels such as fluorophores, chemiluminescent agents or enzymes. The choice of label depends on sensitivity required, ease of conjugation with the probe, stability requirements, and available instrumentation.

The probes and primers necessary for practicing the present invention can be synthesized and labeled using well known techniques. Oligonucleotides used as probes and primers may be chemically synthesized according to the solid phase phosphoramidite triester method first described by Beaucage and Caruthers, *Tetrahedron Letts.*, 22:1859-1862, 1981, using an automated synthesizer, as described in Needham-VanDevanter et al., *Nucleic Acids Res.* 12:6159-6168, 1984. Purification of oligonucleotides is by either native acrylamide gel electrophoresis or by anion-exchange high performance liquid chromatography (HPLC) as described in Pearson and Regnier, *J. Chrom.*, 255:137-149, 1983.

III. Immunoassays for Detection of Amyloid Precursor Protein

One aspect of this invention provides immunoassays used in the detection of APP in order to determine the expression level of the protein for the purpose of diagnosis and prognosis of BA. Antibodies against APP described herein are useful for carrying out these immunological assays.

A. Production of Antibodies Against APP

Methods for producing polyclonal and monoclonal antibodies that react specifically with an immunogen of interest are known to those of skill in the art (see, e.g., Coligan, *Current Protocols in Immunology Wiley*/Greene, NY, 1991; Harlow and Lane, *Antibodies: A Laboratory Manual* Cold Spring Harbor Press, NY, 1989; Stites et al. (eds.) *Basic and Clinical Immunology* (4th ed.) Lange Medical Publications, Los Altos, CA, and references cited therein; Goding, *Monoclonal Antibodies: Principles and Practice* (2d ed.) Academic Press, New York, NY, 1986; and Kohler and Milstein *Nature* 256:495-497, 1975). Such techniques include antibody preparation by selection of antibodies from libraries of recombinant antibodies in phage or similar vectors (see, Huse et al., *Science* 246:1275-1281, 1989; and Ward et al., *Nature* 341:544-546, 1989).

In order to produce antisera containing antibodies with desired specificity, the polypeptide of interest (e.g., APP) or an antigenic fragment thereof can be used to immunize suitable animals, e.g., mice, rabbits, or primates. A standard adjuvant, such as Freund's adjuvant, can be used in accordance with a standard immunization protocol. Alternatively, a synthetic antigenic peptide derived from that particular polypeptide can be conjugated to a carrier protein and subsequently used as an immunogen.

The animal's immune response to the immunogen preparation is monitored by taking test bleeds and determining the titer of reactivity to the antigen of interest. When appropriately high titers of antibody to the antigen are obtained, blood is collected from the animal and antisera are prepared. Further fractionation of the antisera to enrich antibodies specifically reactive to the antigen and purification of the antibodies can be performed subsequently, see, Harlow and Lane, supra, and the general descriptions of protein purification provided above.

Monoclonal antibodies are obtained using various techniques familiar to those of skill in the art. Typically, spleen cells from an animal immunized with a desired antigen are immortalized, commonly by fusion with a myeloma cell (see, Kohler and Milstein, *Eur. J. Immunol.* 6:511-519, 1976). Alternative methods of immortalization include, e.g., transformation with Epstein Barr Virus, oncogenes, or retroviruses, or other methods well known in the art. Colonies arising from single immortalized cells are screened for production of antibodies of the desired specificity and affinity for the antigen, and the yield of the monoclonal antibodies produced by such cells may be enhanced by various techniques, including injection into the peritoneal cavity of a vertebrate host.

Additionally, monoclonal antibodies may also be recombinantly produced upon identification of nucleic acid sequences encoding an antibody with desired specificity or a binding fragment of such antibody by screening a human B cell cDNA library according to the general protocol outlined by Huse et al., supra. The general principles and methods of recombinant polypeptide production discussed above are applicable for antibody production by recombinant methods.

B. Immunoassays for APP

Once antibodies specific for APP are available, the expression of APP at protein level in a sample, e.g., a liver cell or tissue sample, can be measured by a variety of immunoassay methods providing qualitative and quantitative results to a skilled artisan. For a review of immunological and immunoassay procedures in general see, e.g., Stites, supra; U.S. Pat. Nos. 4,366,241; 4,376,110; 4,517,288; and 4,837,168.

1. Labeling in Immunoassays

Immunoassays often utilize a labeling agent to specifically bind to and label the binding complex formed by the antibody and the target protein (e.g., APP). The labeling agent may itself be one of the moieties comprising the antibody/target protein complex, or may be a third moiety, such as another antibody, that specifically binds to the antibody/target protein complex. A label may be detectable by spectroscopic, photochemical, biochemical, immunochemical, electrical, optical or chemical means. Examples include, but are not limited to, magnetic beads (e.g., Dynabeads™), fluorescent dyes (e.g., fluorescein isothiocyanate, Texas red, rhodamine, and the like), radiolabels (e.g., ^3H, ^{125}I, ^{35}S, ^{14}C, or ^{32}P), enzymes (e.g., horse radish peroxidase, alkaline phosphatase, and others commonly used in an ELISA), and colorimetric labels such as colloidal gold or colored glass or plastic (e.g., polystyrene, polypropylene, latex, etc.) beads.

In some cases, the labeling agent is a second antibody bearing a detectable label. Alternatively, the second antibody may lack a label, but it may, in turn, be bound by a labeled third antibody specific to antibodies of the species from which the second antibody is derived. The second antibody can be modified with a detectable moiety, such as biotin, to which a third labeled molecule can specifically bind, such as enzyme-labeled streptavidin.

Other proteins capable of specifically binding immunoglobulin constant regions, such as protein A or protein G, can also be used as the label agents. These proteins are normal constituents of the cell walls of streptococcal bacteria. They exhibit a strong non-immunogenic reactivity with immunoglobulin constant regions from a variety of species (see, generally, Kronval, et al. *J. Immunol.,* 111:1401-1406 (1973); and Akerstrom, et al., *J. Immunol.,* 135:2589-2542 (1985)).

2. Immunoassay Formats

Immunoassays for detecting a target protein of interest (e.g., APP) from samples may be either competitive or noncompetitive. Noncompetitive immunoassays are assays in which the amount of captured target protein is directly measured. In one preferred "sandwich" assay, for example, the antibody specific for the target protein can be bound directly to a solid substrate where the antibody is immobilized. It then captures the target protein in test samples. The antibody/target protein complex thus immobilized is then bound by a labeling agent, such as a second or third antibody bearing a label, as described above.

In competitive assays, the amount of target protein in a sample is measured indirectly by measuring the amount of an added (exogenous) target protein displaced (or competed away) from an antibody specific for the target protein by the target protein present in the sample. In a typical example of such an assay, the antibody is immobilized and the exogenous target protein is labeled. Since the amount of the exogenous target protein bound to the antibody is inversely proportional to the concentration of the target protein present in the sample, the target protein level in the sample can thus be determined based on the amount of exogenous target protein bound to the antibody and thus immobilized.

In some cases, western blot (immunoblot) analysis is used to detect and quantify the presence of APP in the samples. The technique generally comprises separating sample proteins by gel electrophoresis on the basis of molecular weight, transferring the separated proteins to a suitable solid support (such as a nitrocellulose filter, a nylon filter, or a derivatized nylon filter) and incubating the samples with the antibodies that specifically bind the target protein. These antibodies may be directly labeled or alternatively may be subsequently detected using labeled antibodies (e.g., labeled sheep anti-mouse antibodies) that specifically bind to the antibodies against APP.

Other assay formats include liposome immunoassays (LIA), which use liposomes designed to bind specific molecules (e.g., antibodies) and release encapsulated reagents or markers. The released chemicals are then detected according to standard techniques (see, Monroe et al., *Amer. Clin. Prod. Rev.,* 5:34-41 (1986)).

For these immunoassays, the patient being tested may be an infant who is at risk of developing BA or may be one who has been diagnosed with BA, has undergone treatment such as Kasai operation, and is now being assessed for likelihood of a desirable outcome of the treatment.

IV. Treatment Options

One practical application of this invention is intended for early detection of BA among infants who are suspected of suffering from the disease or are at increased risk of developing the disease. Once they are tested according to any one of the methods of this invention and are determined as at risk of suffering from or later developing this disease, they can be subject to additional diagnostic tests, such as blood tests, ultrasound of the abdomen, liver biopsy, and diagnostic surgery (which may include an operative cholangiogram) in order to confirm whether they indeed suffer from the disease. Upon confirmation of the diagnosis of BA, they can then undergo appropriate treatment (e.g., the Kasai operation) as prescribed by their attending physician.

Another application is intended for predicting the likelihood of success among infants who have been diagnosed with BA and have been given appropriate treatment such as the Kasai operation. As it is known that a minority of all infant BA patients who underwent the operation will fail to properly recover, and the symptoms and damaging effects of BA will continue and even worsen. The claimed method of this invention allows physicians to quickly obtain an assessment of the likelihood of treatment outcome (e.g., shortly after the Kasai procedure such as 1 day, 2 days, 3 days, 4 days, 5 days, 6 days, 7 days, or up to 10 days or 2 weeks post-operation) when the infant may or may not yet to show any signs of persistent or deteriorating BA symptoms: whether it will result in desirable outcome, where the infant patient will fully recover from the disease, or it will result in failure, where the infant patient will require additional treatment, such as another Kasai procedure or even liver transplant.

A further aspect of the present invention is the treatment of BA. Since the expression of APP, both at mRNA level and at protein level, has been found to increase in the liver of infants who suffer from BA, or who are at increased risk of later developing BA while exhibiting no BA symptoms for the time being, or who have been diagnosed with BA and given appropriate treatment such as the Kasai procedure but suffer from undesirable outcome (i.e., failure of treatment to alleviate BA symptoms and damages to liver), one treatment method for BA is to target APP by suppressing its expression (both at mRNA and protein level) and/or activity. For instance, specific inhibitors targeting APP expression and/or activity may be administered to an infant who suffers from BA, or who is at heightened risk of later developing BA, or who has received a diagnosis of BA and received treatment (e.g., the Kasai procedure) but is at risk of suffering from treatment failure. Such specific inhibitors include a broad spectrum of possible compounds of distinct chemical and structural features such as a dominant negative APP mutant or its encoding nucleic acid, a nucleic acid encoding an antisense or miRNA, miniRNA, long non-coding RNA targeting APP, an inactivating anti-APP antibody, small chemicals, peptides, proteins, natural extract compounds from herbs, etc., and they are useful in both prophylactic and therapeutic applications for treating BA. Also, APP expression may be suppressed by genetic manipulation techniques including CRISPR.

V. Establishing Standard Controls

For the application of this invention for the purpose of diagnosing BA or assessing risk of developing BA at a later time, in order to establish a standard control, a group of healthy infants without any liver disorders, especially BA, should first be selected before liver samples are obtained. These infants should be of similar age, which is within the appropriate time frame (for example, aged from about 2 weeks to about 8 weeks) when infants may be tested using the methods of the present invention. The health status of the selected infants should be confirmed by well established, routinely employed methods including but not limited to blood tests and abdominal ultrasound.

The selected group of healthy infants must be of a reasonable size, such that the average amount of APP mRNA and protein found in the liver tissue samples (e.g., liver biopsy) calculated from the group can be reasonably regarded as representative of the normal or average amount among the general population of healthy infants not suffering from BA and not at heightened risk of developing BA. Preferably, the selected group comprises at least 10 healthy infants.

For the application of this invention for the purpose of providing a prognosis of BA treatment (especially the Kasai procedure) outcome among infants who have already been diagnosed with BA and received appropriate treatment such as the Kasai procedure, the same standard control described above may be used. In the alternative, a slightly different standard control can be established: a group of infants who suffered from BA and have undergone the same BA treatment such as Kasai operation and successfully recovered should be selected for collection of liver samples. Similar to the description above, these infants should be of similar age and make up a group of a reasonable size, e.g., at least 10 in the group.

Once an average value is established for the amount of APP mRNA or protein based on the individual values found in the liver tissue of each infant of the selected group, this value is considered a standard control value for APP mRNA or protein. Any liver tissue sample that contains a similar amount of APP mRNA or protein can thus be used as a standard control sample.

VI. Kits

The invention also provides kits for detecting APP mRNA or protein, especially in a liver sample taken from an infant, according to the method of the present invention for the purpose of detecting the presence of BA, assessing the risk of later developing BA, or assessing likelihood of a successful BA treatment (such as the Kasai operation) outcome. The kits typically include a first container that contains a first container containing at least one, possibly more, reagent for detecting expression level of APP and a second container containing a composition having a standard control level of APP expression (or a standard control sample). The one or more reagents in the first container may be used for measuring APP mRNA level, such as oligonucleotide primers useful for RT-PCR and/or oligonucleotide probe(s) for specific hybridization with an APP-specific polynucleotide sequence. Other possible reagents may include the necessary enzyme(s) and buffer(s) for performing assays to detect APP mRNA such as RT-PCR or nucleotide hybridization.

For the purpose of determining the APP protein level in a liver sample, the one or more reagents contained in the first container may be those useful for immunoassays capable of specifically detecting and quantifying APP level as a protein, such as ELISA or western blot analysis. One useful reagent is an anti-APP antibody, which may be a polyclonal antibody or a monoclonal antibody capable of specific binding to APP. Further included may be a secondary antibody, for example, an antibody against the anti-APP antibody, preferably conjugated with a detectable label.

In addition, the kit may further include informational material containing instructions for a user on how to use the kit for performing an assay and determining whether increased APP mRNA or protein is present in a patient sample.

EXAMPLES

The following examples are provided by way of illustration only and not by way of limitation. Those of skill in the art will readily recognize a variety of non-critical parameters that could be changed or modified to yield essentially the same or similar results.

Example 1: Amyloid Precursor Protein Expression in Biliary Atresia

This study reveals, for the first time, the specific expression of amyloid precursor protein and/or its processed forms in the bile duct of livers in patients suffering from biliary atresia (BA). This discovery allows one to devise new and reliable diagnostic methods for early detection of BA as well as for predicting the therapeutic outcome of BA treatment such as Kasai operation.

Introduction

Biliary atresia (BA [OMIM 210500]) is characterized by progressive fibro-obliterative cholangiopathy (disease of the bile duct) affecting both the intra- and extra-hepatic bile ducts and resulting in obstructive bile flow, cholestasis, and jaundice in neonates. BA occurs in some infants two to six weeks after birth, and symptoms of BA are usually evident between two and six weeks after birth, in that babies with BA develop progressive cholestasis, a condition in which the liver is unable to excrete bilirubin through the bile ducts in the form of bile. Bilirubin builds up inside of liver and begins to accumulate in the blood, causing symptoms including yellowing of the skin, itchiness, poor absorption of nutrients, pale stools, dark urine, and a swollen abdomen.

The differential diagnosis of BA allows for identification of patients suffering from BA as opposed to other disorders such as neonatal cholestasis (NC) like neonatal hepatitis (NH), paucity of interlobular bile ducts (PILBD), progressive familial intrahepatic cholestasis (PFIC), and various metabolic diseases like galactosemia and a-1 antitrypsin deficiency. The most important objective in such cases is to distinguish obstructive cholestasis from non-obstructive causes. The diagnosis of BA, particularly distinguishing it from other causes of liver injury in the neonatal period, is challenging as there is a high degree of overlap in clinical, biochemical, imaging, and histological characteristics of BA and other causes of NC.

Without surgical treatment (Kasai operation) to re-establish the bile drainage from microscopic residual bile ductules within the liver, progressive hepatic fibrosis leads to cirrhosis, portal vein hypertension, liver failure and death by the age of two (1, 2). However, postsurgical complications, including cholangitis (50%) and portal hypertension (>60%), remain a problem (3-7). Furthermore, regardless of drainage after successful surgery, patients will often develop inflammation and sclerosis of the biliary tree, leading to secondary biliary cirrhosis. To these patients and those who failed the surgical intervention, liver transplantation becomes the only treatment option. Indeed, BA has been the most common indication for liver transplantation during childhood for the past 20 years. Due to liver graft scarcity, many patients die before transplantation.

Urgently, there exists a distinct need to address the following problems: first, diagnosis of BA needs to be improved, in particular at early stages of the disease. The optimal age for Kasai operation is 60-75 days; delayed in diagnosis and surgery will lead to poor outcome. In fact, early diagnosis allows the family and surgeon for better planning of disease management. Second, despite the same Kasai operation, the outcome can vary from patient to patient, suggesting the existence of subsets of patient but there are no reliable markers for prognosis to guide stratified treatments for optimal outcomes. Third, regardless of drainage after Kasai operation, patients will often develop bile duct inflammation and sclerosis, leading to secondary biliary cirrhosis. To these patients and those who failed surgery, liver transplantation is the only option. For those BA patients who are predicted to have poor outcome after Kasai operation may be selected to undergo liver transplantation as the treatment from the start without first undergoing Kasai operation. There is an urgent need to find new and effective therapies that ameliorate BA.

Materials and Methods

Liver Tissues

Control liver: "Quasi-normal" human liver tissues were obtained from non-tumour margin of hepatoblastoma (HB). BA liver: Liver biopsies of BA patients taken during Kasai operation or at transplantation. All tissues were obtained during operations with full informed consent from parents or patients, and the study was approved by Hong Kong West Cluster-Hong Kong University Cluster Research Ethics Committee/Institutional Review Board (UW 16-052).

Antibodies

Anti-β-Amyloid (4G8; Biolegend) reacts to precursor forms, as well as abnormally processed isoforms of amyloid precursor protein. Anti-CK19 (ab52625; abcam) reacts to human cytokeratin 19.

Liver Biopsies

Wedge biopsies (50×50 mm) were obtained from non-syndromic BA patients during laparoscopic cholangiography, hepatoblastoma (HB)).

RNA Sequencing of Liver Tissues

Total RNAs were prepared from 30 mg of control livers (HB; n=2) and BA livers (n=4) using RNeasy Mini Kit (Qiagen) following manufacturer's protocol. Reverse transcription, amplification of 50 ng total RNA of each liver samples, and library construction were performed using single cell RNA-seq technology (Smart-seq 2.0) with minor modifications (8). Qualities of the pre-amplified products of normal and BA livers were confirmed to be optimal by Bioanalyzer. Library construction was performed using Nextera XT Kit following manufacturer's protocol. Libraries were pooled and sequenced by pair ends of 100 base pairs (PE100) on illumina HiSeq 2500 System.

Immuno-Fluorescence Staining

Tissues were fixed in 4% paraformaldehyde (w/v) in PBS (phosphate-buffered saline, pH 7.2) for 48 h at 4° C., dehydrated in graded series of alcohol, cleared in xylene before being embedded in paraffin. Sections (8 μm in thickness) were prepared, mounted onto TESPA-coated microscope glass. Sections were dewaxed in xylene, hydrated in a graded series of alcohol and finally in distilled water. Antigen was retrieved by two steps: (i) incubating in 70% Formic acid for 10 minutes and washing in water; (ii) incubating in 10 mM sodium citrate buffer (pH 6.0) at 95° C. for 10 min and washing in water. After blocking in PBS-T (PBS with 0.1% Triton) supplemented with 1% Bovine Serum Albumin for 1 h at room temperature, sections were incubated with anti-β-Amyloid (4G8, 1:200; Biolegend) and anti-CK19 (ab52625, 1:250; abcam) diluted in PBS-T/BSA for overnight at 4° C. After washing in PBS-T, sections were incubated with appropriate fluorescent tagged secondary antibodies in PBS-T/BSA at 37° C. for 1 h. After PBS-T washings, sections were mounted in Dapi-containing anti-fade mounting fluid. Images were taken with Nikon Eclipse 80i microscope mounted with SPOT RT3 microscope digital camera under fluorescence illumination. Photos were compiled using Adobe Photoshop 7.

Results

Transcriptome Analysis of Normal and BA Liver by RNAseq

The RNA-seq reads were first subjected to quality check using FastQC version 0.11.1. Further the adapter contamination and low-quality regions were filtered using Cutadapt version 1.8.3 with the parameter −q=33 and retained only reads with length≥30. The percentage of high quality bases in the filtered raw reads were greater than 90% for all the samples, with the quality score cut-off=20. Subsequently, sequencing reads were filtered for rRNA sequence by aligning to human rRNA sequences using Bowtie 2 (default parameters). The remaining reads were mapped to the reference genome. The transcriptome mapping/alignment and identification of exon-exon splice junctions with the human genome reference (GRCh38, downloaded from Ensembl database) was done by using TopHat version 2.0.10 (default parameters). All the samples had an overall alignment of >80% with human reference. Counting of aligned reads per gene were done using HTSeq version 0.9.1 for further differential expression analysis. The counts for each gene per samples were presented as table in DESeq2 to accurately detect significant differentially expressed genes across the conditions. Visualization was done using R and Bioconductor (FIG. **5**c?).

Principal component analysis (PCA) of liver biopsies revealed the following observations:

1. Normal liver and BA liver displayed distinctive molecular signatures.
2. The two normal liver controls showed differences among themselves, which may indicate the genetic differences between the patients or the tissue stages during isolation.
3. Three of the BA livers were clustering very closely to each other while the other patient showed enormous difference, which may indicate progression of disease.

TABLE 1

Pathways involvement of the 1558 differentially expressed genes	
Integrin signalling pathway (P00034)	28
Gonadotropin-releasing hormone receptor pathway (P06664)	25
Inflammation mediated by chemokine and cytokine signaling pathway (P00031)	21
CCKR signaling map (P06959)	20
Wnt signaling pathway (P00057)	17
Angiogenesis (P00005)	12
PDGF signaling pathway (P00047)	12
Apoptosis signaling pathway (P00006)	11
Nicotinic acetylcholine receptor signaling pathway (P00044)	10
Huntington disease (P00029)	9
Alzheimer disease-presenilin pathway (P00004)	8
Heterotrimeric G-protein signaling pathway-Gq alpha and Go alpha mediated pathway (P00027)	8
Heterotrimeric G-protein signaling pathway-Gi alpha and Gs alpha mediated pathway (P00026)	8
Parkinson disease (P00049)	7

TABLE 1-continued

Pathways involvement of the 1558 differentially expressed genes	
Interleukin signaling pathway (P00036)	7
EGF receptor signaling pathway (P00018)	7
T cell activation (P00053)	6
TGF-beta signaling pathway (P00052)	6
FGF signaling pathway (P00021)	6
FAS signaling pathway (P00020)	6
Cadherin signaling pathway (P00012)	6
B cell activation (P00010)	6
Axon guidance mediated by Slit/Robo (P00008)	5
Toll receptor signaling pathway (P00054)	5
Oxidative stress response (P00046)	5
Muscarinic acetylcholine receptor 1 and 3 signaling pathway (P00042)	5
Insulin/IGF pathway-protein kinase B signaling cascade (P00033)	5
Insulin/IGF pathway-mitogen activated protein kinase kinase/MAP kinase cascade (P00032)	5
Dopamine receptor mediated signaling pathway (P05912)	5
Pyrimidine Metabolism (P02771)	5
Endothelin signaling pathway (P00019)	5
Blood coagulation (P00011)	5
Adrenaline and noradrenaline biosynthesis (P00001)	4
De novo purine biosynthesis (P02738)	4
PI3 kinase pathway (P00048)	4
Notch signaling pathway (P00045)	4
Muscarinic acetylcholine receptor 2 and 4 signaling pathway (P00043)	4
Nicotine degradation (P05914)	4
5-Hydroxytryptamine degredation (P04372)	4
Axon guidance mediated by semaphorins (P00007)	3
Alpha adrenergic receptor signaling pathway (P00002)	3
Ubiquitin proteasome pathway (P00060)	3
p53 pathway (P00059)	3
Formyltetrahydroformate biosynthesis (P02743)	3
Plasminogen activating cascade (P00050)	3
Synaptic vesicle trafficking (P05734)	3
Metabotropic glutamate receptor group III pathway (P00039)	3
Ionotropic glutamate receptor pathway (P00037)	3
Thyrotropin-releasing hormone receptor signaling pathway (P04394)	3
2-arachidonoylglycerol biosynthesis (P05726)	3
Serine glycine biosynthesis (P02776)	3
Nicotine pharmacodynamics pathway (P06587)	3
Cytoskeletal regulation by Rho GTPase (P00016)	3
Purine metabolism (P02769)	3
5HT2 type receptor mediated signaling pathway (P04374)	3
Axon guidance mediated by netrin (P00009)	2
Pyridoxal-5-phosphate biosynthesis (P02759)	2
Alzheimer disease-amyloid secretase pathway (P00003)	2
Heme biosynthesis (P02746)	2
VEGF signaling pathway (P00056)	2
Tetrahydrofolate biosynthesis (P02742)	2
De novo pyrimidine ribonucleotides biosythesis (P02740)	2
Androgen/estrogene/progesterone biosynthesis (P02727)	2
JAK/STAT signaling pathway (P00038)	2
Vitamin B6 metabolism (P02787)	2
p53 pathway by glucose deprivation (P04397)	2
Hypoxia response via HIF activation (P00030)	2
Vitamin D metabolism and pathway (P04396)	2
Ras Pathway (P04393)	2
Oxytocin receptor mediated signaling pathway (P04391)	2
p38 MAPK pathway (P05918)	2
Hedgehog signaling pathway (P00025)	2
Glycolysis (P00024)	2
General transcription by RNA polymerase I (P00022)	2
Pyruvate metabolism (P02772)	2
Histamine H1 receptor mediated signaling pathway (P04385)	2
Cortocotropin releasing factor receptor signaling pathway (P04380)	2
Circadian clock system (P00015)	2
5HT4 type receptor mediated signaling pathway (P04376)	2
5HT3 type receptor mediated signaling pathway (P04375)	2
5HT1 type receptor mediated signaling pathway (P04373)	2
Toll pathway-drosophila (P06217)	1
SCW signaling pathway (P06216)	1
GBB signaling pathway (P06214)	1
DPP signaling pathway (P06213)	1
DPP-SCW signaling pathway (P06212)	1
BMP/activin signaling pathway-drosophila (P06211)	1
N-acetylglucosamine metabolism (P02756)	1
Methylmalonyl pathway (P02755)	1
Methionine biosynthesis (P02753)	1
Leucine biosynthesis (P02749)	1

TABLE 1-continued

Pathways involvement of the 1558 differentially expressed genes	
Isoleucine biosynthesis (P02748)	1
mRNA splicing (P00058)	1
Histidine biosynthesis (P02747)	1
Transcription regulation by bZIP transcription factor (P00055)	1
Fructose galactose metabolism (P02744)	1
De novo pyrimidine deoxyribonucleotide biosynthesis (P02739)	1
Metabotropic glutamate receptor group I pathway (P00041)	1
Asparagine and aspartate biosynthesis (P02730)	1
Metabotropic glutamate receptor group Il pathway (P00040)	1
GABA-B receptor Il signaling (P05731)	1
Alanine biosynthesis (P02724)	1
Interferon-gamma signaling pathway (P00035)	1
Adenine and hypoxanthine salvage pathway (P02723)	1
p53 pathway feedback loops 2 (P04398)	1
Valine biosynthesis (P02785)	1
Threonine biosynthesis (P02781)	1
P53 pathway feedback loops 1 (P04392)	1
Bupropion degradation (P05729)	1
Heterotrimeric G-protein signaling pathway-rod outer segment phototransduction (P00028)	1
Opioid proopiomelanocortin pathway (P05917)	1
Sulfate assimilation (P02778)	1
Opioid prodynorphin pathway (P05916)	1
Succinate to proprionate conversion (P02777)	1
Opioid proenkephalin pathway (P05915)	1
General transcription regulation (P00023)	1
Salvage pyrimidine ribonucleotides (P02775)	1
Salvage pyrimidine deoxyribonucleotides (P02774)	1
Angiotensin II-stimulated signaling through G proteins and beta-arrestin (P05911)	1
Pyridoxal phosphate salvage pathway (P02770)	1
DNA replication (P00017)	1
Cholesterol biosynthesis (P00014)	1
Beta3 adrenergic receptor signaling pathway (P04379)	1
Beta2 adrenergic receptor signaling pathway (P04378)	1
Beta1 adrenergic receptor signaling pathway (P04377)	1

TABLE 2

List of amyloid pathway genes being differentially expressed

Gene id	$\log_2$ Fold Change
A2M	−1.13
ADAM9	1.53
APBB3	0.98
APH1B	2.06
APOL1	−1.66
ATP2A3	1.89
CAPN1	1.13
FAS	−1.98
FSTL1	0.96
ITPR3	2.22
LPL	2.74
MMP19	3.21
MMP2	2.76
MMP7	5.75
NDUFA6	−1.57
NDUFB1	−0.78
NDUFB3	−0.75
NDUFV3	−0.86
NOTCH3	2.66
TCF3	1.34
TTR	−2.11

Genes Involved in Amyloid Precursor Protein (APP) Metabolism were Differentially Expressed in BA Liver

Pathway analysis for differentially expressed genes (1558) were done using PANTHER (website: www.pantherdb.org/), KEGG Mapper (https://www.genome.jp/kegg/mapper.html), DAVID (website: david.ncifcrf.gov/) and also from literatures. Among the list of pathways (Table 1), apart from hormonal and metabolic pathways, few disease related pathways were observed. Alzheimer disease presenilin and amyloid secretase pathway genes (10 nos.) were observed from panther pathways. KEGG pathway analysis also revealed few more genes to be involved in amyloid pathway genes from whole liver RNA-seq analysis. There were a total of 21 genes from total differentially expressed genes involved in amyloid pathway. The list of 21 genes along with log 2FoldChange are displayed in Table 2. Among which, MMP7 (Matrilysin, involved in amyloid-presenilin pathway) being the highest upregulated gene (log 2FoldChange=5.752) among the total (1558) differentially expressed genes.

Expression of APP and its Processed Forms in BA Liver

Immuno-fluorescence analysis for APP on liver sections of BA (n=18) and control (HB, n=5; CC, n=13; CS, n=2) patients revealed elevated expression of APP and/or its processed forms in the bile ducts of all the BA livers but not in all the control livers (FIG. **2**).

All patents, patent applications, and other publications, including GenBank Accession Numbers or equivalent sequence identification numbers, cited in this application are incorporated by reference in the entirety of their contents for all purposes.

TABLE 3

APP_transcrips_final		
Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
1	ENSP00000284981	>TCONS_00180767|ENST00000346798 (SEQ ID NO: 1) GGTACCCACTGATGGTAATGCTGGCCTGCTGGCTGAACC CCAGATTGCCATGTTCTGTGGCAGACTGAACATGCACAT GAATGTCCAGAATGGGAAGTGGGATTCAGATCCATCAGG GACCAAAACCTGCATTGATACCAAGGAAGGCATCCTGCA GTATTGCCAAGAAGTCTACCCTGAACTGCAGATCACCAAT GTGGTAGAAGCCAACCAACCAGTGACCATCCAGAACTGG TGCAAGCGGGGCCGCAAGCAGTGCAAGACCCATCCCCA CTTTGTGATTCCCTACCGCTGCTTAGTTGGTGAGTTTGTA AGTGATGCCCTTCTCGTTCCTGACAAGTGCAAATTCTTAC ACCAGGAGAGGATGGATGTTTGCGAAACTCATCTTCACT GGCACACCGTCGCCAAAGAGACATGCAGTGAGAAGAGTA CCAACTTGCATGACTACGGCATGTTGCTGCCCTGCGGAA TTGACAAGTTCCGAGGGGTAGAGTTTGTGTGTTGCCCAC TGGCTGAAGAAAGTGACAATGTGGATTCTGCTGATGCGG AGGAGGATGACTCGGATGTCTGGTGGGGCGGAGCAGAC ACAGACTATGCAGATGGGAGTGAAGACAAAGTAGTAGAA GTAGCAGAGGAGGAAGAAGTGGCTGAGGTGGAAGAAGA AGAAGCCGATGATGACGAGGACGATGAGGATGGTGATGA GGTAGAGGAAGAGGCTGAGGAACCCTACGAAGAAGCCA CAGAGAGAACCACCAGCATTGCCACCACCACCACCACCA CCACAGAGTCTGTGGAAGAGGTGGTTCGAGAGGTGTGCT CTGAACAAGCCGAGACGGGGCCGTGCCGAGCAATGATC TCCCGCTGGTACTTTGATGTGACTGAAGGGAAGTGTGCC CCATTCTTTTACGGCGGATGTGGCGGCAACCGGAACAAC TTTGACACAGAAGAGTACTGCATGGCCGTGTGTGGCAGC GCCATGTCCCAAAGTTTACTCAAGACTACCCAGGAACCTC TTGCCCGAGATCCTGTTAAACTTCCTACAACAGCAGCCAG TACCCCTGATGCCGTTGACAAGTATCTCGAGACACCTGG GGATGAGAATGAACATGCCCATTTCCAGAAAGCCAAAGA GAGGCTTGAGGCCAAGCACCGAGAGAGAATGTCCCAGG TCATGAGAGAATGGGAAGAGGCAGAACGTCAAGCAAAGA ACTTGCCTAAAGCTGATAAGAAGGCAGTTATCCAGCATTT CCAGGAGAAAGTGGAATCTTTGGAACAGGAAGCAGCCAA CGAGAGACAGCAGCTGGTGGAGACACACATGGCCAGAG TGGAAGCCATGCTCAATGACCGCCGCCGCCTGGCCCTG GAGAACTACATCACCGCTCTGCAGGCTGTTCCTCCTCGG CCTCGTCACGTGTTCAATATGCTAAAGAAGTATGTCCGCG CAGAACAGAAGGACAGACAGCACACCCTAAAGCATTTCG AGCATGTGCGCATGGTGGATCCCAAGAAAGCCGCTCAGA TCCGGTCCCAGGTTATGACACACCTCCGTGTGATTTATGA GCGCATGAATCAGTCTCTCTCCCTGCTCTACAACGTGCCT GCAGTGGCCGAGGAGATTCAGGATGAAGTTGATGAGCTG CTTCAGAAAGAGCAAAACTATTCAGATGACGTCTTGGCCA ACATGATTAGTGAACCAAGGATCAGTTACGGAAACGATG CTCTCATGCCATCTTTGACCGAAACGAAAACCACCGTGG AGCTCCTTCCCGTGAATGGAGAGTTCAGCCTGGACGATC TCCAGCCGTGGCATTCTTTTGGGGCTGACTCTGTGCCAG CCAACACAGAAAACGAAGTTGAGCCTGTTGATGCCCGCC CTGCTGCCGACCGAGGACTGACCACTCGACCAGGTTCTG GGTTGACAAATATCAAGACGGAGGAGATCTCTGAAGTGA AGATGGATGCAGAATTCCGACATGACTCAGGATATGAAG TTCATCATCAAAAATTGGTGTTCTTTGCAGAAGATGTGGG TTCAAACAAAGGTGCAATCATTGGACTCATGGTGGGCGG TGTTGTCATAGCGACAGTGATCGTCATCACCTTGGTGATG CTGAAGAAGAAACAGTACACATCCATTCATCATGGTGTGG TGGAGGTTGACGCCGCTGTCACCCCAGAGGAGCGCCAC CTGTCCAAGATGCAGCAGAACGGCTACGAAAATCCAACC TACAAGTTCTTTGAGCAGATGCAGAACTAGACCCCCGCC ACAGCAGCCTCTGAAGTTGGACAGCAAAACCATTGCTTC ACTACCCATCGGTGTCCATTTATAGAATAATGTGGGAAGA AACAAACCCGTTTTATGATTTACTCATTATCGCCTTTTGAC AGCTGTGCTGTAACACAAGTAGATGCCTGAACTTGAATTA ATCCACACATCAGTAATGTATTCTATCTCTCTTTACATTTT GGTCTCTATACTACATTATTAATGGGTTTTGTGTACTGTAA AGAATTTAGCTGTATCAAACTAGTGCATGAATAGATTCTCT CCTGATTATTTATCACATAGCCCCTTAGCCAGTTGTATATT ATTCTTGTGGTTTGTGACCCAATTAAGTCCTACTTTACATA TGCTTTAAGAATCGATGGGGGATGCTTCATGTGAACGTG GGAGTTCAGCTGCTTCTCTTGCCTAAGTATTCCTTTCCTG ATCACTATGCATTTTAAAGTTAAACATTTTTAAGTATTTCA GATGCTTTAGAGAGATTTTTTTTCCATGACTGCATTTTACT GTACAGATTGCTGCTTCTGCTATATTTGTGATATAGGAATT AAGAGGATACACACGTTTGTTTCTTCGTGCCTGTTTTATG TGCACACATTAGGCATTGAGACTTCAAGCTTTTCTTTTTTT

TABLE 3-continued

APP_transcrips_final		
Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
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		GTGCTCTGCTGGTCTTCAATTACCAAGAATTCTCCAAAAC
		AATTTTCTGCAGGATGATTGTACAGAATCATTGCTTATGA
		CATGATCGCTTTCTACACTGTATTACATAAATAAATTAAAT
		AAAATAACCCCGGGCAAGACTTTTCTTTGAAGGATGACTA
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		AGGCAGATTCAATTTTCTTTAACCAGTCTGAAGTTTCATTT
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		GGGATGAGGAAGGCATGCCTGGACAAACCCTTCTTTTAA
		GATGTGTCTTCAATTTGTATAAAATGGTGTTTTCATGTAAA
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		AGTGCAGCAAAGAAAAGCTGTGTGGAGGCTGCAGCCTAG
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		CTGGCCTGCTGGCTGAACCCCAGATTGCCATGTTCTGTG
		GCAGACTGAACATGCACATGAATGTCCAGAATGGGAAGT
		GGGATTCAGATCCATCAGGGACCAAAACCTGCATTGATA
		CCAAGGAAGGCATCCTGCAGTATTGCCAAGAAGTCTACC
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		CAGTGACCATCCAGAACTGGTGCAAGCGGGGCCGCAAG
		CAGTGCAAGACCCATCCCCACTTTGTGATTCCCTACCGCT
		GCTTAGTTGGTGAGTTTGTAAGTGATGCCCTTCTCGTTCC
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		ACATGCAGTGAGAAGAGTACCAACTTGCATGACTACGGC
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		GAGTTTGTGTGTTGCCCACTGGCTGAAGAAAGTGACAAT
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		ACGATGAGGATGGTGATGAGGTAGAGGAAGAGGCTGAG
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		GCCACCACCACCACCACCACCACAGAGTCTGTGGAAGAG
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		ACCTACAAGTTCTTTGAGCAGATGCAGAACTAGACCCCC
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TABLE 3-continued

APP_transcrips_final

Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
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3	ENSP00000284981	>TCONS_00180769|ENST00000346798 (SEQ ID NO: 3) CGCCGCGCTCGGGCTCCGTCAGTTTCCTCGGCAGCGGT AGGCGAGAGCACGCGGAGGAGCGTGCGCGGGGGCCCC GGGAGACGGCGGCGGTGGCGGCGCGGGCAGAGCAAGG ACGCGGCGGATCCCACTCGCACAGCAGCGCACTCGGTG CCCCGCGCAGGGTCGCGATGCTGCCCGGTTTGGCACTG CTCCTGCTGGCCGCCTGGACGGCTCGGGCGCTGGAGGT CTACCCTGAACTGCAGATCACCAATGTGGTAGAAGCCAA CCAACCAGTGACCATCCAGAACTGGTGCAAGCGGGGCC GCAAGCAGTGCAAGACCCATCCCCACTTTGTGATTCCCT ACCGCTGCTTAGTTGGTGAGTTTGTAAGTGATGCCCTTCT CGTTCCTGACAAGTGCAAATTCTTACACCAGGAGAGGAT GGATGTTTGCGAAACTCATCTTCACTGGCACACCGTCGC CAAAGAGACATGCAGTGAGAAGAGTACCAACTTGCATGA CTACGGCATGTTGCTGCCCTGCGGAATTGACAAGTTCCG AGGGGTAGAGTTTGTGTGTTGCCCACTGGCTGAAGAAAG TGACAATGTGGATTCTGCTGATGCGGAGGAGGATGACTC GGATGTCTGGTGGGGCGGAGCAGACACAGACTATGCAG ATGGGAGTGAAGACAAAGTAGTAGAAGTAGCAGAGGAGG AAGAAGTGGCTGAGGTGGAAGAAGAAGAAGCCGATGATG ACGAGGACGATGAGGATGGTGATGAGGTAGAGGAAGAG GCTGAGGAACCCTACGAAGAAGCCACAGAGAGAACCACC AGCATTGCCACCACCACCACCACCACCACAGAGTCTGTG GAAGAGGTGGTTCGAGAGGTGTGCTCTGAACAAGCCGA GACGGGGCCGTGCCGAGCAATGATCTCCCGCTGGTACTT TGATGTGACTGAAGGGAAGTGTGCCCCATTCTTTTACGG CGGATGTGGCGGCAACCGGAACAACTTTGACACAGAAGA GTACTGCATGGCCGTGTGTGGCAGCGCCATGTCCCAAAG TTTACTCAAGACTACCCAGGAACCTCTTGCCCGAGATCCT GTTAAACTTCCTACAACAGCAGCCAGTACCCCTGATGCC GTTGACAAGTATCTCGAGACACCTGGGGATGAGAATGAA CATGCCCATTTCCAGAAAGCCAAAGAGAGGCTTGAGGCC AAGCACCGAGAGAGAATGTCCCAGCCTCGTCACGTGTTC AATATGCTAAAGAAGTATGTCCGCGCAGAACAGAAGGAC AGACAGCACACCCTAAAGCATTTCGAGCATGTGCGCATG GTGGATCCCAAGAAAGCCGCTCAGATCCGGTCCCAGGTT ATGACACACCTCCGTGTGATTTATGAGCGCATGAATCAGT CTCTCTCCCTGCTCTACAACGTGCCTGCAGTGGCCGAGG AGATTCAGGATGAAGTTGATGAGCTGCTTCAGAAAGAGC AAAACTATTCAGATGACGTCTTGGCCAACATGATTAGTGA ACCAAGGATCAGTTACGGAAACGATGCTCTCATGCCATCT TTGACCGAAACGAAAACCACCGTGGAGCTCCTTCCCGTG AATGGAGAGTTCAGCCTGGACGATCTCCAGCCGTGGCAT TCTTTTGGGGCTGACTCTGTGCCAGCCAACACAGAAAAC GAAGTTGAGCCTGTTGATGCCCGCCCTGCTGCCGACCGA GGACTGACCACTCGACCAGGTTCTGGGTTGACAAATATC AAGACGGAGGAGATCTCTGAAGTGAAGATGGATGCAGAA TTCCGACATGACTCAGGATATGAAGTTCATCATCAAAAT

TABLE 3-continued

<table>
<tr><th colspan="3">APP_transcrips_final</th></tr>
<tr><th>Number of variants</th><th>Ensembl_protein_id/ UniProt_id</th><th>Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids</th></tr>
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AGTACACATCCATTCATCATGGTGTGGTGGAGGTTGACG
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ATTATTAATGGGTTTTGTGTACTGTAAAGAATTTAGCTGTA
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ACATAGCCCCTTAGCCAGTTGTATATTATTCTTGTGGTTT
GTGACCCAATTAAGTCCTACTTTACATATGCTTTAAGAATC
GATGGGGGATGCTTCATGTGAACGTGGGAGTTCAGCTGC
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CACTTTTACGGGGGGGGTGGGGAGGGGTGCTCTGCTGG
TCTTCAATTACCAAGAATTCTCCAAAACAATTTTCTGCAGG
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ACACTGTATTACATAAATAAATTAAATAAAATAACCCCGGG
CAAGACTTTTCTTTGAAGGATGACTACAGACATTAAATAAT
CGAAGTAATTTTGGGTGGGGAGAAGAGGCAGATTCAATT
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GATGAAAATGGAAGTGGCAATATAAGGGGATGAGGAAGG
CATGCCTGGACAAACCCTTCTTTTAAGATGTGTCTTCAAT
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GAGGAGCACCATTG</td></tr>
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CCCCGCGCAGGGTCGCGATGCTGCCCGGTTTGGCACTG
CTCCTGCTGGCCGCCTGGACGGCTCGGGCGCTGGAGGT
ACCCACTGATGGTAATGCTGGCCTGCTGGCTGAACCCCA
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GGAGAGGATGGATGTTTGCGAAACTCATCTTCACTGGCA
CACCGTCGCCAAAGAGACATGCAGTGAGAAGAGTACCAA
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GGATGACTCGGATGTCTGGTGGGGCGGAGCAGACACAG
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CAGAGGAGGAAGAAGTGGCTGAGGTGGAAGAAGAAGAA
GCCGATGATGACGAGGACGATGAGGATGGTGATGAGGT
AGAGGAAGAGGCTGAGGAACCCTACGAAGAAGCCACAG
AGAGAACCACCAGCATTGCCACCACCACCACCACCACCA
CAGAGTCTGTGGAAGAGGTGGTTCGAGAGGTGTGCTCTG
AACAAGCCGAGACGGGGCCGTGCCGAGCAATGATCTCC
CGCTGGTACTTTGATGTGACTGAAGGGAAGTGTGCCCCA
TTCTTTTACGGCGGATGTGGCGGCAACCGGAACAACTTT
GACACAGAAGAGTACTGCATGGCCGTGTGTGGCAGCGC
CATTCCTACAACAGCAGCCAGTACCCCTGATGCCGTTGA
CAAGTATCTCGAGACACCTGGGGATGAGAATGAACATGC
CCATTTCCAGAAAGCCAAAGAGAGGCTTGAGGCCAAGCA
CCGAGAGAGAATGTCCCAGGTCATGAGAGAATGGGAAGA
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</table>

TABLE 3-continued

APP_transcrips_final		
Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
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		CCGCCGCCGCCTGGCCCTGGAGAACTACATCACCGCTCT
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		GCTAAAGAAGTATGTCCGCGCAGAACAGAAGGACAGACA
		GCACACCCTAAAGCATTTCGAGCATGTGCGCATGGTGGA
		TCCCAAGAAAGCCGCTCAGATCCGGTCCCAGGTTATGAC
		ACACCTCCGTGTGATTTATGAGCGCATGAATCAGTCTCTC
		TCCCTGCTCTACAACGTGCCTGCAGTGGCCGAGGAGATT
		CAGGATGAAGTTGATGAGCTGCTTCAGAAAGAGCAAAAC
		TATTCAGATGACGTCTTGGCCAACATGATTAGTGAACCAA
		GGATCAGTTACGGAAACGATGCTCTCATGCCATCTTTGAC
		CGAAACGAAAACCACCGTGGAGCTCCTTCCCGTGAATGG
		AGAGTTCAGCCTGGACGATCTCCAGCCGTGGCATTCTTT
		TGGGGCTGACTCTGTGCCAGCCAACACAGAAAACGAAGG
		TTCTGGGTTGACAAATATCAAGACGGAGGAGATCTCTGAA
		GTGAAGATGGATGCAGAATTCCGACATGACTCAGGATAT
		GAAGTTCATCATCAAAAATTGGTGTTCTTTGCAGAAGATG
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		GGAGAGGATGGATGTTTGCGAAACTCATCTTCACTGGCA
		CACCGTCGCCAAAGAGACATGCAGTGAGAAGAGTACCAA
		CTTGCATGACTACGGCATGTTGCTGCCCTGCGGAATTGA
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TABLE 3-continued

APP_transcrips_final

Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
		CAGAGGAGGAAGAAGTGGCTGAGGTGGAAGAAGAAGAA GCCGATGATGACGAGGACGATGAGGATGGTGATGAGGT AGAGGAAGAGGCTGAGGAACCCTACGAAGAAGCCACAG AGAGAACCACCAGCATTGCCACCACCACCACCACCACCA CAGAGTCTGTGGAAGAGGTGGTTCGAGAGGTGTGCTCTG AACAAGCCGAGACGGGGCCGTGCCGAGCAATGATCTCC CGCTGGTACTTTGATGTGACTGAAGGGAAGTGTGCCCCA TTCTTTTACGGCGGATGTGGCGGCAACCGGAACAACTTT GACACAGAAGAGTACTGCATGGCCGTGTGTGGCAGCGC CATGTCCCAAAGTTTACTCAAGACTACCCAGGAACCTCTT GCCCGAGATCCTGTTAAACTTCCTACAACAGCAGCCAGT ACCCCTGATGCCGTTGACAAGTATCTCGAGACACCTGGG GATGAGAATGAACATGCCCATTTCCAGAAAGCCAAAGAG AGGCTTGAGGCCAAGCACCGAGAGAGAATGTCCCAGGT CATGAGAGAATGGGAAGAGGCAGAACGTCAAGCAAAGAA CTTGCCTAAAGCTGATAAGAAGGCAGTTATCCAGCATTTC CAGGAGAAAGTGGAATCTTTGGAACAGGAAGCAGCCAAC GAGAGACAGCAGCTGGTGGAGACACACATGGCCAGAGT GGAAGCCATGCTCAATGACCGCCGCCGCCTGGCCCTGG AGAACTACATCACCGCTCTGCAGGCTGTTCCTCCTCGGC CTCGTCACGTGTTCAATATGCTAAAGAAGTATGTCCGCGC AGAACAGAAGGACAGACAGCACACCCTAAAGCATTTOGA GCATGTGCGCATGGTGGATCCCAAGAAAGCCGCTCAGAT CCGGTCCCAGGTTATGACACACCTCCGTGTGATTTATGA GCGCATGAATCAGTCTCTCTCCCTGCTCTACAACGTGCCT GCAGTGGCCGAGGAGATTCAGGATGAAGTTGATGAGCTG CTTCAGAAAGAGCAAAACTATTCAGATGACGTCTTGGCCA ACATGATTAGTGAACCAAGGATCAGTTACGGAAACGATG CTCTCATGCCATCTTTGACCGAAACGAAAACCACCGTGG AGCTCCTTCCCGTGAATGGAGAGTTCAGCCTGGACGATC TCCAGCCGTGGCATTCTTTTGGGGCTGACTCTGTGCCAG CCAACACAGAAAACGAAGGTTCTGGGTTGACAAATATCAA GACGGAGGAGATCTCTGAAGTGAAGATGGATGCAGAATT CCGACATGACTCAGGATATGAAGTTCATCATCAAAAATTG GTGTTCTTTGCAGAAGATGTGGGTTCAAACAAAGGTGCAA TCATTGGACTCATGGTGGGCGGTGTTGTCATAGCGACAG TGATCGTCATCACCTTGGTGATGCTGAAGAAGAAACAGTA CACATCCATTCATCATGGTGTGGTGGAGGTTGACGCCGC TGTCACCCCAGAGGAGCGCCACCTGTCCAAGATGCAGCA GAACGGCTACGAAAATCCAACCTACAAGTTCTTTGAGCAG ATGCAGAACTAGACCCCCGCCACAGCAGCCTCTGAAGTT GGACAGCAAAACCATTGCTTCACTACCCATCGGTGTCCAT TTATAGAATAATGTGGGAAGAAACAAACCCGTTTTATGAT TTACTCATTATCGCCTTTTGACAGCTGTGCTGTAACACAA GTAGATGCCTGAACTTGAATTAATCCACACATCAGTAATG TATTCTATCTCTCTTTACATTTTGGTCTCTATACTACATTAT TAATGGGTTTTGTGTACTGTAAAGAATTTAGCTGTATCAAA CTAGTGCATGAATAGATTCTCTCCTGATTATTTATCACATA GCCCCTTAGCCAGTTGTATATTATTCTTGTGGTTTGTGAC CCAATTAAGTCCTACTTTACATATGCTTTAAGAATCGATGG GGGATGCTTCATGTGAACGTGGGAGTTCAGCTGCTTCTC TTGCCTAAGTATTCCTTTCCTGATCACTATGCATTTTAAAG TTAAACATTTTTAAGTATTTCAGATGCTTTAGAGAGATTTT TTTTCCATGATCGCTTTCTACACTGTATTACATAAATAAAT TAAATAAAATAACCCCGGGCAAGACTTTTCTTTGAAGGAT GACTACAGACATTAAATAATCGAAGTAATTTTGGGTGGGG AGAAGAGGCAGATTCAATTTTCTTTAACCAGTCTGAAGTT TCATTTATGATACAAAAGAAGATGAAAATGGAAGTGGCAA TATAAGGGGATGAGGAAGGCATGCCTGGACAAACCCTTC TTTTAAGATGTGTCTTCAATTTGTATAAAATGGTGTTTTCA TGTAAATAAATACATTCTTGGAGGAGCACCATTG
6	ENSP00000284981	>TCONS_00180773|ENST00000346798 (SEQ ID NO: 6) CAGCAGCGCACTCGGTGCCCCGCGCAGGGTCGCGATGC TGCCCGGTTTGGCACTGCTCCTGCTGGCCGCCTGGACG GCTCGGGCGCTGGAGGTACCCACTGATGGTAATGCTGG CCTGCTGGCTGAACCCCAGATTGCCATGTTCTGTGGCAG ACTGAACATGCACATGAATGTCCAGAATGGGAAGTGGGA TTCAGATCCATCAGGGACCAAAACCTGCATTGATACCAAG GAAGGCATCCTGCAGTATTGCCAAGAAGTCTACCCTGAA CTGCAGATCACCAATGTGGTAGAAGCCAACCAACCAGTG ACCATCCAGAACTGGTGCAAGCGGGGCCGCAAGCAGTG CAAGACCCATCCCCACTTTGTGATTCCCTACCGCTGCTTA GTTGGTGAGTTTGTAAGTGATGCCCTTCTCGTTCCTGACA AGTGCAAATTCTTACACCAGGAGAGGATGGATGTTTGCG

TABLE 3-continued

APP_transcrips_final

Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
		AAACTCATCTTCACTGGCACACCGTCGCCAAAGAGACAT
		GCAGTGAGAAGAGTACCAACTTGCATGACTACGGCATGT
		TGCTGCCCTGCGGAATTGACAAGTTCCGAGGGGTAGAGT
		TTGTGTGTTGCCCACTGGCTGAAGAAAGTGACAATGTGG
		ATTCTGCTGATGCGGAGGAGGATGACTCGGATGTCTGGT
		GGGGCGGAGCAGACACAGACTATGCAGATGGGAGTGAA
		GACAAAGTAGTAGAAGTAGCAGAGGAGGAAGAAGTGGCT
		GAGGTGGAAGAAGAAGAAGCCGATGATGACGAGGACGA
		TGAGGATGGTGATGAGGTAGAGGAAGAGGCTGAGGAAC
		CCTACGAAGAAGCCACAGAGAGAACCACCAGCATTGCCA
		CCACCACCACCACCACCACAGAGTCTGTGGAAGAGGTGG
		TTCGAGAGGTGTGCTCTGAACAAGCCGAGACGGGGCCG
		TGCCGAGCAATGATCTCCCGCTGGTACTTTGATGTGACT
		GAAGGGAAGTGTGCCCCATTCTTTTACGGCGGATGTGGC
		GGCAACCGGAACAACTTTGACACAGAAGAGTACTGCATG
		GCCGTGTGTGGCAGCGCCATGTCCCAAAGTTTACTCAAG
		ACTACCCAGGAACCTCTTGCCCGAGATCCTGTTAAACTTC
		CTACAACAGCAGCCAGTACCCCTGATGCCGTTGACAAGT
		ATCTCGAGACACCTGGGGATGAGAATGAACATGCCCATT
		TCCAGAAAGCCAAAGAGAGGCTTGAGGCCAAGCACCGA
		GAGAGAATGTCCCAGGTCATGAGAGAATGGGAAGAGGCA
		GAACGTCAAGCAAAGAACTTGCCTAAAGCTGATAAGAAG
		GCAGTTATCCAGCATTTCCAGGAGAAAGTGGAATCTTTGG
		AACAGGAAGCAGCCAACGAGAGACAGCAGCTGGTGGAG
		ACACACATGGCCAGAGTGGAAGCCATGCTCAATGACCGC
		CGCCGCCTGGCCCTGGAGAACTACATCACCGCTCTGCAG
		GCTGTTCCTCCTCGGCCTCGTCACGTGTTCAATATGCTAA
		AGAAGTATGTCCGCGCAGAACAGAAGGACAGACAGCACA
		CCCTAAAGCATTTCGAGCATGTGCGCATGGTGGATCCCA
		AGAAAGCCGCTCAGATCCGGTCCCAGGTTATGACACACC
		TCCGTGTGATTTATGAGCGCATGAATCAGTCTCTCTCCCT
		GCTCTACAACGTGCCTGCAGTGGCCGAGGAGATTCAGGA
		TGAAGTTGATGAGCTGCTTCAGAAAGAGCAAAACTATTCA
		GATGACGTCTTGGCCAACATGATTAGTGAACCAAGGATC
		AGTTACGGAAACGATGCTCTCATGCCATCTTTGACCGAAA
		CGAAAACCACCGTGGAGCTCCTTCCCGTGAATGGAGAGT
		TCAGCCTGGACGATCTCCAGCCGTGGCATTCTTTTGGGG
		CTGACTCTGTGCCAGCCAACACAGAAAACGAAGTTGAGC
		CTGTTGATGCCCGCCCTGCTGCCGACCGAGGACTGACCA
		CTCGACCAGGTTCTGGGTTGACAAATATCAAGACGGAGG
		AGATCTCTGAAGTGAAGATGGATGCAGAATTCCGACATG
		ACTCAGGATATGAAGTTCATCATCAAAAATTGGTGTTCTTT
		GCAGAAGATGTGGGTTCAAACAAAGGTGCAATCATTGGA
		CTCATGGTGGGCGGTGTTGTCATAGCGACAGTGATCGTC
		ATCACCTTGGTGATGCTGAAGAAGAAACAGTACACATCCA
		TTCATCATGGTGTGGTGGAGGTTGACGCCGCTGTCACCC
		CAGAGGAGCGCCACCTGTCCAAGATGCAGCAGAACGGC
		TACGAAAATCCAACCTACAAGTTCTTTGAGCAGATGCAGA
		ACTAGACCCCCGCCACAGCAGCCTCTGAAGTTGGACAGC
		AAAACCATTGCTTCACTACCCATCGGTGTCCATTTATAGA
		ATAATGTGGGAAGAAACAAACCCGTTTTATGATTTACTCA
		TTATCGCCTTTTGACAGCTGTGCTGTAACACAAGTAGATG
		CCTGAACTTGAATTAATCCACACATCAGTAATGTATTCTAT
		CTCTCTTTACATTTTGGTCTCTATACTACATTATTAATGGG
		TTTTGTGTACTGTAAAGAATTTAGCTGTATCAAACTAGTGC
		ATGAATAGATTCTCTCCTGATTATTTATCACATAGCCCCTT
		AGCCAGTTGTATATTATTCTTGTGGTTTGTGACCCAATTAA
		GTCCTACTTTACATATGCTTTAAGAATCGATGGGGGATGC
		TTCATGTGAACGTGGGAGTTCAGCTGCTTCTCTTGCCTAA
		GTATTCCTTTCCTGATCACTATGCATTTTAAAGTTAAACAT
		TTTTAAGTATTTCAGATGCTTTAGAGAGATTTTTTTTCCAT
		GACTGCATTTTACTGTACAGATTGCTGCTTCTGCTATATTT
		GTGATATAGGAATTAAGAGGATACACACGTTTGTTTCTTC
		GTGCCTGTTTTATGTGCACACATTAGGCATTGAGACTTCA
		AGCTTTTCTTTTTTTGTCCACGTATCTTTGGGTCTTTGATA
		AAGAAAAGAATCCCTGTTCATTGTAAGCACTTTTACGGGG
		CGGGTGGGGAGGGGTGCTCTGCTGGTCTTCAATTACCAA
		GAATTCTCCAAAACAATTTTCTGCAGGATGATTGTACAGA
		ATCATTGCTTATGACATGATCGCTTTCTACACTGTATTACA
		TAAATAAATTAAATAAAATAACCCCGGGCAAGACTTTTCTT
		TGAAGGATGACTACAGACATTAAATAATCGAAGTAATTTT
		GGGTGGGGAGAAGAGGCAGATTCAATTTTCTTTAACCAG
		TCTGAAGTTTCATTTATGATACAAAAGAAGATGAAAATGG
		AAGTGGCAATATAAGGGGATGAGGAAGGCATGCCTGGAC

TABLE 3-continued

APP_transcrips_final		
Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
		AAACCCTTCTTTTAAGATGTGTCTTCAATTTGTATAAAATG GTGTTTTCATGTAAATAAATACATTCTTGGAGGAGCA
7	ENSP00000345463	>TCONS_00180775|ENST00000348990 (SEQ ID NO: 7) AGTTTCCTCGGCAGCGGTAGGCGAGAGCACGCGGAGGA GCGTGCGCGGGGGCCCCGGGAGACGGCGGCGGTGGCG GCGCGGGCAGAGCAAGGACGCGGCGGATCCCACTCGCA CAGCAGCGCACTCGGTGCCCCGCGCAGGGTCGCGATGC TGCCCGGTTTGGCACTGCTCCTGCTGGCCGCCTGGACG GCTCGGGCGCTGGAGGTACCCACTGATGGTAATGCTGG CCTGCTGGCTGAACCCCAGATTGCCATGTTCTGTGGCAG ACTGAACATGCACATGAATGTCCAGAATGGGAAGTGGGA TTCAGATCCATCAGGGACCAAAACCTGCATTGATACCAAG GAAGGCATCCTGCAGTATTGCCAAGAAGTCTACCCTGAA CTGCAGATCACCAATGTGGTAGAAGCCAACCAACCAGTG ACCATCCAGAACTGGTGCAAGCGGGGCCGCAAGCAGTG CAAGACCCATCCCCACTTTGTGATTCCCTACCGCTGCTTA GTTGGTGAGTTTGTAAGTGATGCCCTTCTCGTTCCTGACA AGTGCAAATTCTTACACCAGGAGAGGATGGATGTTTGCG AAACTCATCTTCACTGGCACACCGTCGCCAAAGAGACAT GCAGTGAGAAGAGTACCAACTTGCATGACTACGGCATGT TGCTGCCCTGCGGAATTGACAAGTTCCGAGGGGTAGAGT TTGTGTGTTGCCCACTGGCTGAAGAAAGTGACAATGTGG ATTCTGCTGATGCGGAGGAGGATGACTCGGATGTCTGGT GGGGCGGAGCAGACACAGACTATGCAGATGGGAGTGAA GACAAAGTAGTAGAAGTAGCAGAGGAGGAAGAAGTGGCT GAGGTGGAAGAAGAAGAAGCCGATGATGACGAGGACGA TGAGGATGGTGATGAGGTAGAGGAAGAGGCTGAGGAAC CCTACGAAGAAGCCACAGAGAGAACCACCAGCATTGCCA CCACCACCACCACCACCACAGAGTCTGTGGAAGAGGTGG TTCGAGTTCCTACAACAGCAGCCAGTACCCCTGATGCCG TTGACAAGTATCTCGAGACACCTGGGGATGAGAATGAAC ATGCCCATTTCCAGAAAGCCAAAGAGAGGCTTGAGGCCA AGCACCGAGAGAGAATGTCCCAGGTCATGAGAGAATGGG AAGAGGCAGAACGTCAAGCAAAGAACTTGCCTAAAGCTG ATAAGAAGGCAGTTATCCAGCATTTCCAGGAGAAAGTGG AATCTTTGGAACAGGAAGCAGCCAACGAGAGACAGCAGC TGGTGGAGACACACATGGCCAGAGTGGAAGCCATGCTCA ATGACCGCCGCCGCCTGGCCCTGGAGAACTACATCACC GCTCTGCAGGCTGTTCCTCCTCGGCCTCGTCACGTGTTC AATATGCTAAAGAAGTATGTCCGCGCAGAACAGAAGGAC AGACAGCACACCCTAAAGCATTTCGAGCATGTGCGCATG GTGGATCCCAAGAAAGCCGCTCAGATCCGGTCCCAGGTT ATGACACACCTCCGTGTGATTTATGAGCGCATGAATCAGT CTCTCTCCCTGCTCTACAACGTGCCTGCAGTGGCCGAGG AGATTCAGGATGAAGTTGATGAGCTGCTTCAGAAAGAGC AAAACTATTCAGATGACGTCTTGGCCAACATGATTAGTGA ACCAAGGATCAGTTACGGAAACGATGCTCTCATGCCATCT TTGACCGAAACGAAAACCACCGTGGAGCTCCTTCCCGTG AATGGAGAGTTCAGCCTGGACGATCTCCAGCCGTGGCAT TCTTTTGGGGCTGACTCTGTGCCAGCCAACACAGAAAAC GAAGTTGAGCCTGTTGATGCCCGCCCTGCTGCCGACCGA GGACTGACCACTCGACCAGGTTCTGGGTTGACAAATATC AAGACGGAGGAGATCTCTGAAGTGAAGATGGATGCAGAA TTCCGACATGACTCAGGATATGAAGTTCATCATCAAAAAT TGGTGTTCTTTGCAGAAGATGTGGGTTCAAACAAAGGTG CAATCATTGGACTCATGGTGGGCGGTGTTGTCATAGCGA CAGTGATCGTCATCACCTTGGTGATGCTGAAGAAGAAAC AGTACACATCCATTCATCATGGTGTGGTGGAGGTTGACG CCGCTGTCACCCCAGAGGAGCGCCACCTGTCCAAGATG CAGCAGAACGGCTACGAAAATCCAACCTACAAGTTCTTTG AGCAGATGCAGAACTAGACCCCCGCCACAGCAGCCTCTG AAGTTGGACAGCAAAACCATTGCTTCACTACCCATCGGTG TCCATTTATAGAATAATGTGGGAAGAAACAAACCCGTTTT ATGATTTACTCATTATCGCCTTTTGACAGCTGTGCTGTAA CACAAGTAGATGCCTGAACTTGAATTAATCCACACATCAG TAATGTATTCTATCTCTCTTTACATTTTGGTCTCTATACTAC ATTATTAATGGGTTTTGTGTACTGTAAAGAATTTAGCTGTA TCAAACTAGTGCATGAATAGATTCTCTCCTGATTATTTATC ACATAGCCCCTTAGCCAGTTGTATATTATTCTTGTGGTTT GTGACCCAATTAAGTCCTACTTTACATATGCTTTAAGAATC GATGGGGGATGCTTCATGTGAACGTGGGAGTTCAGCTGC TTCTCTTGCCTAAGTATTCCTTTCCTGATCACTATGCATTT TAAAGTTAAACATTTTTAAGTATTTCAGATGCTTTAGAGAG ATTTTTTTTCCATGACTGCATTTTACTGTACAGATTGCTGC

TABLE 3-continued

APP_transcrips_final		
Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
		TTCTGCTATATTTGTGATATAGGAATTAAGAGGATACACA CGTTTGTTTCTTCGTGCCTGTTTTATGTGCACACATTAGG CATTGAGACTTCAAGCTTTTCTTTTTTTGTCCACGTATCTT TGGGTCTTTGATAAAGAAAAGAATCCCTGTTCATTGTAAG CACTTTTACGGGGGGGGTGGGGAGGGGTGCTCTGCTGG TCTTCAATTACCAAGAATTCTCCAAAACAATTTTCTGCAGG ATGATTGTACAGAATCATTGCTTATGACATGATCGCTTTCT ACACTGTATTACATAAATAAATTAAATAAAATAACCCCGGG CAAGACTTTTCTTTGAAGGATGACTACAGACATTAAATAAT CGAAGTAATTTTGGGTGGGGAGAAGAGGCAGATTCAATT TTCTTTAACCAGTCTGAAGTTTCATTTATGATACAAAAGAA GATGAAAATGGAAGTGGCAATATAAGGGGATGAGGAAGG CATGCCTGGACAAACCCTTCTTTTAAGATGTGTCTTCAAT TTGTATAAAATGGTGTTTTCATGTAAATAAATACATTCTTG GAGGAGCA
8	ENSP00000346129	>TCONS_00180774|ENST00000354192 (SEQ ID NO: 8) GCGAGAGCACGCGGAGGAGCGTGCGCGGGGGCCCCGG GAGACGGCGGCGGTGGCGGCGCGGGCAGAGCAAGGAC GCGGCGGATCCCACTCGCACAGCAGCGCACTCGGTGCC CCGCGCAGGGTCGCGATGCTGCCCGGTTTGGCACTGCT CCTGCTGGCCGCCTGGACGGCTCGGGCGCTGGAGGTCT ACCCTGAACTGCAGATCACCAATGTGGTAGAAGCCAACC AACCAGTGACCATCCAGAACTGGTGCAAGCGGGGCCGC AAGCAGTGCAAGACCCATCCCCACTTTGTGATTCCCTACC GCTGCTTAGTTGGTGAGTTTGTAAGTGATGCCCTTCTCGT TCCTGACAAGTGCAAATTCTTACACCAGGAGAGGATGGA TGTTTGCGAAACTCATCTTCACTGGCACACCGTCGCCAAA GAGACATGCAGTGAGAAGAGTACCAACTTGCATGACTAC GGCATGTTGCTGCCCTGCGGAATTGACAAGTTCCGAGGG GTAGAGTTTGTGTGTTGCCCACTGGCTGAAGAAAGTGAC AATGTGGATTCTGCTGATGCGGAGGAGGATGACTCGGAT GTCTGGTGGGGCGGAGCAGACACAGACTATGCAGATGG GAGTGAAGACAAAGTAGTAGAAGTAGCAGAGGAGGAAGA AGTGGCTGAGGTGGAAGAAGAAGAAGCCGATGATGACG AGGACGATGAGGATGGTGATGAGGTAGAGGAAGAGGCT GAGGAACCCTACGAAGAAGCCACAGAGAGAACCACCAG CATTGCCACCACCACCACCACCACCACAGAGTCTGTGGA AGAGGTGGTTCGAGTTCCTACAACAGCAGCCAGTACCCC TGATGCCGTTGACAAGTATCTCGAGACACCTGGGGATGA GAATGAACATGCCCATTTCCAGAAAGCCAAAGAGAGGCT TGAGGCCAAGCACCGAGAGAGAATGTCCCAGGTCATGAG AGAATGGGAAGAGGCAGAACGTCAAGCAAAGAACTTGCC TAAAGCTGATAAGAAGGCAGTTATCCAGCATTTCCAGGAG AAAGTGGAATCTTTGGAACAGGAAGCAGCCAACGAGAGA CAGCAGCTGGTGGAGACACACATGGCCAGAGTGGAAGC CATGCTCAATGACCGCCGCCGCCTGGCCCTGGAGAACTA CATCACCGCTCTGCAGGCTGTTCCTCCTCGGCCTCGTCA CGTGTTCAATATGCTAAAGAAGTATGTCCGCGCAGAACA GAAGGACAGACAGCACACCCTAAAGCATTTCGAGCATGT GCGCATGGTGGATCCCAAGAAAGCCGCTCAGATCCGGTC CCAGGTTATGACACACCTCCGTGTGATTTATGAGCGCAT GAATCAGTCTCTCTCCCTGCTCTACAACGTGCCTGCAGT GGCCGAGGAGATTCAGGATGAAGTTGATGAGCTGCTTCA GAAAGAGCAAAACTATTCAGATGACGTCTTGGCCAACATG ATTAGTGAACCAAGGATCAGTTACGGAAACGATGCTCTCA TGCCATCTTTGACCGAAACGAAAACCACCGTGGAGCTCC TTCCCGTGAATGGAGAGTTCAGCCTGGACGATCTCCAGC CGTGGCATTCTTTTGGGGCTGACTCTGTGCCAGCCAACA CAGAAAACGAAGTTGAGCCTGTTGATGCCCGCCCTGCTG CCGACCGAGGACTGACCACTCGACCAGGTTCTGGGTTGA CAAATATCAAGACGGAGGAGATCTCTGAAGTGAAGATGG ATGCAGAATTCCGACATGACTCAGGATATGAAGTTCATCA TCAAAAATTGGTGTTCTTTGCAGAAGATGTGGGTTCAAAC AAAGGTGCAATCATTGGACTCATGGTGGGCGGTGTTGTC ATAGCGACAGTGATCGTCATCACCTTGGTGATGCTGAAG AAGAAACAGTACACATCCATTCATCATGGTGTGGTGGAG GTTGACGCCGCTGTCACCCCAGAGGAGCGCCACCTGTC CAAGATGCAGCAGAACGGCTACGAAAATCCAACCTACAA GTTCTTTGAGCAGATGCAGAACTAGACCCCCGCCACAGC AGCCTCTGAAGTTGGACAGCAAAACCATTGCTTCACTACC CATCGGTGTCCATTTATAGAATAATGTGGGAAGAAACAAA CCCGTTTTATGATTTACTCATTATCGCCTTTTGACAGCTGT GCTGTAACACAAGTAGATGCCTGAACTTGAATTAATCCAC ACATCAGTAATGTATTCTATCTCTCTTTACATTTTGGTCTC

TABLE 3-continued

APP_transcrips_final

Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
		TATACTACATTATTAATGGGTTTTGTGTACTGTAAAGAATT TAGCTGTATCAAACTAGTGCATGAATAGATTCTCTCCTGA TTATTTATCACATAGCCCCTTAGCCAGTTGTATATTATTCT TGTGGTTTGTGACCCAATTAAGTCCTACTTTACATATGCTT TAAGAATCGATGGGGGATGCTTCATGTGAACGTGGGAGT TCAGCTGCTTCTCTTGCCTAAGTATTCCTTTCCTGATCACT ATGCATTTTAAAGTTAAACATTTTTAAGTATTTCAGATGCT TTAGAGAGATTTTTTTTCCATGACTGCATTTTACTGTACAG ATTGCTGCTTCTGCTATATTTGTGATATAGGAATTAAGAG GATACACACGTTTGTTTCTTCGTGCCTGTTTTATGTGCAC ACATTAGGCATTGAGACTTCAAGCTTTTCTTTTTTTGTCCA CGTATCTTTGGGTCTTTGATAAAGAAAAGAATCCCTGTTC ATTGTAAGCACTTTTACGGGGGGGGTGGGGAGGGGTGC TCTGCTGGTCTTCAATTACCAAGAATTCTCCAAAACAATTT TCTGCAGGATGATTGTACAGAATCATTGCTTATGACATGA TCGCTTTCTACACTGTATTACATAAATAAATTAAATAAAAT AACCCCGGGCAAGACTTTTCTTTGAAGGATGACTACAGA CATTAAATAATCGAAGTAATTTTGGGTGGGGAGAAGAGG CAGATTCAATTTTCTTTAACCAGTCTGAAGTTTCATTTATG ATACAAAAGAAGATGAAAATGGAAGTGGCAATATAAGGG GATGAGGAAGGCATGCCTGGACAAACCCCTTCTTTTAAGAT GTGTCTTCAATTTGTATAAAATGGTGTTTTCATGTAAATAA ATACATTCTTGGAGGAGCA
9	ENSP00000350578	>TCONS_00180776|ENST00000357903 SEQ ID NO: 9) CGCCGCGCTCGGGCTCCGTCAGTTTCCTCGGCAGCGGT AGGCGAGAGCACGCGGAGGAGCGTGCGCGGGGGCCCC GGGAGACGGCGGCGGTGGCGGCGCGGGCAGAGCAAGG ACGCGGCGGATCCCACTCGCACAGCAGCGCACTCGGTG CCCCGCGCAGGGTCGCGATGCTGCCCGGTTTGGCACTG CTCCTGCTGGCCGCCTGGACGGCTCGGGCGCTGGAGGT ACCCACTGATGGTAATGCTGGCCTGCTGGCTGAACCCCA GATTGCCATGTTCTGTGGCAGACTGAACATGCACATGAAT GTCCAGAATGGGAAGTGGGATTCAGATCCATCAGGGACC AAAACCTGCATTGATACCAAGGAAGGCATCCTGCAGTATT GCCAAGAAGTCTACCCTGAACTGCAGATCACCAATGTGG TAGAAGCCAACCAACCAGTGACCATCCAGAACTGGTGCA AGCGGGGCCGCAAGCAGTGCAAGACCCATCCCCACTTT GTGATTCCCTACCGCTGCTTAGTTGGTGAGTTTGTAAGTG ATGCCCTTCTCGTTCCTGACAAGTGCAAATTCTTACACCA GGAGAGGATGGATGTTTGCGAAACTCATCTTCACTGGCA CACCGTCGCCAAAGAGACATGCAGTGAGAAGAGTACCAA CTTGCATGACTACGGCATGTTGCTGCCCTGCGGAATTGA CAAGTTCCGAGGGGTAGAGTTTGTGTGTTGCCCACTGGC TGAAGAAAGTGACAATGTGGATTCTGCTGATGCGGAGGA GGATGACTCGGATGTCTGGTGGGGCGGAGCAGACACAG ACTATGCAGATGGGAGTGAAGACAAAGTAGTAGAAGTAG CAGAGGAGGAAGAAGTGGCTGAGGTGGAAGAAGAAGAA GCCGATGATGACGAGGACGATGAGGATGGTGATGAGGT AGAGGAAGAGGCTGAGGAACCCTACGAAGAAGCCACAG AGAGAACCACCAGCATTGCCACCACCACCACCACCACCA CAGAGTCTGTGGAAGAGGTGGTTCGAGAGGTGTGCTCTG AACAAGCCGAGACGGGGCCGTGCCGAGCAATGATCTCC CGCTGGTACTTTGATGTGACTGAAGGGAAGTGTGCCCCA TTCTTTTACGGCGGATGTGGCGGCAACCGGAACAACTTT GACACAGAAGAGTACTGCATGGCCGTGTGTGGCAGCGC CATTCCTACAACAGCAGCCAGTACCCCTGATGCCGTTGA CAAGTATCTCGAGACACCTGGGGATGAGAATGAACATGC CCATTTCCAGAAAGCCAAAGAGAGGCTTGAGGCCAAGCA CCGAGAGAGAATGTCCCAGGTCATGAGAGAATGGGAAGA GGCAGAACGTCAAGCAAAGAACTTGCCTAAAGCTGATAA GAAGGCAGTTATCCAGCATTTCCAGGAGAAAGTGGAATC TTTGGAACAGGAAGCAGCCAACGAGAGACAGCAGCTGGT GGAGACACACATGGCCAGAGTGGAAGCCATGCTCAATGA CCGCCGCCGCCTGGCCCTGGAGAACTACATCACCGCTCT GCAGGCTGTTCCTCCTCGGCCTCGTCACGTGTTCAATAT GCTAAAGAAGTATGTCCGCGCAGAACAGAAGGACAGACA GCACACCCTAAAGCATTTCGAGCATGTGCGCATGGTGGA TCCCAAGAAAGCCGCTCAGATCCGGTCCCAGGTTATGAC ACACCTCCGTGTGATTTATGAGCGCATGAATCAGTCTCTC TCCCTGCTCTACAACGTGCCTGCAGTGGCCGAGGAGATT CAGGATGAAGTTGATGAGCTGCTTCAGAAAGAGCAAAAC TATTCAGATGACGTCTTGGCCAACATGATTAGTGAACCAA GGATCAGTTACGGAAACGATGCTCTCATGCCATCTTTGAC CGAAACGAAAACCACCGTGGAGCTCCTTCCCGTGAATGG

TABLE 3-continued

APP_transcrips_final		
Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
		AGAGTTCAGCCTGGACGATCTCCAGCCGTGGCATTCTTT TGGGGCTGACTCTGTGCCAGCCAACACAGAAAACGAAGT TGAGCCTGTTGATGCCCGCCCTGCTGCCGACCGAGGACT GACCACTCGACCAGGTTCTGGGTTGACAAATATCAAGAC GGAGGAGATCTCTGAAGTGAAGATGGATGCAGAATTCCG ACATGACTCAGGATATGAAGTTCATCATCAAAAATTGGTG TTCTTTGCAGAAGATGTGGGTTCAAACAAAGGTGCAATCA TTGGACTCATGGTGGGCGGTGTTGTCATAGCGACAGTGA TCGTCATCACCTTGGTGATGCTGAAGAAGAAACAGTACAC ATCCATTCATCATGGTGTGGTGGAGGTTGACGCCGCTGT CACCCCAGAGGAGCGCCACCTGTCCAAGATGCAGCAGA ACGGCTACGAAAATCCAACCTACAAGTTCTTTGAGCAGAT GCAGAACTAGACCCCCGCCACAGCAGCCTCTGAAGTTGG ACAGCAAAACCATTGCTTCACTACCCATCGGTGTCCATTT ATAGAATAATGTGGGAAGAAACAAACCCGTTTTATGATTT ACTCATTATCGCCTTTTGACAGCTGTGCTGTAACACAAGT AGATGCCTGAACTTGAATTAATCCACACATCAGTAATGTA TTCTATCTCTCTTTACATTTTGGTCTCTATACTACATTATTA ATGGGTTTTGTGTACTGTAAAGAATTTAGCTGTATCAAACT AGTGCATGAATAGATTCTCTCCTGATTATTTATCACATAGC CCCTTAGCCAGTTGTATATTATTCTTGTGGTTTGTGACCC AATTAAGTCCTACTTTACATATGCTTTAAGAATCGATGGG GGATGCTTCATGTGAACGTGGGAGTTCAGCTGCTTCTCTT GCCTAAGTATTCCTTTCCTGATCACTATGCATTTTAAAGTT AAACATTTTTAAGTATTTCAGATGCTTTAGAGAGATTTTTT TTCCATGACTGCATTTTACTGTACAGATTGCTGCTTCTGCT ATATTTGTGATATAGGAATTAAGAGGATACACACGTTTGTT TCTTCGTGCCTGTTTTATGTGCACACATTAGGCATTGAGA CTTCAAGCTTTTCTTTTTTTGTCCACGTATCTTTGGGTCTT TGATAAAGAAAAGAATCCCTGTTCATTGTAAGCACTTTTAC GGGGCGGGTGGGGAGGGGTGCTCTGCTGGTCTTCAATT ACCAAGAATTCTCCAAAACAATTTTCTGCAGGATGATTGT ACAGAATCATTGCTTATGACATGATCGCTTTCTACACTGT ATTACATAAATAAATTAAATAAAATAACCCCGGGCAAGACT TTTCTTTGAAGGATGACTACAGACATTAAATAATCGAAGTA ATTTTGGGTGGGGAGAAGAGGCAGATTCAATTTTCTTTAA CCAGTCTGAAGTTTCATTTATGATACAAAAGAAGATGAAA ATGGAAGTGGCAATATAAGGGGATGAGGAAGGCATGCCT GGACAAACCCTTCTTTTAAGATGTGTCTTCAATTTGTATAA AATGGTGTTTTCATGTAAATAAATACATTCTTGGAGGAGC A
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TABLE 3-continued

APP_transcrips_final

Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
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12	ENSP00000350578	>TCONS_00179154|ENST00000357903 (SEQ ID NO: 12) CATGAGGAACTCCTTGGGCCTTGACCATTTTTAGCACTTT CACAGTTTCTGATTTTGATCAAGCCACTTAATCACCCAGG ATTCTGAATCTGAAAACCAGGAGGGTTTAATTCCACTTTC ATGTTCCCTTGAGGAGCTGTTGGCAGCAGGATGAAGGCA GGTCCATACAGAACATGGGAAGGAAGCCAGGAAGCCAG CTTTTCCTTTCACATCAAAGAGATCTAGAAAGCAAAACCT GTCTCACATTTGCATACAATATTAGACTTACATATGGATGC CATGTCTGTCAAAAGACAGGCTAATTAGGGCATTAGTTTC TTTAATGGTTGCGATTTAGTAATTCACAAATAGTCCCACTT AATTTTTCATCCTATACCTAAAGGATCTACTTCACACTTGA AGTTTAAGAAGGCTTCTCCCTTAAGATAGAAAGAAGGCAT TCAAGTATCAAAATACTGGTTTCCTGCCAAAAAATAAGGG ATGAGAAAAAGCAGACACTTAGCTTATCAATCAAAATGCT GGCAGGGAAAGACTACTGGATTACCAAGTTCATTCCCCT AGCCTGCACTGATTCCTCTTTTTCTCTAAAGTTTCTTATTT TTTCAGTTTTTCTCATGATACTGACATTGCCAACCAGCAG TCTGGAAACTGTTCAGGTTGATTCTTAGCAGAAAATCGAG GGGCTCTCCTGTTACTGTTAATATCCTTAAAACACTTAAAT TTGGTTAGTTTGCTTCAAGCATTCTCAGTATATTACAAAAA AAAAAAGTACTCAAGAATTTCTAGACTTTATTTTGACTGAC ATCAGCTACCCTAATGAACAGGAGGGGACAACAGCAAGG TATATTAGGAGCATCTCCTTCCTTTTTAATCAGAATTATAT AGGAATTAAGAACTCTAAGGCCACAGTAGAGTATAGTATC TTGGAAGAAGAAAGCGGAGAATGTCTGACATTTTCACTGA TCGTTTAGGCTGATGGCTTAAACCATTTCCACCCAAGTTT CTTACAAGTTAGCATTTCCAGCCAACATTACCTACTGCAA TTTCTCTATAATCTTAAGGGTATTGAGCCCCCAAATGAGA GAGAGAAAAGAGATGTAAACTAAACAGGAGTCAGAGAAG GGGAAACTGAGTCTGTTGCACATCATTTACCCTTTAACAT GATTTTAAAGGTAATAATGCTTATAAAAATATTAGTAGTAG TAAGGGATATCAGGTGACAAGCAGAAGTGCCCCTCTCCA CAGATATGCCAGTGTATCTGTAGAAATACGGTGCTAAAAT TAGAAAAGACTGAACATTTTAATTTATTAGGTAGACCC
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TABLE 3-continued

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TABLE 3-continued

APP_transcrips_final		
Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
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TABLE 3-continued

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TABLE 3-continued

APP_transcrips_final

Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
		ATGCTGTTGACTCACCCTCCCGAGGATGGAAGCTCTGGC CTGGGTCAAGTTGTGGTCACTGCAGTTAACAGTTTGTTGA TCTCAGGGAGTATTCCACAGTTGCTGATGTAATTGACAAT GATTGGAGCCAGCTCTTCCCCAGATTCAAATGGACCAATT AGAGGACTTGTTGGTTCTGTTTATCAACTATGTACCCACT GATGGTAATGCTGGCCTGCTGGCTGAACCCCAGATTGCC ATGTTCTGTGGCAGACTGAACATGCACATGAATGTCCAGA ATGGGAAGTGGGATTCAGATCCATCAGGGACCAAAACCT GCATTGATACCAAGGAAGGCATCCTGCAGTATTGCCAAG AAGTCTACCCTGAACTGCAGATCACCAATGTGGTAGAAG CCAACCAACCAGTGACCATCCAGAACTGGTGCAAGCGGG GCCGCAAGCAGTGCAAGACCCATCCCCACTTTGTGATTC CCTACCGCTGCTTAGTTGGTGAGTTTGTAAGTGATGCCCT TCTCGTTCCTGACAAGTGCAAATTCTTACACCAGGAGAGG ATGGATGTTTGCGAAACTCATCTTCACTGGCACACCGTCG CCAAAGAGACATGCAGTGAGAAGAGTACCAACTTGCATG ACTACGGCATGTTGCTGCCCTGCGGAATTGACAAGTTCC GAGGGGTAGAGTTTGTGTGTTGCCCACTGGCTGAAGAAA GTGACAATGTGGATTCTGCTGATGCGGAGGAGGATGACT CGGATGTCTGGTGGGGCGGAGCAGACACAGACTATGCA GATGGGAGTGAAGACAAAGTAGTAGAAGTAGCAGAGGAG GAAGAAGTGGCTGAGGTGGAAGAAGAAGAAGCCGATGAT GACGAGGACGATGAGGATGGTGATGAGGTAGAGGAAGA GGCTGAGGAACCCTACGAAGAAGCCACAGAGAGAACCA CCAGCATTGCCACCACCACCACCACCACCACAGAGTCTG TGGAAGAGGTGGTTCGAGAGGTGTGCTCTGAACAAGCCG AGACGGGGCCGTGCCGAGCAATGATCTCCCGCTGGTAC TTTGATGTGACTGAAGGGAAGTGTGCCCCATTCTTTTACG GCGGATGTGGCGGCAACCGGAACAACTTTGACACAGAAG AGTACTGCATGGCCGTGTGTGGCAGCGCCATTCCTACAA CAGCAGCCAGTACCCCTGATGCCGTTGACAAGTATCTCG AGACACCTGGGGATGAGAATGAACATGCCCATTTCCAGA AAGCCAAAGAGAGGCTTGAGGCCAAGCACCGAGAGAGA ATGTCCCAGGTCATGAGAGAATGGGAAGAGGCAGAACGT CAAGCAAAGAACTTGCCTAAAGCTGATAAGAAGGCAGTTA TCCAGCATTTCCAGGAGAAAGTGGAATCTTTGGAACAGG AAGCAGCCAACGAGAGACAGCAGCTGGTGGAGACACAC ATGGCCAGAGTGGAAGCCATGCTCAATGACCGCCGCCG CCTGGCCCTGGAGAACTACATCACCGCTCTGCAGGCTGT TCCTCCTCGGCCTCGTCACGTGTTCAATATGCTAAAGAAG TATGTCCGCGCAGAACAGAAGGACAGACAGCACACCCTA AAGCATTTCGAGCATGTGCGCATGGTGGATCCCAAGAAA GCCGCTCAGATCCGGTCCCAGGTTATGACACACCTCCGT GTGATTTATGAGCGCATGAATCAGTCTCTCTCCCTGCTCT ACAACGTGCCTGCAGTGGCCGAGGAGATTCAGGATGAAG TTGATGAGCTGCTTCAGAAAGAGCAAAACTATTCAGATGA CGTCTTGGCCAACATGATTAGTGAACCAAGGATCAGTTAC GGAAACGATGCTCTCATGCCATCTTTGACCGAAACGAAAA CCACCGTGGAGCTCCTTCCCGTGAATGGAGAGTTCAGCC TGGACGATCTCCAGCCGTGGCATTCTTTTGGGGCTGACT CTGTGCCAGCCAACACAGAAAACGAAGTTGAGCCTGTTG ATGCCCGCCCTGCTGCCGACCGAGGACTGACCACTCGA CCAGGTTCTGGGTTGACAAATATCAAGACGGAGGAGATC TCTGAAGTGAAGATGGATGCAGAATTCCGACATGACTCA GGATATGAAGTTCATCATCAAAAATTGGTGTTCTTTGCAG AAGATGTGGGTTCAAACAAAGGTGCAATCATTGGACTCAT GGTGGGCGGTGTTGTCATAGCGACAGTGATCGTCATCAC CTTGGTGATGCTGAAGAAGAAACAGTACACATCCATTCAT CATGGTGTGGTGGAGGTTGACGCCGCTGTCACCCCAGA GGAGCGCCACCTGTCCAAGATGCAGCAGAACGGCTACG AAAATCCAACCTACAAGTTCTTTGAGCAGATGCAGAACTA GACCCCCGCCACAGCAGCCTCTGAAGTTGGACAGCAAAA CCATTGCTTCACTACCCATCGGTGTCCATTTATAGAATAA TGTGGGAAGAAACAAACCCGTTTTATGATTTACTCATTAT CGCCTTTTGACAGCTGTGCTGTAACACAAGTAGATGCCT GAACTTGAATTAATCCACACATCAGTAATGTATTCTATCTC TCTTTACATTTTGGTCTCTATACTACATTATTAATGGGTTTT GTGTACTGTAAAGAATTTAGCTGTATCAAACTAGTGCATG AATAGATTCTCTCCTGATTATTTATCACATAGCCCCTTAGC CAGTTGTATATTATTCTTGTGGTTTGTGACCCAATTAAGTC CTACTTTACATATGCTTTAAGAATCGATGGGGGATGCTTC ATGTGAACGTGGGAGTTCAGCTGCTTCTCTTGCCTAAGTA TTCCTTTCCTGATCACTATGCATTTTAAAGTTAAACATTTTT AAGTATTTCAGATGCTTTAGAGAGATTTTTTTTCCATGACT GCATTTTACTGTACAGATTGCTGCTTCTGCTATATTTGTGA

TABLE 3-continued

APP_transcrips_final		
Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
		TATAGGAATTAAGAGGATACACACGTTTGTTTCTTCGTGC CTGTTTTATGTGCACACATTAGGCATTGAGACTTCAAGCT TTTCTTTTTTTGTCCACGTATCTTTGGGTCTTTGATAAAGA AAAGAATCCCTGTTCATTGTAAGCACTTTTACGGGGCGG GTGGGGAGGGGTGCTCTGCTGGTCTTCAATTACCAAGAA TTCTCCAAAACAATTTTCTGCAGGATGATTGTACAGAATC ATTGCTTATGACATGATCGCTTTCTACACTGTATTACATAA ATAAATTAAATAAAATAACCCCGGGCAAGACTTTTCTTTGA AGGATGACTACAGACATTAAATAATCGAAGTAATTTTGGG TGGGGAGAAGAGGCAGATTCAATTTTCTTTAACCAGTCTG AAGTTTCATTTATGATACAAAAGAAGATGAAAATGGAAGT GGCAATATAAGGGGATGAGGAAGGCATGCCTGGACAAAC CCTTCTTTTAAGATGTGTCTTCAATTTGTATAAAATGGTGT TTTCATGTAAATAAATACATTCTTGGAGGAGCACCATTG
17	ENSP00000387483	>TCONS_00180778|ENST00000440126 (SEQ ID NO: 17) AGAGTTCCGGAGGCTTTGGTTGTCCATTGTTAGCTTAGAA GTTGGCGCAGTGTGCGTGTGATCCACGCCTAAATAGCAC AGCCTTGCTGTGCGTGGTAGAAGTTGGGTTAGTGTTGAC ATGCTGTTGACTCACCCTCCCGAGGATGGAAGCTCTGGC CTGGGTCAAGTTGTGGTCACTGCAGTTAACAGTTTGTTGA TCTCAGGGAGTATTCCACAGTTGCTGATGTAATTGACAAT GATTGGAGCCAGCTCTTCCCCAGATTCAAATGGACCAATT AGAGGACTTGTTGGTTCTGTTTATCAACTATGTACCCACT GATGGTAATGCTGGCCTGCTGGCTGAACCCCAGATTGCC ATGTTCTGTGGCAGACTGAACATGCACATGAATGTCCAGA ATGGGAAGTGGGATTCAGATCCATCAGGGACCAAAACCT GCATTGATACCAAGGAAGGCATCCTGCAGTATTGCCAAG AAGTCTACCCTGAACTGCAGATCACCAATGTGGTAGAAG CCAACCAACCAGTGACCATCCAGAACTGGTGCAAGCGGG GCCGCAAGCAGTGCAAGACCCATCCCCACTTTGTGATTC CCTACCGCTGCTTAGTTGGTGAGTTTGTAAGTGATGCCCT TCTCGTTCCTGACAAGTGCAAATTCTTACACCAGGAGAGG ATGGATGTTTGCGAAACTCATCTTCACTGGCACACCGTCG CCAAAGAGACATGCAGTGAGAAGAGTACCAACTTGCATG ACTACGGCATGTTGCTGCCCTGCGGAATTGACAAGTTCC GAGGGGTAGAGTTTGTGTGTTGCCCACTGGCTGAAGAAA GTGACAATGTGGATTCTGCTGATGCGGAGGAGGATGACT CGGATGTCTGGTGGGGCGGAGCAGACACAGACTATGCA GATGGGAGTGAAGACAAAGTAGTAGAAGTAGCAGAGGAG GAAGAAGTGGCTGAGGTGGAAGAAGAAGAAGCCGATGAT GACGAGGACGATGAGGATGGTGATGAGGTAGAGGAAGA GGCTGAGGAACCCTACGAAGAAGCCACAGAGAGAACCA CCAGCATTGCCACCACCACCACCACCACCACAGAGTCTG TGGAAGAGGTGGTTCGAGAGGTGTGCTCTGAACAAGCCG AGACGGGGCCGTGCCGAGCAATGATCTCCCGCTGGTAC TTTGATGTGACTGAAGGGAAGTGTGCCCCATTCTTTTACG GCGGATGTGGCGGCAACCGGAACAACTTTGACACAGAAG AGTACTGCATGGCCGTGTGTGGCAGCGCCATTCCTACAA CAGCAGCCAGTACCCCTGATGCCGTTGACAAGTATCTCG AGACACCTGGGGATGAGAATGAACATGCCCATTTCCAGA AAGCCAAAGAGAGGCTTGAGGCCAAGCACCGAGAGAGA ATGTCCCAGGTCATGAGAGAATGGGAAGAGGCAGAACGT CAAGCAAAGAACTTGCCTAAAGCTGATAAGAAGGCAGTTA TCCAGCATTTCCAGGAGAAAGTGGAATCTTTGGAACAGG AAGCAGCCAACGAGAGACAGCAGCTGGTGGAGACACAC ATGGCCAGAGTGGAAGCCATGCTCAATGACCGCCGCCG CCTGGCCCTGGAGAACTACATCACCGCTCTGCAGGCTGT TCCTCCTCGGCCTCGTCACGTGTTCAATATGCTAAAGAAG TATGTCCGCGCAGAACAGAAGGACAGACAGCACACCCTA AAGCATTTCGAGCATGTGCGCATGGTGGATCCCAAGAAA GCCGCTCAGATCCGGTCCCAGGTTATGACACACCTCCGT GTGATTTATGAGCGCATGAATCAGTCTCTCTCCCTGCTCT ACAACGTGCCTGCAGTGGCCGAGGAGATTCAGGATGAAG TTGATGAGCTGCTTCAGAAAGAGCAAAACTATTCAGATGA CGTCTTGGCCAACATGATTAGTGAACCAAGGATCAGTTAC GGAAACGATGCTCTCATGCCATCTTTGACCGAAACGAAAA CCACCGTGGAGCTCCTTCCCGTGAATGGAGAGTTCAGCC TGGACGATCTCCAGCCGTGGCATTCTTTTGGGGCTGACT CTGTGCCAGCCAACACAGAAAACGAAGTTGAGCCTGTTG ATGCCCGCCCTGCTGCCGACCGAGGACTGACCACTCGA CCAGGTTCTGGGTTGACAAATATCAAGACGGAGGAGATC TCTGAAGTGAAGATGGATGCAGAATTCCGACATGACTCA GGATATGAAGTTCATCATCAAAAATTGGTGTTCTTTGCAG

TABLE 3-continued

APP_transcrips_final		
Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
		AAGATGTGGGTTCAAACAAAGGTGCAATCATTGGACTCAT GGTGGGCGGTGTTGTCATAGCGACAGTGATCGTCATCAC CTTGGTGATGCTGAAGAAGAAACAGTACACATCCATTCAT CATGGTGTGGTGGAGGTTGACGCCGCTGTCACCCCAGA GGAGCGCCACCTGTCCAAGATGCAGCAGAACGGCTACG AAAATCCAACCTACAAGTTCTTTGAGCAGATGCAGAACTA GACCCCCGCCACAGCAGCCTCTGAAGTTGGACAGCAAAA CCATTGCTTCACTACCCATCGGTGTCCATTTATAGAATAA TGTGGGAAGAAACAAACCCGTTTTATGATTTACTCATTAT CGCCTTTTGACAGCTGTGCTGTAACACAAGTAGATGCCT GAACTTGAATTAATCCACACATCAGTAATGTATTCTATCTC TCTTTACATTTTGGTCTCTATACTACATTATTAATGGGTTTT GTGTACTGTAAAGAATTTAGCTGTATCAAACTAGTGCATG AATAGATTCTCTCCTGATTATTTATCACATAGCCCCTTAGC CAGTTGTATATTATTCTT
18	ENSP00000396923	>TCONS_00180783|ENST00000448850 (SEQ ID NO: 18) ACTGCAGATCACCAATGTGGTAGAAGCCAACCAACCAGT GACCATCCAGAACTGGTGCAAGCGGGGCCGCAAGCAGT GCAAGACCCATCCCCACTTTGTGATTCCCTACCGCTGCTT AGTTGGTGAGTTTGTAAGTGATGCCCTTCTCGTTCCTGAC AAGTGCAAATTCTTACACCAGGAGAGGATGGATGTTTGC GAAACTCATCTTCACTGGCACACCGTCGCCAAAGAGACA TGCAGTGAGAAGAGTACCAACTTGCATGACTACGGCATG TTGCTGCCCTGCGGAATTGACAAGTTCCGAGGGGTAGAG TTTGTGTGTTGCCCACTGGCTGAAGAAAGTGACAATGTG GATTCTGCTGATGCGGAGGAGGATGACTCGGATGTCTGG TGGGGCGGAGCAGACACAGACTATGCAGATGGGAGTGA AGACAAAGTAGTAGAAGTAGCAGAGGAGGAAGAAGTGGC TGAGGTGGAAGAAGAAGAAGCCGATGATGACGAGGACG ATGAGGATGGTGATGAGGTAGAGGAAGAGGCTGAGGAA CCCTACGAAGAAGCCACAGAGAGAACCACCAGCATTGCC ACCACCACCACCACCACCACAGAGTCTGTGGAAGAGGTG GTTCGAGAGGTGTGCTCTGAACAAGCCGAGACGGGGCC GTGCCGAGCAATGATCTCCCGCTGGTACTTTGATGTGAC TGAAGGGAAGTGTGCCCCATTCTTTTACGGCGGATGTGG CGGCAACCGGAACAACTTTGACACAGAAGAGTACTGCAT GGCCGTGTGTGGCAGCGCCATTCCTACAACAGCAGCCA GTACCCCTGATGCCGTTGACAAGTATCTCGAGACACCTG GGGATGAGAATGAACATGCCCATTTCCAGAAAGCCAAAG AGAGGCTTGAGGCCAAGCACCGAGAGAGAATGTCCCAG GTCATGAGAGAATGGGAAGAGGCAGAACGTCAAGCAAAG AACTTGCCTAAAGCTGATAAGAAGGCAGTTATCCAGCATT TCCAGGAGAAAGTGGAATCTTTGGAACAGGAAGCAGCCA ACGAGAGACAGCAGCTGGTGGAGACACACATGGCCAGA GTGGAAGCCATGCTCAATGACCGCCGCCGCCTGGCCCT GGAGAACTACATCACCGCTCTGCAGGCTGTTCCTCCTCG GCCTCGTCACGTGTTCAATATGCTAAAGAAGTATGTCCGC GCAGAACAGAAGGACAGACAGCACACCCTAAAGCATTTC GAGCATGTGCGCATGGTGGATCCCAAGAAAGCCGCTCAG ATCCGGTCCCAGGTTATGACACACCTCCGTGTGATTTATG AGCGCATGAATCAGTCTCTCTCCCTGCTCTACAACGTGC CTGCAGTGGCCGAGGAGATTCAGGATGAAGTTGGTGCAG TGGCTCATGCCTGTAATTCCAGCATTTTGGGAGGCCAAG GTGGGCAGATGACTTGAGCCCAGAAGTTCAAGACCAGAA TGGGAAACATGGCAAGACCACATTTCTACAAAAAAATTAT CCAGGCATGATAACATCTATTTGTAGTCCCAGCTACTCAG GAGGCTGTGGTGGGAGGATCTCCTGAGCCTGGGGTGGC TGAGGCTGCAGTGAGCCTTGATCACGCCACCTGGGCAAT AGAGCAAGACCCTGTCTCAAAAAAAGGAAGAAAAAGACT ATTATTTCCCCCATTGAATGGTCTTGGCACTATTACACAAA ATCAATTGTCCATAGATAATATGGGTTTATTTCTTAATTCT TAGTTCTTTTCTTTGATCTGTGTGCCTGTGCTTACTGTAGT ACCACACTGTTTTGATTATTGTAGCTTTGTAGTAAATTTTG AAATCAGC
19	ENSP00000398879	>TCONS_00180779|ENST00000439274 (SEQ ID NO: 19) CAGCAGCGCACTCGGTGCCCCGCGCAGGGTCGCGATGC TGCCCGGTTTGGCACTGCTCCTGCTGGCCGCCTGGACG GCTCGGGCGCTGGAGGTCTACCCTGAACTGCAGATCACC AATGTGGTAGAAGCCAACCAACCAGTGACCATCCAGAAC TGGTGCAAGCGGGGCCGCAAGCAGTGCAAGACCCATCC CCACTTTGTGATTCCCTACCGCTGCTTAGTTGGTGAGTTT

TABLE 3-continued

APP_transcrips_final		
Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
		GTAAGTGATGCCCTTCTCGTTCCTGACAAGTGCAAATTCT TACACCAGGAGAGGATGGATGTTTGCGAAACTCATCTTCA CTGGCACACCGTCGCCAAAGAGACATGCAGTGAGAAGAG TACCAACTTGCATGACTACGGCATGTTGCTGCCCTGCGG AATTGACAAGTTCCGAGGGGTAGAGTTTGTGTGTTGCCC ACTGGCTGAAGAAAGTGACAATGTGGATTCTGCTGATGC GGAGGAGGATGACTCGGATGTCTGGTGGGGCGGAGCAG ACACAGACTATGCAGATGGGAGTGAAGACAAAGTAGTAG AAGTAGCAGAGGAGGAAGAAGTGGCTGAGGTGGAAGAA GAAGAAGCCGATGATGACGAGGACGATGAGGATGGTGAT GAGGTAGAGGAAGAGGCTGAGGAACCCTACGAAGAAGC CACAGAGAGAACCACCAGCATTGCCACCACCACCACCAC CACCACAGAGTCTGTGGAAGAGGTGGTTCGAGAGGTGTG CTCTGAACAAGCCGAGACGGGGCCGTGCCGAGCAATGA TCTCCCGCTGGTACTTTGATGTGACTGAAGGGAAGTGTG CCCCATTCTTTTACGGCGGATGTGGCGGCAACCGGAACA ACTTTGACACAGAAGAGTACTGCATGGCCGTGTGTGGCA GCGCCATGTCCCAAAGTTTACTCAAGACTACCCAGGAAC CTCTTGCCCGAGATCCTGTTAAACTTCCTACAACAGCAGC CAGTACCCCTGATGCCGTTGACAAGTATCTCGAGACACC TGGGGATGAGAATGAACATGCCCATTTCCAGAAAGCCAA AGAGAGGCTTGAGGCCAAGCACCGAGAGAGAATGTCCC AGGTCATGAGAGAATGGGAAGAGGCAGAACGTCAAGCAA AGAACTTGCCTAAAGCTGATAAGAAGGCAGTTATCCAGCA TTTCCAGGAGAAAGTGGAATCTTTGGAACAGGAAGCAGC CAACGAGAGACAGCAGCTGGTGGAGACACACATGGCCA GAGTGGAAGCCATGCTCAATGACCGCCGCCGCCTGGCC CTGGAGAACTACATCACCGCTCTGCAGGCTGTTCCTCCT CGGCCTCGTCACGTGTTCAATATGCTAAAGAAGTATGTCC GCGCAGAACAGAAGGACAGACAGCACACCCTAAAGCATT TCGAGCATGTGCGCATGGTGGATCCCAAGAAAGCCGCTC AGATCCGGTCCCAGGTTATGACACACCTCCGTGTGATTTA TGAGCGCATGAATCAGTCTCTCTCCCTGCTCTACAACGTG CCTGCAGTGGCCGAGGAGATTCAGGATGAAGTTGATGAG CTGCTTCAGAAAGAGCAAAACTATTCAGATGACGTCTTGG CCAACATGATTAGTGAACCAAGGATCAGTTACGGAAACG ATGCTCTCATGCCATCTTTGACCGAAACGAAAACCACCGT GGAGCTCCTTCCCGTGAATGGAGAGTTCAGCCTGGACGA TCTCCAGCCGTGGCATTCTTTTGGGGCTGACTCTGTGCC AGCCAACACAGAAAACGAAGTTGAGCCTGTTGATGCCCG CCCTGCTGCCGACCGAGGACTGACCACTCGACCAGGTTC TGGGTTGACAAATATCAAGACGGAGGAGATCTCTGAAGT GAAGATGGATGCAGAATTCCGACATGACTCAGGATATGA AGTTCATCATCAAAAATTGGTGTTCTTTGCAGAAGATGTG GGTTCAAACAAAGGTGCAATCATTGGACTCATGGTGGGC GGTGTTGTCATAGCGACAGTGATCGTCATCACCTTGGTG ATGCTGAAGAAGAAACAGTACACATCCATTCATCATGGTG TGGTGGAGGTTGACGCCGCTGTCACCCCAGAGGAGCGC CACCTGTCCAAGATGCAGCAGAACGGCTACGAAAATCCA ACCTACAAGTTCTTTGAGCAGATGCAGAACTAGACCCCC GCCACAGCAGCCTCTGAAGTTGGACAGCAAAACCATTGC TTCACTACCCATCGGTGTCCATTTATAGAATAATGTGGGA AGAAACAAACCCGTTTTATGATTTACTCATTATCGCCTTTT GACAGCTGTGCTGTAACACAAGTAGATGCCTGAACTTGA ATTAATCCACACATCAGTAATGTATTCTATCTCTCTTTACA TTTTGGTCTCTATACTACATTATTAATGGGTTTTGTGTACT GTAAAGAATTTAGCTGTATCAAACTAGTGCATGAATAGAT TCTCTCCTGATTATTTATCACATAGCCCCTTAGCCAGTTGT ATATTATTCT
20	ENSP00000406539	>TCONS_00180784|ENST00000415997 (SEQ ID NO: 20) GGGCGGAGCAGACACAGACTATGCAGATGGGAGTGAAG ACAAAGTAGTAGAAGTAGCAGAGGAGGAAGAAGTGGCTG AGGTGGAAGAAGAAGAAGCCGATGATGACGAGGACGAT GAGGATGGTGATGAGGTAGAGGAAGAGGCTGAGGAACC CTACGAAGAAGCCACAGAGAGAACCACCAGCATTGCCAC CACCACCACCACCACCACAGAGTCTGTGGAAGAGGTGGT TCGAGTGTCCCAAAGTTTACTCAAGACTACCCAGGAACCT CTTGCCCGAGATCCTGTTAAACTTCCTACAACAGCAGCCA GTACCCCTGATGCCGTTGACAAGTATCTCGAGACACCTG GGGATGAGAATGAACATGCCCATTTCCAGAAAGCCAAAG AGAGGCTTGAGGCCAAGCACCGAGAGAGAATGTCCCAG GTCATGAGAGAATGGGAAGAGGCAGAACGTCAAGCAAAG

TABLE 3-continued

APP_transcrips_final		
Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
		AACTTGCCTAAAGCTGATAAGAAGGCAGTTATCCAGCATT TCCAGGAGAAAGTGGAATCTTTGGAACAGGAAGCA
21	Uniprot_id: P05067	>TCONS_00180780|ENST00000464867 (SEQ ID NO: 21) ATCCTTGCCAACCTCTCAACCAGGATTTAACTTCTGCTTTT CCCCCATTTTCAAAAATTATAGCATGTATTTAAAGGCAGC AGAAGCCTTACTTTCAGGTTTCCCTTACCCTTTCATTTCTT TTTGTTCAAAATAGGTAGTAATTGAAGTTTTAAATATAGGG TATCATTTTTCTTTAAGAGTCATTTATCAATTTTCTTCTAAC TTCAGGCCTAGAAAGAAGTTTTGGGTAGGCTTTGTCTTAC AGTGTTATTATTTATGAGTAAAACTAATTGGTTGTCCTGCA TACTTTAATTATGATGTAATACAGGTTCTGGGTTGACAAAT ATCAAGACGGAGGAGATCTCTGAAGTGAAGATGGATGCA GAATTCCGACATGACTCAGGATATGAAGTTCATCATCAAA AATTGGTGTTCTTTGCAGAAGATGTGGGTTCAAACAAAGG TGCAATCATTGGACTCATGGTGGGCGGTGTTGTCATAGC GACAGTGATCGTCATCACCTTGGTGATGCTGAAGAAGAA ACAGTACACATCCATTCATCATGGTGTGGTGGAGGTTGA CGCCGCTGTCACCCCAGAGGAGCGCCACCTGTCCAAGA TGCAGCAGAACGGCTACGAAAATCCAACCTACAAGTTCTT TGAGCAGATGCAGAACTAGACCCCCGCCACAGCAGCCTC TGAAGTTGGACAGCAAAACCATTGCTTCACTACCCATCGG TGTCCATTTATAGAATAATGTGGGAAGAAACAAACCCGTT TTATGATTTACTCATTATCGCCTTTTGACAGCTGTGCTGTA ACACAAGTAGATGCC
22	Uniprot_id: P05067	>TCONS_00180785|ENST00000491395 (SEQ ID NO: 22) CAGACTATGCAGATGGGAGTGAAGACAAAGTAGTAGAAG TAGCAGAGGAGGAAGAAGTGGCTGAGGTGGAAGAAGAA GAAGCCGATGATGACGAGGACGATGAGGATGGTGATGA GGTAGAGGAAGAGGCTGAGGAACCCTACGAAGAAGCCA CAGAGAGAACCACCAGCATTGCCACCACCACCACCACCA CCACAGAGTCTGTGGAAGAGGTGGTTCGAGAGGTGTGCT CTGAACAAGCCGAGACGGGGCCGTGCCGAGCAATGATC TCCCGCTGGTACTTTGATGTGACTGAAGGGAAGTGTGCC CCATTCTTTTACGGCGGATGTGGCGGCAACCGGAACAAC TTTGACACAGAAGAGTACTGCATGGCCGTGTGTGGCAGC GCCATGTCCCAAAGTTTACTCAAGACTACCCAGGAACCTC TTGCCCGAGATCCTGTTAAACGTACGTTGTCATTCACCTG AGGGAAGGGAAGAGGGGAGGAGGATGCTGCTTGGTTCA CATAACTCCAGCATCATCACCTTCTTTGCATGGTTTTGTG TTTCTTGAACACCTGTCTTAGTAAAATGTTTCTTCCCATTA CCTTGCTTGTAATTACATCTGATTTTGCCAGACA
23	Uniprot_id: P05067	>TCONS_00180786|ENST00000474136 (SEQ ID NO: 23) CGAGAGCACGCGGAGGAGCGTGCGCGGGGGCCCCGGG AGACGGCGGCGGTGGCGGCGCGGGCAGAGCAAGGACG CGGCGGATCCCACTCGCACAGCAGCGCACTCGGTGCCC CGCGCAGGGTCGCGATGCTGCCCGGTTTGGCACTGCTC CTGCTGGCCGCCTGGACGGCTCGGGCGCTGGAGGTACC CACTGATGGTAATGCTGGCCTGCTGGCTGAACCCCAGAT TGCCATGTTCTGTGGCAGACTGAACATGCACATGAATGTC CAGAATGGGAAGTGGGATTCAGATCCATCAGGGACCAAA ACCTGCATTGATACCAAGGAAGGCATCCTGCAGTATTGC CAAGAAGTCTACCCTGAACTGCAGATCACCAATGTGGTA GAAGCCAACCAACCAGTGACCATCCAGAACTGGTGCAAG CGGGGCCGCAAGCAGTGCAAGACCCATCCCCACTTTGTG ATTCCCTACCGCTGCTTAGTTGGTGAGTTTGTAAGTGATG CCCTTCTCGTTCCTGACAAGTGCAAATTCTTACACCAGGA GAGGATGGATGTTTGCGAAACTCATCTTCACTGGCACAC CGTCGCCAAAGAGACATGCAGTGAGAAGAGTACCAACTT GCATGACTACGGCATGTTGCTGCCCTGCGGAATTGACAA GTTCCGAGGGGTAGAGTTTGTGTGTTGCCCACTGGCTGA AGAAAGTGACAATGTGGATTCTGCTGATGCGGAGGAGGA TGACTCGGATGTCTGGTGGGGCGGAGCAGACACAGACT ATGCAGATGGGAGTGAAGACAAAGTAGTAGAAGTAGCAG AGGAGGAAGAAGTGGCTGAGGTGGAAGAAGAAGAAGCC GATGATGACGAGGACGATGAGGATGGTGATGAGGTAGA GGAAGAGGCTGAGGAACCCTACGAAGAAGCCACAGAGA GAACCACCAGCATTGCCACCACCACCACCACCACCACAG AGTCTGTGGAAGAGGTGGTTCGAGAGAAGTGGTATAAGG AAGTACATTCTGGCCAGGCACGATGGCTCATGCTGTAAT

TABLE 3-continued

APP_transcrips_final		
Number of variants	Ensembl_protein_id/ UniProt_id	Fasta_Formatted_DNA_sequences_with_cufflinks_ transcript_ids_and_corres-ponding_Ensembl_ids
		CCCAGCACTTTGGGAGGCCGAGGTGGGTGCATCACCTG AGGTCAGGAGTTTGAGACCAGCCTGGCCAACATGGTGAA ACCCCTCGCTACTAAAAATACAAAAATTAGCCGGGCGTG GTGGCACACACCTGTGGTCCCAGCTACTCGGGAGGCTG AAGCAGGAGAATCGCTTGAACCCGGGAGACGGAGGTTG CAGTAAGCCGAGTTCACTCCATTGTACTCTAGCCTGGGT GACAGAGCGAGATTCGTCTCAAAAAAAAAAAAA
24	Uniprot_id: P05067	>TCONS_00180787|ENST00000463070 (SEQ ID NO: 24) GTAACATTCTAAAGGTAGTAGGGTCTTGATTGGGTTGCTT AGGCATTAAAAGGCTGTTTAACTTGTCTTGAAGTCTATCTT TCCTTGATGTCTTCTGCGGTAAGAACACTGTGATACAGAT GGAATGACGGGAAGTGGTTTTCCTTTCTTTCAGTTGGTGA GTTTGTAAGTGATGCCCTTCTCGTTCCTGACAAGTGCAAA TTCTTACACCAGGAGAGGATGGATGTTTGCGAAACTCATC TTCACTGGCACACCGTCGCCAAAGAGACATGCAGTGAGA AGAGTACCAACTTGCATGACTACGGCATGTTGCTGCCCT GCGGAATTGACAAGTTCCGAGGGGTAGAGTTTGTGTGTT GCCCACTGGCTGAAGAAAGTGACAATGTGGATTCTGCTG ATGCGGAGGAGGATGACTCGGATGTCTGGTGGGGCGGA GCAGACACAGACTATGCAGATGGGAGTGAAGACAAAGTA GTAGAAGTAGCAGAGGAGGAAGAAGTGGCTGAGGTGGA AGAAGAAGAAGCCGATGATGACGAGGACGATGAGGATG GTGATGAGGTAGAGGAAGAGGCTGAGGAACCC
25	Uniprot_id: P05067	>TCONS_00180788|ENST00000548570 (SEQ ID NO: 25) ACTGGAGGGCTGAGAAGAGACTGATGGCATTTGTTGTTC TTGACCTTGAAAGAAGAGTTGCAGATTGTTGGAGCAAGG CCAGATGGTAATAGGTTGGAAAGAACAAGTGAGGGGCGT GAGAGTGACAGTCTACAACCCCGTTAAGAAGTTATCTGTG AAAATGCCTCTTCCTGTCTTGATTATAGCCTCCCTCGCAC ATGGCTTTCTGAGTATGTTGGTGAGTTTGTAAGTGATGCC CTTCTCGTTCCTGACAAGTGCAAATTCTTACACCAGGAGA GGATGGATGTTTGCGAAACTCATCTTCACTGGCACACCG TCGCCAAAGAGACATGCAGTGAGAAGAGTACCAACTTGC ATGACT
26	Uniprot_id: P05067	>TCONS_00180789|ENST00000462267 (SEQ ID NO: 26) CTGAACATGCACATGAATGTCCAGAATGGGAAGTGGGAT TCAGATCCATCAGGGACCAAAACCTGCATTGATACCAAG GAAGGCATCCTGCAGTATTGCCAAGAAGTCTACCCTGAA CTGCAGATCACCAATGTGGTAGAAGCCAACCAACCAGTG ACCATCCAGAACTGGTGCAAGCGGGGCCGCAAGCAGTG CAAGACCCATCCCCACTTTGTGATTCCCTACCGCTGCTTA GGTGAGCCGGCCGGCCGTGGGGCTGGTGTTGATTGGGG GCCTGGTCTTGAGGGAAGAAAAAGAGGATGCTCCTGTTA GGTCACATACACAGACTTGTTCTTCAGCACATTGCCACTC TGTGTTGTACTGTGTTTTGGACTCTTGCAGTTACATTCTGT GCACTGACCCTATAGGAGCAGTATTTTTGAGTTCCCTGCC TCAGAATGAATTTACCCAGGGTGTATATTGAAATTACAAAT TCCTGGGCCAGTTCCAGGACTCCTGAATGAAAAATGCCT ATAGTAGCGGATCCGGGAATTCTTATTTTACCGTATCGCA TAGATGATTCTCATGAACAGGGGCCTTGTGTGTTTCTTCA CATAGACTTTCTAGAAGAAAGAATCTAATGTGAAGCTGCA GCATTTTGTTAATTTCTAAAAAAAAAAAAAAAAAA
27	Uniprot_id: P05067	>TCONS_00180790|ENST00000466453 (SEQ ID NO: 27) GGCGCGGCCTCTTCCCTGGCAGCTCTGGGGACTCTGGT TTAGTTCCCCTGGGGGCACAGGATGCTGGGGAGGGTCC GAAGGGTCTTTTTTTTAGGGTGCAGATAAAAGGATCGAAT TGAGTGAAGATTAAGACGGAGAAGATGGCGCCTCTGCAG TGCAGCAAAGAAAAGCTGTGTGGAGGCTGCAGCCTAGTG AAATCCACCCACCACTAGGTACCCACTGATGGTAATGCT GGCCTGCTGGCTGAACCCCAGATTGCCATGTTCTGTGGC AGACTGAACATGCACATGAATGTCCAGAATGGGAAGTGG GATTCAGATCCATCAGGGACCAAAACCTGCATTGATACCA AGGAAGGCATCCTGCAGTATTGC

REFERENCES

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SEQUENCE LISTING

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aagttcatca tcaaaaattg gtgttctttg cagaagatgt gggttcaaac aaaggtgcaa   2160

tcattggact catggtgggc ggtgttgtca tagcgacagt gatcgtcatc accttggtga   2220

tgctgaagaa gaaacagtac acatccattc atcatggtgt ggtggaggtt gacgccgctg   2280

tcaccccaga ggagcgccac ctgtccaaga tgcagcagaa cggctacgaa aatccaacct   2340

acaagttctt tgagcagatg cagaactaga cccccgccac agcagcctct gaagttggac   2400

agcaaaacca ttgcttcact acccatcggt gtccatttat agaataatgt gggaagaaac   2460

aaacccgttt tatgatttac tcattatcgc cttttgacag ctgtgctgta acacaagtag   2520

atgcctgaac ttgaattaat ccacacatca gtaatgtatt ctatctctct ttacattttg   2580

gtctctatac tacattatta atgggttttg tgtactgtaa agaatttagc tgtatcaaac   2640

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ttattcttgt ggtttgtgac ccaattaagt cctactttac atatgcttta agaatcgatg   2760

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tttttccat gactgcattt tactgtacag attgctgctt ctgctatatt tgtgatatag   2940

gaattaagag gatacacacg tttgtttctt cgtgcctgtt ttatgtgcac acattaggca   3000

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aatccctgtt cattgtaagc acttttacgg ggcgggtggg gaggggtgct ctgctggtct   3120

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tatgacatga tcgctttcta cactgtatta cataaataaa ttaaataaaa taaccccggg   3240

caagactttt ctttgaagga tgactacaga cattaaataa tcgaagtaat tttgggtggg   3300

gagaagaggc agattcaatt ttctttaacc agtctgaagt ttcatttatg atacaaaaga   3360

agatgaaaat ggaagtggca atataagggg atgaggaagg catgcctgga caaacccttc   3420

ttttaagatg tgtcttcaat ttgtataaaa tggtgttttc atgtaaataa atacattctt   3480

ggaggagcac cattg                                                    3495
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<210> SEQ ID NO 5
<211> LENGTH: 3254
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor
      protein coding sequence

<400> SEQUENCE: 5

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gtgcgcgggg gccccgggag acggcggcgg tggcggcgcg ggcagagcaa ggacgcggcg    120

gatcccactc gcacagcagc gcactcggtg ccccgcgcag ggtcgcgatg ctgcccggtt    180

tggcactgct cctgctggcc gcctggacgg ctcgggcgct ggaggtaccc actgatggta    240

atgctggcct gctggctgaa ccccagattg ccatgttctg tggcagactg aacatgcaca    300

tgaatgtcca gaatgggaag tgggattcag atccatcagg gaccaaaacc tgcattgata    360

ccaaggaagg catcctgcag tattgccaag aagtctaccc tgaactgcag atcaccaatg    420

tggtagaagc caaccaacca gtgaccatcc agaactggtg caagcggggc cgcaagcagt    480

gcaagaccca tccccacttt gtgattccct accgctgctt agttggtgag tttgtaagtg    540

atgcccttct cgttcctgac aagtgcaaat tcttacacca ggagaggatg gatgtttgcg    600
```

```
aaactcatct tcactggcac accgtcgcca aagagacatg cagtgagaag agtaccaact    660
tgcatgacta cggcatgttg ctgccctgcg gaattgacaa gttccgaggg gtagagtttg    720
tgtgttgccc actggctgaa gaaagtgaca atgtggattc tgctgatgcg gaggaggatg    780
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ccacagagag aaccaccagc attgccacca ccaccaccac caccacagag tctgtggaag   1020
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aagcagccaa cgagagacag cagctggtgg agacacacat ggccagagtg gaagccatgc   1560
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tttccatgat cgctttctac actgtattac ataaataaat taaataaaat aaccccgggc   3000

aagacttttc tttgaaggat gactacagac attaaataat cgaagtaatt ttgggtgggg   3060

agaagaggca gattcaattt tctttaacca gtctgaagtt tcatttatga tacaaaagaa   3120

gatgaaaatg gaagtggcaa tataagggga tgaggaaggc atgcctggac aaacccttct   3180

tttaagatgt gtcttcaatt tgtataaaat ggtgttttca tgtaaataaa tacattcttg   3240

gaggagcacc attg                                                     3254


<210> SEQ ID NO 6
<211> LENGTH: 3467
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor
      protein coding sequence

<400> SEQUENCE: 6

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gctggccgcc tggacggctc gggcgctgga ggtacccact gatggtaatg ctggcctgct    120

ggctgaaccc cagattgcca tgttctgtgg cagactgaac atgcacatga atgtccagaa    180

tgggaagtgg gattcagatc catcagggac caaaacctgc attgatacca aggaaggcat    240

cctgcagtat tgccaagaag tctaccctga actgcagatc accaatgtgg tagaagccaa    300

ccaaccagtg accatccaga actggtgcaa gcggggccgc aagcagtgca agacccatcc    360

ccactttgtg attccctacc gctgcttagt tggtgagttt gtaagtgatg cccttctcgt    420

tcctgacaag tgcaaattct tacaccagga gaggatggat gtttgcgaaa ctcatcttca    480

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ggctgaagaa agtgacaatg tggattctgc tgatgcggag gaggatgact cggatgtctg    660

gtggggcgga gcagacacag actatgcaga tgggagtgaa gacaaagtag tagaagtagc    720

agaggaggaa gaagtggctg aggtggaaga agaagaagcc gatgatgacg aggacgatga    780

ggatggtgat gaggtagagg aagaggctga ggaaccctac gaagaagcca cagagagaac    840

caccagcatt gccaccacca ccaccaccac cacagagtct gtggaagagg tggttcgaga    900

ggtgtgctct gaacaagccg agacggggcc gtgccgagca atgatctccc gctggtactt    960

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ccgcctggcc ctggagaact acatcaccgc tctgcaggct gttcctcctc ggcctcgtca   1500

cgtgttcaat atgctaaaga agtatgtccg cgcagaacag aaggacagac agcacaccct   1560

aaagcatttc gagcatgtgc gcatggtgga tcccaagaaa gccgctcaga tccggtccca   1620

ggttatgaca cacctccgtg tgatttatga gcgcatgaat cagtctctct ccctgctcta   1680
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caacgtgcct gcagtggccg aggagattca ggatgaagtt gatgagctgc ttcagaaaga 1740

gcaaaactat tcagatgacg tcttggccaa catgattagt gaaccaagga tcagttacgg 1800

aaacgatgct ctcatgccat ctttgaccga aacgaaaacc accgtggagc tccttcccgt 1860

gaatggagag ttcagcctgg acgatctcca gccgtggcat tcttttgggg ctgactctgt 1920

gccagccaac acagaaaacg aagttgagcc tgttgatgcc cgccctgctg ccgaccgagg 1980

actgaccact cgaccaggtt ctgggttgac aaatatcaag acggaggaga tctctgaagt 2040

gaagatggat gcagaattcc gacatgactc aggatatgaa gttcatcatc aaaaattggt 2100

gttctttgca gaagatgtgg gttcaaacaa aggtgcaatc attggactca tggtgggcgg 2160

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gaactagacc cccgccacag cagcctctga agttggacag caaaaccatt gcttcactac 2400

ccatcggtgt ccatttatag aataatgtgg gaagaaacaa acccgtttta tgatttactc 2460

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ggagttcagc tgcttctctt gcctaagtat tcctttcctg atcactatgc attttaaagt 2820

taaacatttt taagtatttc agatgcttta gagagatttt ttttccatga ctgcatttta 2880

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ctgtattaca taaataaatt aaataaaata accccgggca agacttttct ttgaaggatg 3240

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ctttaaccag tctgaagttt catttatgat acaaaagaag atgaaaatgg aagtggcaat 3360

ataaggggat gaggaaggca tgcctggaca aacccttctt ttaagatgtg tcttcaattt 3420

gtataaaatg gtgttttcat gtaaataaat acattcttgg aggagca 3467

<210> SEQ ID NO 7
<211> LENGTH: 3355
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor protein coding sequence

<400> SEQUENCE: 7

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gcactcggtg ccccgcgcag ggtcgcgatg ctgcccggtt tggcactgct cctgctggcc 180

gcctggacgg ctcgggcgct ggaggtaccc actgatggta atgctggcct gctggctgaa 240

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ccccagattg ccatgttctg tggcagactg aacatgcaca tgaatgtcca gaatgggaag    300
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gtgaccatcc agaactggtg caagcggggc cgcaagcagt gcaagaccca tccccacttt    480
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accgtcgcca aagagacatg cagtgagaag agtaccaact tgcatgacta cggcatgttg    660
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gcattttact gtacagattg ctgcttctgc tatatttgtg atataggaat taagaggata 2820

cacacgtttg tttcttcgtg cctgttttat gtgcacacat taggcattga gacttcaagc 2880

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gaaggatgac tacagacatt aaataatcga agtaattttg ggtggggaga agaggcagat 3180

tcaattttct ttaaccagtc tgaagtttca tttatgatac aaaagaagat gaaaatggaa 3240

gtggcaatat aaggggatga ggaaggcatg cctggacaaa cccttctttt aagatgtgtc 3300

ttcaatttgt ataaaatggt gttttcatgt aaataaatac attcttggag gagca 3355

<210> SEQ ID NO 8
<211> LENGTH: 3167
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor protein coding sequence

<400> SEQUENCE: 8

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ggcagagcaa ggacgcggcg gatcccactc gcacagcagc gcactcggtg ccccgcgcag 120

ggtcgcgatg ctgcccggtt tggcactgct cctgctggcc gcctggacgg ctcgggcgct 180

ggaggtctac cctgaactgc agatcaccaa tgtggtagaa gccaaccaac cagtgaccat 240

ccagaactgg tgcaagcggg gccgcaagca gtgcaagacc catccccact ttgtgattcc 300

ctaccgctgc ttagttggtg agtttgtaag tgatgccctt ctcgttcctg acaagtgcaa 360

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caaagagaca tgcagtgaga agagtaccaa cttgcatgac tacggcatgt tgctgccctg 480

cggaattgac aagttccgag ggtagagtt tgtgtgttgc ccactggctg aagaaagtga 540

caatgtggat tctgctgatg cggaggagga tgactcggat gtctggtggg gcggagcaga 600

cacagactat gcagatggga gtgaagacaa agtagtagaa gtagcagagg aggaagaagt 660

ggctgaggtg gaagaagaag aagccgatga tgacgaggac gatgaggatg gtgatgaggt 720

agaggaagag gctgaggaac cctacgaaga agccacagag agaaccacca gcattgccac 780

caccaccacc accaccacag agtctgtgga agaggtggtt cgagttccta caacagcagc 840

cagtacccct gatgccgttg acaagtatct cgagacacct ggggatgaga atgaacatgc 900

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ccgcctggcc ctggagaact acatcaccgc tctgcaggct gttcctcctc ggcctcgtca 1200

cgtgttcaat atgctaaaga agtatgtccg cgcagaacag aaggacagac agcacaccct 1260

aaagcatttc gagcatgtgc gcatggtgga tcccaagaaa gccgctcaga tccggtccca 1320

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ggttatgaca cacctccgtg tgatttatga gcgcatgaat cagtctctct ccctgctcta   1380

caacgtgcct gcagtggccg aggagattca ggatgaagtt gatgagctgc ttcagaaaga   1440

gcaaaactat tcagatgacg tcttggccaa catgattagt gaaccaagga tcagttacgg   1500

aaacgatgct ctcatgccat ctttgaccga aacgaaaacc accgtggagc tccttcccgt   1560

gaatggagag ttcagcctgg acgatctcca gccgtggcat tcttttgggg ctgactctgt   1620

gccagccaac acagaaaacg aagttgagcc tgttgatgcc cgccctgctg ccgaccgagg   1680

actgaccact cgaccaggtt ctgggttgac aaatatcaag acggaggaga tctctgaagt   1740

gaagatggat gcagaattcc gacatgactc aggatatgaa gttcatcatc aaaaattggt   1800

gttctttgca gaagatgtgg gttcaaacaa aggtgcaatc attggactca tggtgggcgg   1860

tgttgtcata gcgacagtga tcgtcatcac cttggtgatg ctgaagaaga aacagtacac   1920

atccattcat catggtgtgg tggaggttga cgccgctgtc accccagagg agcgccacct   1980

gtccaagatg cagcagaacg gctacgaaaa tccaacctac aagttctttg agcagatgca   2040

gaactagacc cccgccacag cagcctctga agttggacag caaaaccatt gcttcactac   2100

ccatcggtgt ccatttatag aataatgtgg gaagaaacaa acccgtttta tgatttactc   2160

attatcgcct tttgacagct gtgctgtaac acaagtagat gcctgaactt gaattaatcc   2220

acacatcagt aatgtattct atctctcttt acattttggt ctctatacta cattattaat   2280

gggttttgtg tactgtaaag aatttagctg tatcaaacta gtgcatgaat agattctctc   2340

ctgattattt atcacatagc cccttagcca gttgtatatt attcttgtgg tttgtgaccc   2400

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taaacatttt taagtatttc agatgcttta gagagatttt ttttccatga ctgcatttta   2580

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tgtttcttcg tgcctgtttt atgtgcacac attaggcatt gagacttcaa gcttttcttt   2700

ttttgtccac gtatctttgg gtctttgata aagaaaagaa tccctgttca ttgtaagcac   2760

ttttacgggg cgggtgggga ggggtgctct gctggtcttc aattaccaag aattctccaa   2820

aacaatttttc tgcaggatga ttgtacagaa tcattgctta tgacatgatc gctttctaca   2880

ctgtattaca taaataaatt aaataaaata accccgggca agacttttct ttgaaggatg   2940

actacagaca ttaaataatc gaagtaattt tgggtgggga gaagaggcag attcaatttt   3000

ctttaaccag tctgaagttt catttatgat acaaaagaag atgaaaatgg aagtggcaat   3060

ataaggggat gaggaaggca tgcctggaca aacccttctt ttaagatgtg tcttcaattt   3120

gtataaaatg gtgttttcat gtaaataaat acattcttgg aggagca                 3167
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<210> SEQ ID NO 9
<211> LENGTH: 3543
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor
      protein coding sequence

<400> SEQUENCE: 9

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gtgcgcgggg gccccgggag acggcggcgg tggcggcgcg ggcagagcaa ggacgcggcg    120

gatcccactc gcacagcagc gcactcggtg ccccgcgcag ggtcgcgatg ctgcccggtt    180
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tggcactgct cctgctggcc gcctggacgg ctcgggcgct ggaggtaccc actgatggta    240
atgctggcct gctggctgaa ccccagattg ccatgttctg tggcagactg aacatgcaca    300
tgaatgtcca gaatgggaag tgggattcag atccatcagg gaccaaaacc tgcattgata    360
ccaaggaagg catcctgcag tattgccaag aagtctaccc tgaactgcag atcaccaatg    420
tggtagaagc caaccaacca gtgaccatcc agaactggtg caagcggggc cgcaagcagt    480
gcaagaccca tccccacttt gtgattccct accgctgctt agttggtgag tttgtaagtg    540
atgcccttct cgttcctgac aagtgcaaat tcttacacca ggagaggatg gatgtttgcg    600
aaactcatct tcactggcac accgtcgcca aagagacatg cagtgagaag agtaccaact    660
tgcatgacta cggcatgttg ctgccctgcg gaattgacaa gttccgaggg gtagagtttg    720
tgtgttgccc actggctgaa gaaagtgaca atgtggattc tgctgatgcg gaggaggatg    780
actcggatgt ctggtggggc ggagcagaca cagactatgc agatgggagt gaagacaaag    840
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acgaggacga tgaggatggt gatgaggtag aggaagaggc tgaggaaccc tacgaagaag    960
ccacagagag aaccaccagc attgccacca ccaccaccac caccacagag tctgtggaag   1020
aggtggttcg agaggtgtgc tctgaacaag ccgagacggg gccgtgccga gcaatgatct   1080
cccgctggta ctttgatgtg actgaaggga agtgtgcccc attcttttac ggcggatgtg   1140
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ttcctacaac agcagccagt acccctgatg ccgttgacaa gtatctcgag acacctgggg   1260
atgagaatga acatgcccat ttccagaaag ccaaagagag gcttgaggcc aagcaccgag   1320
agagaatgtc ccaggtcatg agagaatggg aagaggcaga acgtcaagca aagaacttgc   1380
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aggaagcagc caacgagaga cagcagctgg tggagacaca catggccaga gtggaagcca   1500
tgctcaatga ccgccgccgc ctggccctgg agaactacat caccgctctg caggctgttc   1560
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acagacagca caccctaaag catttcgagc atgtgcgcat ggtggatccc aagaaagccg   1680
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caaggatcag ttacggaaac gatgctctca tgccatcttt gaccgaaacg aaaaccaccg   1920
tggagctcct tcccgtgaat ggagagttca gcctggacga tctccagccg tggcattctt   1980
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atcatcaaaa attggtgttc tttgcagaag atgtgggttc aaacaaaggt gcaatcattg   2220
gactcatggt gggcggtgtt gtcatagcga cagtgatcgt catcaccttg gtgatgctga   2280
agaagaaaca gtacacatcc attcatcatg gtgtggtgga ggttgacgcc gctgtcaccc   2340
cagaggagcg ccacctgtcc aagatgcagc agaacggcta cgaaaatcca acctacaagt   2400
tctttgagca gatgcagaac tagacccccg ccacagcagc ctctgaagtt ggacagcaaa   2460
accattgctt cactacccat cggtgtccat ttatagaata atgtgggaag aaacaaaccc   2520
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gttttatgat ttactcatta tcgccttttg acagctgtgc tgtaacacaa gtagatgcct   2580

gaacttgaat taatccacac atcagtaatg tattctatct ctctttacat tttggtctct   2640

atactacatt attaatgggt tttgtgtact gtaaagaatt tagctgtatc aaactagtgc   2700

atgaatagat tctctcctga ttatttatca catagcccct tagccagttg tatattattc   2760

ttgtggtttg tgacccaatt aagtcctact ttacatatgc tttaagaatc gatgggggat   2820

gcttcatgtg aacgtgggag ttcagctgct tctcttgcct aagtattcct ttcctgatca   2880

ctatgcattt taaagttaaa catttttaag tatttcagat gctttagaga gatttttttt   2940

ccatgactgc attttactgt acagattgct gcttctgcta tatttgtgat ataggaatta   3000

agaggataca cacgtttgtt tcttcgtgcc tgttttatgt gcacacatta ggcattgaga   3060

cttcaagctt ttcttttttt gtccacgtat ctttgggtct ttgataaaga aaagaatccc   3120

tgttcattgt aagcactttt acggggcggg tggggagggg tgctctgctg gtcttcaatt   3180

accaagaatt ctccaaaaca attttctgca ggatgattgt acagaatcat tgcttatgac   3240

atgatcgctt tctacactgt attacataaa taaattaaat aaaataaccc cgggcaagac   3300

ttttctttga aggatgacta cagacattaa ataatcgaag taattttggg tggggagaag   3360

aggcagattc aatttttcttt aaccagtctg aagtttcatt tatgatacaa aagaagatga   3420

aaatggaagt ggcaatataa ggggatgagg aaggcatgcc tggacaaacc cttcttttaa   3480

gatgtgtctt caatttgtat aaaatggtgt tttcatgtaa ataaatacat tcttggagga   3540

gca                                                                  3543
```

<210> SEQ ID NO 10
<211> LENGTH: 690
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor protein coding sequence

<400> SEQUENCE: 10

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ctttatcact ttactatgca atttttatga ctatgcttaa ggagagtaaa tttttcgaag     60

ttattgaggc aatggagagt ctttggaatg aggataatta ggcctgagga cacagaggaa    120

tcatgaggaa gaattctcca gtttcattcc tttttctggg tacagtttgt ttctccttct    180

aagtaagttc ctagatatag aatgaattgg aaaaaatgaa acgtgaggtt tgctacgtct    240

ataacagtat cacatttcat tttttaaaac tgccaatgct ttcagtgagg accagaaagt    300

acagtgagaa aaaaaaattc ctcaaatatt agttttcatg ctcttgcacg catttttata    360

aaggcaaaag tcattctggt gcctgtatac aatctaaagg cataatctcc tggagccttc    420

agtgctggtt ttggggtttt ctggagatca atccacagtg tcccattttt tctgctggag    480

ctctgaaccc actaagagag agcaagaaga gatgtaaacc tctcctttgc ttctgataaa    540

gccaagccct tactagtcca catgatgctt tctctgggga gtgagtcaca tacaggagac    600

atggcttgtc cagctgcgtg ctggactgaa tttcagctcc aacctgaacc ttccaacagg    660

acaagggagg gaggagaatg ggccgtagca                                     690
```

<210> SEQ ID NO 11
<211> LENGTH: 905
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor protein coding sequence

<400> SEQUENCE: 11

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catgttgccc aggctggtct taaactcctg ggctcaaatg atcctcctgc ctcagcctcc    120
catagggcta ggattacagg tgtgagccac catgcctggc cttttatact gtttattccc    180
tttagataca gttaatggac attaataagc agtttatgcc aatccctttt catagctaat    240
aagaagtctt atcttggata aataaaacct agccactaat gctgccacac ccaaataaac    300
tctctatgca tctgaacttc ttgaggttaa taatgcttca ttgtaatatt ttataaaatc    360
accactttgg aagtgaacac cactagaaat tcacatggcc agtttataga tggggactgg    420
gggaagcagg gaggatctct gcaatccagt agttaggctc tgccagaatg tatttagagt    480
tgtctctaaa tacattctga gtctctgctt catttttttt ttcaatgaaa atgacaatgt    540
ctgtcatccc atgttcaaga caaatataac aagtttctaa gaatccttat atttttgtta    600
ttgcatataa gcatgagttt taaaaacctc tgaatgttta aaggatcacc tgggggaata    660
agaaaaatgc agatctctgt aggtctgagg tagagcccaa gaatctgtat ttttaactag    720
aattcctcta aattactgtg cttcacttgg ccccaattac acattaggag acaatgattt    780
caggatgaca atcagccttt tccatccaag gacttaaagc agctaaaccg aagacaccga    840
caaataccag atacttttct cagtctactg gctgatggca caaaagtgca gaagtcacgt    900
agggc                                                                905
```

<210> SEQ ID NO 12
<211> LENGTH: 1320
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor protein coding sequence

<400> SEQUENCE: 12

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aagccactta atcacccagg attctgaatc tgaaaaccag gagggtttaa ttccactttc    120
atgttccctt gaggagctgt tggcagcagg atgaaggcag gtccatacag aacatgggaa    180
ggaagccagg aagccagctt ttcctttcac atcaaagaga tctagaaagc aaaacctgtc    240
tcacatttgc atacaatatt agacttacat atggatgcca tgtctgtcaa aagacaggct    300
aattagggca ttagtttctt taatggttgc gatttagtaa ttcacaaata gtcccactta    360
atttttcatc ctatacctaa aggatctact tcacacttga agtttaagaa ggcttctccc    420
ttaagataga aagaaggcat tcaagtatca aaatactggt ttcctgccaa aaaataaggg    480
atgagaaaaa gcagacactt agcttatcaa tcaaaatgct ggcagggaaa gactactgga    540
ttaccaagtt cattccccta gcctgcactg attcctcttt ttctctaaag tttcttattt    600
tttcagtttt tctcatgata ctgacattgc caaccagcag tctggaaact gttcaggttg    660
attcttagca gaaaatcgag gggctctcct gttactgtta atatccttaa aacacttaaa    720
tttggttagt ttgcttcaag cattctcagt atattacaaa aaaaaaaagt actcaagaat    780
ttctagactt tattttgact gacatcagct accctaatga acaggagggg acaacagcaa    840
ggtatattag gagcatctcc ttccttttta atcagaatta tataggaatt aagaactcta    900
aggccacagt agagtatagt atcttggaag aagaaagcgg agaatgtctg acattttcac    960
tgatcgttta ggctgatggc ttaaaccatt tccacccaag tttcttacaa gttagcattt   1020
```

ccagccaaca ttacctactg caatttctct ataatcttaa gggtattgag cccccaaatg 1080

agagagagaa aagagatgta aactaaacag gagtcagaga aggggaaact gagtctgttg 1140

cacatcattt accctttaac atgattttaa aggtaataat gcttataaaa atattagtag 1200

tagtaaggga tatcaggtga caagcagaag tgcccctctc cacagatatg ccagtgtatc 1260

tgtagaaata cggtgctaaa attagaaaag actgaacatt ttaatttatt aggtagaccc 1320

<210> SEQ ID NO 13
<211> LENGTH: 3552
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor
protein coding sequence

<400> SEQUENCE: 13

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gatcccactc gcacagcagc gcactcggtg ccccgcgcag ggtcgcgatg ctgcccggtt 180

tggcactgct cctgctggcc gcctggacgg ctcgggcgct ggaggtaccc actgatggta 240

atgctggcct gctggctgaa ccccagattg ccatgttctg tggcagactg aacatgcaca 300

tgaatgtcca gaatgggaag tgggattcag atccatcagg gaccaaaacc tgcattgata 360

ccaaggaagg catcctgcag tattgccaag aagtctaccc tgaactgcag atcaccaatg 420

tggtagaagc caaccaacca gtgaccatcc agaactggtg caagcggggc cgcaagcagt 480

gcaagaccca tccccacttt gtgattccct accgctgctt agttggtgag tttgtaagtg 540

atgcccttct cgttcctgac aagtgcaaat tcttacacca ggagaggatg gatgtttgcg 600

aaactcatct tcactggcac accgtcgcca aagagacatg cagtgagaag agtaccaact 660

tgcatgacta cggcatgttg ctgccctgcg gaattgacaa gttccgaggg gtagagtttg 720

tgtgttgccc actggctgaa gaaagtgaca atgtggattc tgctgatgcg gaggaggatg 780

actcggatgt ctggtggggc ggagcagaca cagactatgc agatgggagt gaagacaaag 840

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acgaggacga tgaggatggt gatgaggtag aggaagaggc tgaggaaccc tacgaagaag 960

ccacagagag aaccaccagc attgccacca ccaccaccac caccacagag tctgtggaag 1020

aggtggttcg agaggtgtgc tctgaacaag ccgagacggg gccgtgccga gcaatgatct 1080

cccgctggta ctttgatgtg actgaaggga agtgtgcccc attcttttac ggcggatgtg 1140

gcggcaaccg gaacaacttt gacacagaag agtactgcat ggccgtgtgt ggcagcgcca 1200

tgtcccaaag tttactcaag actacccagg aacctcttgc ccgagatcct gttaaacttc 1260

ctacaacagc agccagtacc cctgatgccg ttgacaagta tctcgagaca cctggggatg 1320

agaatgaaca tgcccatttc cagaaagcca aagagaggct tgaggccaag caccgagaga 1380

gaatgtccca ggtcatgaga gaatgggaag aggcagaacg tcaagcaaag aacttgccta 1440

aagctgataa gaaggcagtt atccagcatt tccaggagaa agtggaatct ttggaacagg 1500

aagcagccaa cgagagacag cagctggtgg agacacacat ggccagagtg gaagccatgc 1560

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ctcggcctcg tcacgtgttc aatatgctaa agaagtatgt ccgcgcagaa cagaaggaca 1680

gacagcacac cctaaagcat ttcgagcatg tgcgcatggt ggatcccaag aaagccgctc 1740 agatccggtc ccaggttatg acacacctcc gtgtgattta tgagcgcatg aatcagtctc 1800

tctccctgct ctacaacgtg cctgcagtgg ccgaggagat tcaggatgaa gttgatgagc 1860

tgcttcagaa agagcaaaac tattcagatg acgtcttggc caacatgatt agtgaaccaa 1920

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gggctgactc tgtgccagcc aacacagaaa acgaaggttc tgggttgaca aatatcaaga 2100

cggaggagat ctctgaagtg aagatggatg cagaattccg acatgactca ggatatgaag 2160

ttcatcatca aaaattggtg ttctttgcag aagatgtggg ttcaaacaaa ggtgcaatca 2220

ttggactcat ggtgggcggt gttgtcatag cgacagtgat cgtcatcacc ttggtgatgc 2280

tgaagaagaa acagtacaca tccattcatc atggtgtggt ggaggttgac gccgctgtca 2340

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agttctttga gcagatgcag aactagaccc ccgccacagc agcctctgaa gttggacagc 2460

aaaaccattg cttcactacc catcggtgtc catttataga ataatgtggg aagaaacaaa 2520

cccgttttat gatttactca ttatcgcctt ttgacagctg tgctgtaaca caagtagatg 2580

cctgaacttg aattaatcca cacatcagta atgtattcta tctctcttta catttttggtc 2640

tctatactac attattaatg ggttttgtgt actgtaaaga atttagctgt atcaaactag 2700

tgcatgaata gattctctcc tgattattta tcacatagcc ccttagccag ttgtatatta 2760

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tcactatgca ttttaaagtt aaacattttt aagtatttca gatgctttag agagattttt 2940

tttccatgac tgcattttac tgtacagatt gctgcttctg ctatatttgt gatataggaa 3000

ttaagaggat acacacgttt gtttcttcgt gcctgtttta tgtgcacaca ttaggcattg 3060

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gacatgatcg ctttctacac tgtattacat aaataaatta aataaaataa ccccgggcaa 3300

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aagaggcaga ttcaattttc tttaaccagt ctgaagtttc atttatgata caaaagaaga 3420

tgaaaatgga agtggcaata taaggggatg aggaaggcat gcctggacaa acccttcttt 3480

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ggagcaccat tg 3552

<210> SEQ ID NO 14
<211> LENGTH: 2366
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor
protein coding sequence

<400> SEQUENCE: 14

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atgctggcct gctggctgaa ccccagattg ccatgttctg tggcagactg aacatgcaca 180

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```
tgaatgtcca gaatgggaag tgggattcag atccatcagg gaccaaaacc tgcattgata    240
ccaaggaagg catcctgcag tattgccaag aagtctaccc tgaactgcag atcaccaatg    300
tggtagaagc caaccaacca gtgaccatcc agaactggtg caagcggggc cgcaagcagt    360
gcaagaccca tccccacttt gtgattccct accgctgctt agttggtgag tttgtaagtg    420
atgcccttct cgttcctgac aagtgcaaat tcttacacca ggagaggatg gatgtttgcg    480
aaactcatct tcactggcac accgtcgcca aagagacatg cagtgagaag agtaccaact    540
tgcatgacta cggcatgttg ctgccctgcg gaattgacaa gttccgaggg gtagagtttg    600
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tagtagaagt agcagaggag gaagaagtgg ctgaggtgga agaagaagaa gccgatgatg    780
acgaggacga tgaggatggt gatgaggtag aggaagaggc tgaggaaccc tacgaagaag    840
ccacagagag aaccaccagc attgccacca ccaccaccac caccacagag tctgtggaag    900
aggtggttcg agaggtgtgc tctgaacaag ccgagacggg gccgtgccga gcaatgatct    960
cccgctggta ctttgatgtg actgaaggga agtgtgcccc attcttttac ggcggatgtg   1020
gcggcaaccg gaacaacttt gacacagaag agtactgcat ggccgtgtgt ggcagcgcca   1080
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ctacaacagc agccagtacc cctgatgccg ttgacaagta tctcgagaca cctggggatg   1200
agaatgaaca tgcccatttc cagaaagcca aagagaggct tgaggccaag caccgagaga   1260
gaatgtccca ggtcatgaga gaatgggaag aggcagaacg tcaagcaaag aacttgccta   1320
aagctgataa gaaggcagtt atccagcatt tccaggagaa agtggaatct ttggaacagg   1380
aagcagccaa cgagagacag cagctggtgg agacacacat ggccagagtg gaagccatgc   1440
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ctcggcctcg tcacgtgttc aatatgctaa agaagtatgt ccgcgcagaa cagaaggaca   1560
gacagcacac cctaaagcat ttcgagcatg tgcgcatggt ggatcccaag aaagccgctc   1620
agatccggtc ccaggttatg acacacctcc gtgtgattta tgagcgcatg aatcagtctc   1680
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cggaggagat ctctgaagtg aagatggatg cagaattccg acatgactca ggatatgaag   2040
ttcatcatca aaaattggtg ttctttgcag aagatgtggg ttcaaacaaa ggtgcaatca   2100
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tgaagaagaa acagtacaca tccattcatc atggtgtggt ggaggttgac gccgctgtca   2220
ccccagagga gcgccacctg tccaagatgc agcagaacgg ctacgaaaat ccaacctaca   2280
agttctttga gcagatgcag aactagaccc ccgccacagc agcctctgaa gttggacagc   2340
aaaaccattg cttcactacc catcgg                                        2366
```

<210> SEQ ID NO 15
<211> LENGTH: 3294
<212> TYPE: DNA

<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor
      protein coding sequence

<400> SEQUENCE: 15

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gcgtcctagg actcaccttt ccctgatcct gcaccgtccc tctcctggcc ccagactctc    120

cctcccactg ttcacgaagc ccaggtaccc actgatggta atgctggcct gctggctgaa    180

ccccagattg ccatgttctg tggcagactg aacatgcaca tgaatgtcca gaatgggaag    240

tgggattcag atccatcagg gaccaaaacc tgcattgata ccaaggaagg catcctgcag    300

tattgccaag aagtctaccc tgaactgcag atcaccaatg tggtagaagc caaccaacca    360

gtgaccatcc agaactggtg caagcggggc cgcaagcagt gcaagaccca tccccacttt    420

gtgattccct accgctgctt agttggtgag tttgtaagtg atgcccttct cgttcctgac    480

aagtgcaaat tcttacacca ggagaggatg gatgtttgcg aaactcatct tcactggcac    540

accgtcgcca aagagacatg cagtgagaag agtaccaact tgcatgacta cggcatgttg    600

ctgccctgcg gaattgacaa gttccgaggg gtagagtttg tgtgttgccc actggctgaa    660

gaaagtgaca atgtggattc tgctgatgcg gaggaggatg actcggatgt ctggtggggc    720

ggagcagaca cagactatgc agatgggagt gaagacaaag tagtagaagt agcagaggag    780

gaagaagtgg ctgaggtgga agaagaagaa gccgatgatg acgaggacga tgaggatggt    840

gatgaggtag aggaagaggc tgaggaaccc tacgaagaag ccacagagag aaccaccagc    900

attgccacca ccaccaccac caccacagag tctgtggaag aggtggttcg agttcctaca    960

acagcagcca gtacccctga tgccgttgac aagtatctcg agacacctgg ggatgagaat   1020

gaacatgccc atttccagaa agccaaagag aggcttgagg ccaagcaccg agagagaatg   1080

tcccaggtca tgagagaatg ggaagaggca gaacgtcaag caaagaactt gcctaaagct   1140

gataagaagg cagttatcca gcatttccag gagaaagtgg aatctttgga acaggaagca   1200

gccaacgaga gacagcagct ggtggagaca cacatggcca gagtggaagc catgctcaat   1260

gaccgccgcc gcctggccct ggagaactac atcaccgctc tgcaggctgt tcctcctcgg   1320

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cggtcccagg ttatgacaca cctccgtgtg atttatgagc gcatgaatca gtctctctcc   1500

ctgctctaca acgtgcctgc agtggccgag gagattcagg atgaagttga tgagctgctt   1560

cagaaagagc aaaactattc agatgacgtc ttggccaaca tgattagtga accaaggatc   1620

agttacggaa acgatgctct catgccatct ttgaccgaaa cgaaaaccac cgtggagctc   1680

cttcccgtga atggagagtt cagcctggac gatctccagc cgtggcattc ttttggggct   1740

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gaccgaggac tgaccactcg accaggttct gggttgacaa atatcaagac ggaggagatc   1860

tctgaagtga agatggatgc agaattccga catgactcag gatatgaagt tcatcatcaa   1920

aaattggtgt tctttgcaga agatgtgggt tcaaacaaag gtgcaatcat tggactcatg   1980

gtgggcggtg ttgtcatagc gacagtgatc gtcatcacct tggtgatgct gaagaagaaa   2040

cagtacacat ccattcatca tggtgtggtg gaggttgacg ccgctgtcac cccagaggag   2100

cgccacctgt ccaagatgca gcagaacggc tacgaaaatc caacctacaa gttctttgag   2160
```

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cagatgcaga actagacccc cgccacagca gcctctgaag ttggacagca aaaccattgc   2220

ttcactaccc atcggtgtcc atttatagaa taatgtggga agaaacaaac ccgttttatg   2280

atttactcat tatcgccttt tgacagctgt gctgtaacac aagtagatgc ctgaacttga   2340

attaatccac acatcagtaa tgtattctat ctctctttac attttggtct ctatactaca   2400

ttattaatgg gttttgtgta ctgtaaagaa tttagctgta tcaaactagt gcatgaatag   2460

attctctcct gattatttat cacatagccc cttagccagt tgtatattat tcttgtggtt   2520

tgtgacccaa ttaagtccta ctttacatat gctttaagaa tcgatggggg atgcttcatg   2580

tgaacgtggg agttcagctg cttctcttgc ctaagtattc ctttcctgat cactatgcat   2640

tttaaagtta aacattttta agtatttcag atgctttaga gagatttttt ttccatgact   2700

gcattttact gtacagattg ctgcttctgc tatatttgtg atataggaat taagaggata   2760

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gtaagcactt ttacggggcg ggtggggagg ggtgctctgc tggtcttcaa ttaccaagaa   2940

ttctccaaaa caattttctg caggatgatt gtacagaatc attgcttatg acatgatcgc   3000

tttctacact gtattacata aataaattaa ataaaataac cccgggcaag acttttcttt   3060

gaaggatgac tacagacatt aaataatcga agtaattttg ggtggggaga agaggcagat   3120

tcaattttct ttaaccagtc tgaagtttca tttatgatac aaaagaagat gaaaatggaa   3180

gtggcaatat aaggggatga ggaaggcatg cctggacaaa cccttctttt aagatgtgtc   3240

ttcaatttgt ataaaatggt gttttcatgt aaataaatac attcttggag gagc         3294
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<210> SEQ ID NO 16
<211> LENGTH: 3633
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor protein coding sequence

<400> SEQUENCE: 16

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atccacgcct aaatagcaca gccttgctgt gcgtggtaga agttgggtta gtgttgacat    120

gctgttgact caccctcccg aggatggaag ctctggcctg ggtcaagttg tggtcactgc    180

agttaacagt ttgttgatct cagggagtat tccacagttg ctgatgtaat tgacaatgat    240

tggagccagc tcttccccag attcaaatgg accaattaga ggacttgttg gttctgttta    300

tcaactatgt acccactgat ggtaatgctg gcctgctggc tgaaccccag attgccatgt    360

tctgtggcag actgaacatg cacatgaatg tccagaatgg gaagtgggat tcagatccat    420

cagggaccaa aacctgcatt gataccaagg aaggcatcct gcagtattgc caagaagtct    480

accctgaact gcagatcacc aatgtggtag aagccaacca accagtgacc atccagaact    540

ggtgcaagcg gggccgcaag cagtgcaaga cccatcccca ctttgtgatt ccctaccgct    600

gcttagttgg tgagtttgta agtgatgccc ttctcgttcc tgacaagtgc aaattcttac    660

accaggagag gatggatgtt tgcgaaactc atcttcactg gcacaccgtc gccaaagaga    720

catgcagtga gaagagtacc aacttgcatg actacggcat gttgctgccc tgcggaattg    780

acaagttccg aggggtagag tttgtgtgtt gcccactggc tgaagaaagt gacaatgtgg    840

attctgctga tgcggaggag gatgactcgg atgtctggtg ggcggagca gacacagact    900
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-continued

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atgcagatgg gagtgaagac aaagtagtag aagtagcaga ggaggaagaa gtggctgagg    960
tggaagaaga agaagccgat gatgacgagg acgatgagga tggtgatgag gtagaggaag   1020
aggctgagga accctacgaa gaagccacag agagaaccac cagcattgcc accaccacca   1080
ccaccaccac agagtctgtg gaagaggtgg ttcgagaggt gtgctctgaa caagccgaga   1140
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ccccattctt ttacggcgga tgtggcggca accggaacaa ctttgacaca gaagagtact   1260
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aggagaaagt ggaatctttg gaacaggaag cagccaacga gagacagcag ctggtggaga   1560
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acatcaccgc tctgcaggct gttcctcctc ggcctcgtca cgtgttcaat atgctaaaga   1680
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tgatttatga gcgcatgaat cagtctctct ccctgctcta caacgtgcct gcagtggccg   1860
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tcttggccaa catgattagt gaaccaagga tcagttacgg aaacgatgct ctcatgccat   1980
ctttgaccga aacgaaaacc accgtggagc tccttcccgt gaatggagag ttcagcctgg   2040
acgatctcca gccgtggcat tcttttgggg ctgactctgt gccagccaac acagaaaacg   2100
aagttgagcc tgttgatgcc cgccctgctg ccgaccgagg actgaccact cgaccaggtt   2160
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gacatgactc aggatatgaa gttcatcatc aaaaattggt gttctttgca gaagatgtgg   2280
gttcaaacaa aggtgcaatc attggactca tggtgggcgg tgttgtcata gcgacagtga   2340
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cagcctctga agttggacag caaaaccatt gcttcactac ccatcggtgt ccatttatag   2580
aataatgtgg gaagaaacaa acccgtttta tgatttactc attatcgcct tttgacagct   2640
gtgctgtaac acaagtagat gcctgaactt gaattaatcc acacatcagt aatgtattct   2700
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aatttagctg tatcaaacta gtgcatgaat agattctctc ctgattattt atcacatagc   2820
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atgtgcacac attaggcatt gagacttcaa gcttttcttt ttttgtccac gtatctttgg   3180
gtctttgata aagaaaagaa tccctgttca ttgtaagcac ttttacgggg cgggtgggga   3240
ggggtgctct gctggtcttc aattaccaag aattctccaa aacaattttc tgcaggatga   3300
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ttgtacagaa tcattgctta tgacatgatc gctttctaca ctgtattaca taaataaatt 3360

aaataaaata accccgggca agacttttct ttgaaggatg actacagaca ttaaataatc 3420

gaagtaattt tgggtgggga gaagaggcag attcaatttt ctttaaccag tctgaagttt 3480

catttatgat acaaaagaag atgaaaatgg aagtggcaat ataaggggat gaggaaggca 3540

tgcctggaca aacccttctt ttaagatgtg tcttcaattt gtataaaatg gtgttttcat 3600

gtaaataaat acattcttgg aggagcacca ttg 3633

<210> SEQ ID NO 17
<211> LENGTH: 2846
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor protein coding sequence

<400> SEQUENCE: 17

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gctgttgact caccctcccg aggatggaag ctctggcctg ggtcaagttg tggtcactgc 180

agttaacagt ttgttgatct cagggagtat tccacagttg ctgatgtaat tgacaatgat 240

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tcaactatgt acccactgat ggtaatgctg gcctgctggc tgaaccccag attgccatgt 360

tctgtggcag actgaacatg cacatgaatg tccagaatgg gaagtgggat tcagatccat 420

cagggaccaa aacctgcatt gataccaagg aaggcatcct gcagtattgc caagaagtct 480

accctgaact gcagatcacc aatgtggtag aagccaacca accagtgacc atccagaact 540

ggtgcaagcg gggccgcaag cagtgcaaga cccatcccca ctttgtgatt ccctaccgct 600

gcttagttgg tgagtttgta agtgatgccc ttctcgttcc tgacaagtgc aaattcttac 660

accaggagag gatggatgtt tgcgaaactc atcttcactg gcacaccgtc gccaaagaga 720

catgcagtga gaagagtacc aacttgcatg actacggcat gttgctgccc tgcggaattg 780

acaagttccg aggggtagag tttgtgtgtt gcccactggc tgaagaaagt gacaatgtgg 840

attctgctga tgcggaggag gatgactcgg atgtctggtg ggcggagca gacacagact 900

atgcagatgg gagtgaagac aaagtagtag aagtagcaga ggaggaagaa gtggctgagg 960

tggaagaaga agaagccgat gatgacgagg acgatgagga tggtgatgag gtagaggaag 1020

aggctgagga accctacgaa gaagccacag agagaaccac cagcattgcc accaccacca 1080

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cggggccgtg ccgagcaatg atctcccgct ggtactttga tgtgactgaa gggaagtgtg 1200

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cacacatggc cagagtggaa gccatgctca atgaccgccg ccgcctggcc ctggagaact 1620

acatcaccgc tctgcaggct gttcctcctc ggcctcgtca cgtgttcaat atgctaaaga 1680

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agtatgtccg cgcagaacag aaggacagac agcacaccct aaagcatttc gagcatgtgc   1740
gcatggtgga tcccaagaaa gccgctcaga tccggtccca ggttatgaca cacctccgtg   1800
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tggaggttga cgccgctgtc accccagagg agcgccacct gtccaagatg cagcagaacg   2460
gctacgaaaa tccaacctac aagttctttg agcagatgca gaactagacc cccgccacag   2520
cagcctctga agttggacag caaaaccatt gcttcactac ccatcggtgt ccatttatag   2580
aataatgtgg gaagaaacaa acccgtttta tgatttactc attatcgcct tttgacagct   2640
gtgctgtaac acaagtagat gcctgaactt gaattaatcc acacatcagt aatgtattct   2700
atctctcttt acattttggt ctctatacta cattattaat gggttttgtg tactgtaaag   2760
aatttagctg tatcaaacta gtgcatgaat agattctctc ctgattattt atcacatagc   2820
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<210> SEQ ID NO 18
<211> LENGTH: 1846
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor protein coding sequence

<400> SEQUENCE: 18

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tggtgagttt gtaagtgatg cccttctcgt tcctgacaag tgcaaattct tacaccagga    180
gaggatggat gtttgcgaaa ctcatcttca ctggcacacc gtcgccaaag agacatgcag    240
tgagaagagt accaacttgc atgactacgg catgttgctg ccctgcggaa ttgacaagtt    300
ccgaggggta gagtttgtgt gttgcccact ggctgaagaa agtgacaatg tggattctgc    360
tgatgcggag gaggatgact cggatgtctg gtggggcgga gcagacacag actatgcaga    420
tgggagtgaa gacaaagtag tagaagtagc agaggaggaa gaagtggctg aggtggaaga    480
agaagaagcc gatgatgacg aggacgatga ggatggtgat gaggtagagg aagaggctga    540
ggaaccctac gaagaagcca cagagagaac caccagcatt gccaccacca ccaccaccac    600
cacagagtct gtggaagagg tggttcgaga ggtgtgctct gaacaagccg agacggggcc    660
gtgccgagca atgatctccc gctggtactt tgatgtgact gaagggaagt gtgcccccatt    720
cttttacggc ggatgtggcg gcaaccggaa caactttgac acagaagagt actgcatggc    780
cgtgtgtggc agcgccattc ctacaacagc agccagtacc cctgatgccg ttgacaagta    840
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tgaggccaag caccgagaga gaatgtccca ggtcatgaga gaatgggaag aggcagaacg 960

tcaagcaaag aacttgccta aagctgataa gaaggcagtt atccagcatt tccaggagaa 1020

agtggaatct ttggaacagg aagcagccaa cgagagacag cagctggtgg agacacacat 1080

ggccagagtg gaagccatgc tcaatgaccg ccgccgcctg gccctggaga actacatcac 1140

cgctctgcag gctgttcctc ctcggcctcg tcacgtgttc aatatgctaa agaagtatgt 1200

ccgcgcagaa cagaaggaca gacagcacac cctaaagcat ttcgagcatg tgcgcatggt 1260

ggatcccaag aaagccgctc agatccggtc ccaggttatg acacacctcc gtgtgattta 1320

tgagcgcatg aatcagtctc tctccctgct ctacaacgtg cctgcagtgg ccgaggagat 1380

tcaggatgaa gttggtgcag tggctcatgc ctgtaattcc agcattttgg gaggccaagg 1440

tgggcagatg acttgagccc agaagttcaa gaccagaatg ggaaacatgg caagaccaca 1500

tttctacaaa aaaattatcc aggcatgata acatctattt gtagtcccag ctactcagga 1560

ggctgtggtg ggaggatctc ctgagcctgg ggtggctgag gctgcagtga gccttgatca 1620

cgccacctgg gcaatagagc aagaccctgt ctcaaaaaaa ggaagaaaaa gactattatt 1680

tcccccattg aatggtcttg gcactattac acaaaatcaa ttgtccatag ataatatggg 1740

tttatttctt aattcttagt tcttttcttt gatctgtgtg cctgtgctta ctgtagtacc 1800

acactgtttt gattattgta gctttgtagt aaattttgaa atcagc 1846

<210> SEQ ID NO 19
<211> LENGTH: 2517
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor protein coding sequence

<400> SEQUENCE: 19

cagcagcgca ctcggtgccc cgcgcagggt cgcgatgctg cccggtttgg cactgctcct 60

gctggccgcc tggacggctc gggcgctgga ggtctaccct gaactgcaga tcaccaatgt 120

ggtagaagcc aaccaaccag tgaccatcca gaactggtgc aagcggggcc gcaagcagtg 180

caagacccat ccccactttg tgattcccta ccgctgctta gttggtgagt ttgtaagtga 240

tgcccttctc gttcctgaca agtgcaaatt cttacaccag gagaggatgg atgtttgcga 300

aactcatctt cactggcaca ccgtcgccaa agagacatgc agtgagaaga gtaccaactt 360

gcatgactac ggcatgttgc tgccctgcgg aattgacaag ttccgagggg tagagtttgt 420

gtgttgccca ctggctgaag aaagtgacaa tgtggattct gctgatgcgg aggaggatga 480

ctcggatgtc tggtggggcg gagcagacac agactatgca gatgggagtg aagacaaagt 540

agtagaagta gcagaggagg aagaagtggc tgaggtggaa gaagaagaag ccgatgatga 600

cgaggacgat gaggatggtg atgaggtaga ggaagaggct gaggaaccct acgaagaagc 660

cacagagaga accaccagca ttgccaccac caccaccacc accacagagt ctgtggaaga 720

ggtggttcga gaggtgtgct ctgaacaagc cgagacgggg ccgtgccgag caatgatctc 780

ccgctggtac tttgatgtga ctgaagggaa gtgtgcccca ttcttttacg gcggatgtgg 840

cggcaaccgg aacaactttg acacagaaga gtactgcatg gccgtgtgtg gcagcgccat 900

gtcccaaagt ttactcaaga ctacccagga acctcttgcc cgagatcctg ttaaacttcc 960

tacaacagca gccagtaccc ctgatgccgt tgacaagtat ctcgagacac ctggggatga 1020

```
gaatgaacat gcccatttcc agaaagccaa agagaggctt gaggccaagc accgagagag   1080
aatgtcccag gtcatgagag aatgggaaga ggcagaacgt caagcaaaga acttgcctaa   1140
agctgataag aaggcagtta tccagcattt ccaggagaaa gtggaatctt tggaacagga   1200
agcagccaac gagagacagc agctggtgga gacacacatg gccagagtgg aagccatgct   1260
caatgaccgc cgccgcctgg ccctggagaa ctacatcacc gctctgcagg ctgttcctcc   1320
tcggcctcgt cacgtgttca atatgctaaa gaagtatgtc cgcgcagaac agaaggacag   1380
acagcacacc ctaaagcatt tcgagcatgt gcgcatggtg gatcccaaga aagccgctca   1440
gatccggtcc caggttatga cacacctccg tgtgatttat gagcgcatga atcagtctct   1500
ctccctgctc tacaacgtgc ctgcagtggc cgaggagatt caggatgaag ttgatgagct   1560
gcttcagaaa gagcaaaact attcagatga cgtcttggcc aacatgatta gtgaaccaag   1620
gatcagttac ggaaacgatg ctctcatgcc atctttgacc gaaacgaaaa ccaccgtgga   1680
gctccttccc gtgaatggag agttcagcct ggacgatctc cagccgtggc attcttttgg   1740
ggctgactct gtgccagcca acacagaaaa cgaagttgag cctgttgatg cccgccctgc   1800
tgccgaccga ggactgacca ctcgaccagg ttctgggttg acaaatatca agacggagga   1860
gatctctgaa gtgaagatgg atgcagaatt ccgacatgac tcaggatatg aagttcatca   1920
tcaaaaattg gtgttctttg cagaagatgt gggttcaaac aaaggtgcaa tcattggact   1980
catggtgggc ggtgttgtca tagcgacagt gatcgtcatc accttggtga tgctgaagaa   2040
gaaacagtac acatccattc atcatggtgt ggtggaggtt gacgccgctg tcaccccaga   2100
ggagcgccac ctgtccaaga tgcagcagaa cggctacgaa aatccaacct acaagttctt   2160
tgagcagatg cagaactaga cccccgccac agcagcctct gaagttggac agcaaaacca   2220
ttgcttcact acccatcggt gtccatttat agaataatgt gggaagaaac aaacccgttt   2280
tatgatttac tcattatcgc cttttgacag ctgtgctgta acacaagtag atgcctgaac   2340
ttgaattaat ccacacatca gtaatgtatt ctatctctct ttacattttg gtctctatac   2400
tacattatta atgggttttg tgtactgtaa agaatttagc tgtatcaaac tagtgcatga   2460
atagattctc tcctgattat ttatcacata gccccttagc cagttgtata ttattct      2517
```

<210> SEQ ID NO 20
<211> LENGTH: 541
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor protein coding sequence

<400> SEQUENCE: 20

```
gggcggagca gacacagact atgcagatgg gagtgaagac aaagtagtag aagtagcaga     60
ggaggaagaa gtggctgagg tggaagaaga agaagccgat gatgacgagg acgatgagga    120
tggtgatgag gtagaggaag aggctgagga accctacgaa gaagccacag agagaaccac    180
cagcattgcc accaccacca ccaccaccac agagtctgtg gaagaggtgg ttcgagtgtc    240
ccaaagttta ctcaagacta cccaggaacc tcttgcccga gatcctgtta aacttcctac    300
aacagcagcc agtacccctg atgccgttga caagtatctc gagacacctg gggatgagaa    360
tgaacatgcc catttccaga aagccaaaga gaggcttgag gccaagcacc gagagagaat    420
gtcccaggtc atgagagaat gggaagaggc agaacgtcaa gcaaagaact tgcctaaagc    480
tgataagaag gcagttatcc agcatttcca ggagaaagtg gaatctttgg aacaggaagc    540
```

```
a                                                                  541


<210> SEQ ID NO 21
<211> LENGTH: 816
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor
      protein coding sequence

<400> SEQUENCE: 21

atccttgcca acctctcaac caggatttaa cttctgcttt tcccccattt tcaaaaatta     60

tagcatgtat ttaaaggcag cagaagcctt actttcaggt ttcccttacc ctttcatttc    120

tttttgttca aaataggtag taattgaagt tttaaatata gggtatcatt tttctttaag    180

agtcatttat caattttctt ctaacttcag gcctagaaag aagttttggg taggctttgt    240

cttacagtgt tattatttat gagtaaaact aattggttgt cctgcatact ttaattatga    300

tgtaatacag gttctgggtt gacaaatatc aagacggagg agatctctga agtgaagatg    360

gatgcagaat tccgacatga ctcaggatat gaagttcatc atcaaaaatt ggtgttcttt    420

gcagaagatg tgggttcaaa caaaggtgca atcattggac tcatggtggg cggtgttgtc    480

atagcgacag tgatcgtcat caccttggtg atgctgaaga agaaacagta cacatccatt    540

catcatggtg tggtggaggt tgacgccgct gtcaccccag aggagcgcca cctgtccaag    600

atgcagcaga acggctacga aaatccaacc tacaagttct ttgagcagat gcagaactag    660

acccccgcca cagcagcctc tgaagttgga cagcaaaacc attgcttcac tacccatcgg    720

tgtccattta tagaataatg tgggaagaaa caaacccgtt ttatgattta ctcattatcg    780

ccttttgaca gctgtgctgt aacacaagta gatgcc                              816


<210> SEQ ID NO 22
<211> LENGTH: 619
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor
      protein coding sequence

<400> SEQUENCE: 22

cagactatgc agatgggagt gaagacaaag tagtagaagt agcagaggag gaagaagtgg     60

ctgaggtgga agaagaagaa gccgatgatg acgaggacga tgaggatggt gatgaggtag    120

aggaagaggc tgaggaaccc tacgaagaag ccacagagag aaccaccagc attgccacca    180

ccaccaccac caccacagag tctgtggaag aggtggttcg agaggtgtgc tctgaacaag    240

ccgagacggg gccgtgccga gcaatgatct cccgctggta ctttgatgtg actgaaggga    300

agtgtgcccc attcttttac ggcggatgtg gcggcaaccg gaacaacttt gacacagaag    360

agtactgcat ggccgtgtgt ggcagcgcca tgtcccaaag tttactcaag actacccagg    420

aacctcttgc ccgagatcct gttaaacgta cgttgtcatt cacctgaggg aagggaagag    480

gggaggagga tgctgcttgg ttcacataac tccagcatca tcaccttctt tgcatggttt    540

tgtgtttctt gaacacctgt cttagtaaaa tgtttcttcc cattaccttg cttgtaatta    600

catctgattt tgccagaca                                                 619


<210> SEQ ID NO 23
<211> LENGTH: 1309
<212> TYPE: DNA
```

<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor protein coding sequence

<400> SEQUENCE: 23

```
cgagagcacg cggaggagcg tgcgcggggg ccccgggaga cggcggcggt ggcggcgcgg     60

gcagagcaag gacgcggcgg atcccactcg cacagcagcg cactcggtgc cccgcgcagg    120

gtcgcgatgc tgcccggttt ggcactgctc ctgctggccg cctggacggc tcgggcgctg    180

gaggtaccca ctgatggtaa tgctggcctg ctggctgaac cccagattgc catgttctgt    240

ggcagactga acatgcacat gaatgtccag aatgggaagt gggattcaga tccatcaggg    300

accaaaacct gcattgatac caaggaaggc atcctgcagt attgccaaga agtctaccct    360

gaactgcaga tcaccaatgt ggtagaagcc aaccaaccag tgaccatcca gaactggtgc    420

aagcggggcc gcaagcagtg caagacccat ccccactttg tgattcccta ccgctgctta    480

gttggtgagt ttgtaagtga tgcccttctc gttcctgaca agtgcaaatt cttacaccag    540

gagaggatgg atgtttgcga aactcatctt cactggcaca ccgtcgccaa agagacatgc    600

agtgagaaga gtaccaactt gcatgactac ggcatgttgc tgccctgcgg aattgacaag    660

ttccgagggg tagagtttgt gtgttgccca ctggctgaag aaagtgacaa tgtggattct    720

gctgatgcgg aggaggatga ctcggatgtc tggtggggcg gagcagacac agactatgca    780

gatgggagtg aagacaaagt agtagaagta gcagaggagg aagaagtggc tgaggtggaa    840

gaagaagaag ccgatgatga cgaggacgat gaggatggtg atgaggtaga ggaagaggct    900

gaggaaccct acgaagaagc cacagagaga accaccagca ttgccaccac caccaccacc    960

accacagagt ctgtggaaga ggtggttcga gagaagtggt ataaggaagt acattctggc   1020

caggcacgat ggctcatgct gtaatcccag cactttggga ggccgaggtg ggtgcatcac   1080

ctgaggtcag gagtttgaga ccagcctggc caacatggtg aaacccctcg ctactaaaaa   1140

tacaaaaatt agccgggcgt ggtggcacac acctgtggtc ccagctactc gggaggctga   1200

agcaggagaa tcgcttgaac ccgggagacg gaggttgcag taagccgagt tcactccatt   1260

gtactctagc ctgggtgaca gagcgagatt cgtctcaaaa aaaaaaaaa               1309
```

<210> SEQ ID NO 24
<211> LENGTH: 582
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor protein coding sequence

<400> SEQUENCE: 24

```
gtaacattct aaaggtagta gggtcttgat tgggttgctt aggcattaaa aggctgttta     60

acttgtcttg aagtctatct ttccttgatg tcttctgcgg taagaacact gtgatacaga    120

tggaatgacg ggaagtggtt ttcctttctt tcagttggtg agtttgtaag tgatgccctt    180

ctcgttcctg acaagtgcaa attcttacac caggagagga tggatgtttg cgaaactcat    240

cttcactggc acaccgtcgc caaagagaca tgcagtgaga agagtaccaa cttgcatgac    300

tacggcatgt tgctgccctg cggaattgac aagttccgag ggtagagtt tgtgtgttgc    360

ccactggctg aagaaagtga caatgtggat tctgctgatg cggaggagga tgactcggat    420

gtctggtggg gcggagcaga cacagactat gcagatggga gtgaagacaa agtagtagaa    480

gtagcagagg aggaagaagt ggctgaggtg gaagaagaag aagccgatga tgacgaggac    540
```

```
gatgaggatg gtgatgaggt agaggaagag gctgaggaac cc                    582


<210> SEQ ID NO 25
<211> LENGTH: 361
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor
      protein coding sequence

<400> SEQUENCE: 25

actggagggc tgagaagaga ctgatggcat ttgttgttct tgaccttgaa agaagagttg     60

cagattgttg gagcaaggcc agatggtaat aggttggaaa gaacaagtga ggggcgtgag    120

agtgacagtc tacaaccccg ttaagaagtt atctgtgaaa atgcctcttc ctgtcttgat    180

tatagcctcc ctcgcacatg gctttctgag tatgttggtg agtttgtaag tgatgccctt    240

ctcgttcctg acaagtgcaa attcttacac caggagagga tggatgtttg cgaaactcat    300

cttcactggc acaccgtcgc caaagagaca tgcagtgaga agagtaccaa cttgcatgac    360

t                                                                    361


<210> SEQ ID NO 26
<211> LENGTH: 667
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor
      protein coding sequence

<400> SEQUENCE: 26

ctgaacatgc acatgaatgt ccagaatggg aagtgggatt cagatccatc agggaccaaa     60

acctgcattg ataccaagga aggcatcctg cagtattgcc aagaagtcta ccctgaactg    120

cagatcacca atgtggtaga agccaaccaa ccagtgacca tccagaactg gtgcaagcgg    180

ggccgcaagc agtgcaagac ccatccccac tttgtgattc cctaccgctg cttaggtgag    240

ccggccggcc gtggggctgg tgttgattgg gggcctggtc ttgagggaag aaaaagagga    300

tgctcctgtt aggtcacata cacagacttg ttcttcagca cattgccact ctgtgttgta    360

ctgtgttttg gactcttgca gttacattct gtgcactgac cctataggag cagtattttt    420

gagttccctg cctcagaatg aatttaccca gggtgtatat tgaaattaca aattcctggg    480

ccagttccag gactcctgaa tgaaaaatgc ctatagtagc ggatccggga attcttattt    540

taccgtatcg catagatgat tctcatgaac aggggccttg tgtgtttctt cacatagact    600

ttctagaaga aagaatctaa tgtgaagctg cagcattttg ttaatttcta aaaaaaaaaa    660

aaaaaaa                                                              667


<210> SEQ ID NO 27
<211> LENGTH: 374
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amyloid precursor
      protein coding sequence

<400> SEQUENCE: 27

ggcgcggcct cttccctggc agctctgggg actctggttt agttcccctg gggcacagg     60

atgctgggga gggtccgaag ggtctttttt ttagggtgca gataaaagga tcgaattgag    120

tgaagattaa gacggagaag atggcgcctc tgcagtgcag caaagaaaag ctgtgtggag    180
```

-continued

```
gctgcagcct agtgaaatcc acccaccact aggtacccac tgatggtaat gctggcctgc    240

tggctgaacc ccagattgcc atgttctgtg gcagactgaa catgcacatg aatgtccaga    300

atgggaagtg ggattcagat ccatcaggga ccaaaacctg cattgatacc aaggaaggca    360

tcctgcagta ttgc                                                      374
```

What is claimed is:

1. A method for diagnosing biliary atresia (BA) or assessing risk of developing BA in an infant, comprising:

(i) determining expression level of amyloid precursor protein (APP) in a liver sample taken from the infant;

(ii) detecting an increase in the APP expression level from step (i) when compared with a standard control value; and

(iii) determining the infant as having BA or at risk of developing BA.

2. The method of claim **1**, wherein the APP expression level is APP mRNA level.

3. The method of claim **1**, wherein the APP expression level is APP protein level.

4. The method of claim **2**, wherein step (i) comprises a reverse transcription polymerase chain reaction (RT-PCR).

5. The method of claim **3**, wherein step (i) comprises an immunoassay.

6. The method of claim **1**, wherein the liver sample is a liver biopsy.

7. The method of claim **1**, further comprising a step, following step (iii), of performing at least one additional diagnostic test for BA.

8. The method of claim **7**, further comprising a step, following step (iii), of treating the infant with Kasai procedure.

9. A method for assessing effectiveness of Kasai procedure in an infant who has BA and has undergone Kasai procedure, comprising:

(i) determining expression level of amyloid precursor protein (APP) in a liver sample taken from the infant;

(ii) detecting an increase in the APP expression level from step (i) when compared with a standard control value; and

(iii) determining the Kasai procedure as ineffective.

10. The method of claim **9**, wherein the APP expression level is APP mRNA level.

11. The method of claim **9**, wherein the APP expression level is APP protein level.

12. The method of claim **10**, wherein step (i) comprises an RT-PCR.

13. The method of claim **11**, wherein step (i) comprises an immunoassay.

14. The method of claim **9**, wherein the liver sample is a liver biopsy.

15. The method of claim **9**, further comprising a step, following step (iii), of performing liver transplant.

* * * * *